Model Analysis

Table of contents

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```
    cols = list(dict.fromkeys(target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    return df
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r"
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r")as
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
```

```
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)



    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	7340

Nombre de lignes df clean	7340
Nombre de features	77
% de NaN	0.0
Taille dataset entraînement	5138
Taille dataset test	2202
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_1h_mean, r_NH3_st_1h_std, r_NH3_st_1h_rms, r_NH3_st_1h_median, r_NH3_st_1h_min, r_NH3_st_1h_max, r_NH3_st_1h_diff_last, r_NH3_st_1h_slope, r_NH3_st_2h_mean, r_NH3_st_2h_std, r_NH3_st_2h_rms, r_NH3_st_2h_median, r_NH3_st_2h_min, r_NH3_st_2h_max, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_NH3_st_3h_mean, r_NH3_st_3h_std, r_NH3_st_3h_rms, r_NH3_st_3h_median, r_NH3_st_3h_min, r_NH3_st_3h_max, r_NH3_st_3h_diff_last, r_NH3_st_3h_slope, r_temperature_st_1h_mean, r_temperature_st_1h_std, r_temperature_st_1h_rms, r_temperature_st_1h_median, r_temperature_st_1h_min, r_temperature_st_1h_max, r_temperature_st_1h_diff_last, r_temperature_st_1h_slope, r_temperature_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_rms, r_temperature_st_2h_median, r_temperature_st_2h_min, r_temperature_st_2h_max, r_temperature_st_2h_diff_last, r_temperature_st_2h_slope, r_temperature_st_3h_mean, r_temperature_st_3h_std, r_temperature_st_3h_rms, r_temperature_st_3h_median, r_temperature_st_3h_min, r_temperature_st_3h_max, r_temperature_st_3h_diff_last, r_temperature_st_3h_slope, r_umidity_st_1h_mean, r_umidity_st_1h_std, r_umidity_st_1h_rms, r_umidity_st_1h_median, r_umidity_st_1h_min, r_umidity_st_1h_max, r_umidity_st_1h_diff_last, r_umidity_st_1h_slope, r_umidity_st_2h_mean, r_umidity_st_2h_std, r_umidity_st_2h_rms, r_umidity_st_2h_median, r_umidity_st_2h_min, r_umidity_st_2h_max, r_umidity_st_2h_diff_last, r_umidity_st_2h_slope, r_umidity_st_3h_mean, r_umidity_st_3h_std, r_umidity_st_3h_rms, r_umidity_st_3h_median, r_umidity_st_3h_min, r_umidity_st_3h_max, r_umidity_st_3h_diff_last, r_umidity_st_3h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

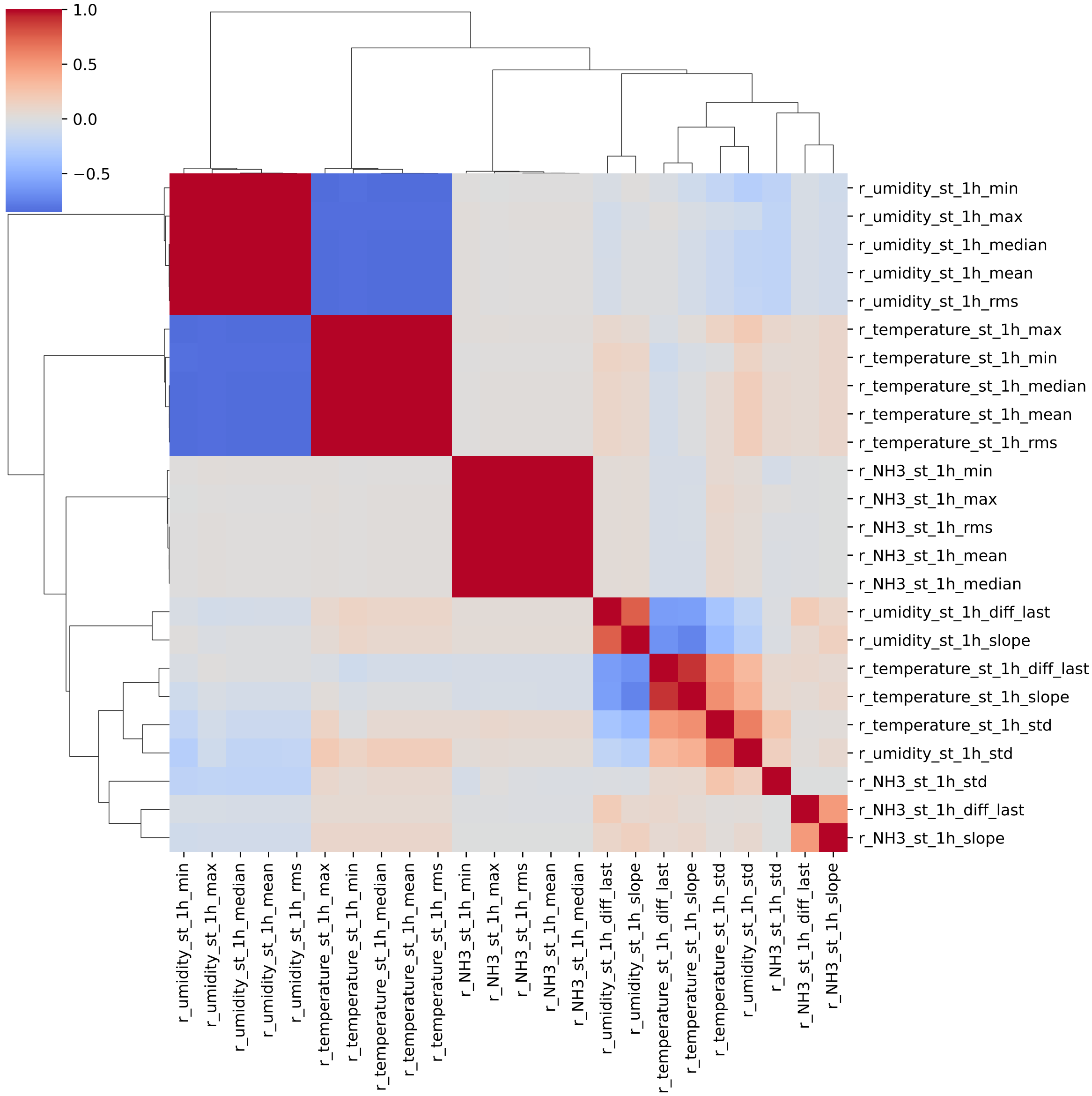
The following section has been done with a threshold choosen of : 0.8

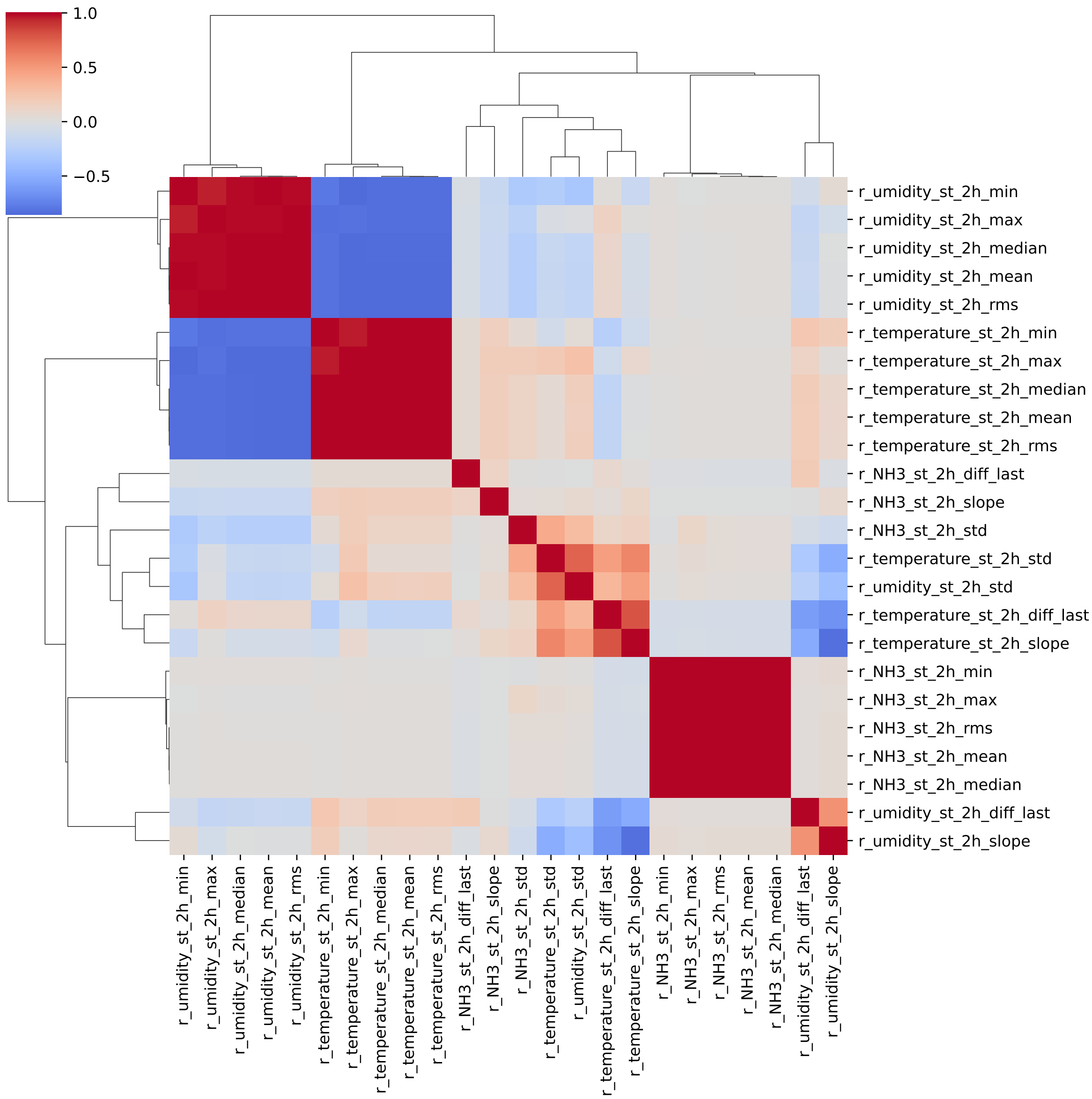
cluster 1: r, *, N, H, 3 ; **cluster 2:** r,* , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, *, u, m, i, d, i, t, y ; **cluster 4:** r,* , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_st_1h_mean, r_NH3_st_1h_median, r_NH3_st_1h_rms, r_NH3_st_1h_max, r_NH3_st_1h_min ; **cluster 7:** r_NH3_st_1h_std ; **cluster 8:** r_NH3_st_1h_diff_last ; **cluster 9:** r_NH3_st_1h_slope ; **cluster 10:** r_temperature_st_1h_median, r_umidity_st_1h_rms, r_temperature_st_1h_min, r_temperature_st_1h_mean, r_umidity_st_1h_mean, r_umidity_st_1h_max, r_temperature_st_1h_max, r_umidity_st_1h_min, r_temperature_st_1h_rms, r_umidity_st_1h_median ; **cluster 11:** r_temperature_st_1h_std ; **cluster 12:** r_temperature_st_1h_slope, r_temperature_st_1h_diff_last ; **cluster 13:** r_umidity_st_1h_std ; **cluster 14:** r_umidity_st_1h_diff_last ; **cluster 15:** r_umidity_st_1h_slope ; **cluster 16:** r_NH3_st_2h_min, r_NH3_st_2h_max, r_NH3_st_2h_mean, r_NH3_st_2h_median, r_NH3_st_2h_rms ; **cluster 17:** r_NH3_st_2h_std ; **cluster 18:** r_NH3_st_2h_diff_last ; **cluster 19:** r_NH3_st_2h_slope ; **cluster 20:** r_umidity_st_2h_min, r_umidity_st_2h_max, r_temperature_st_2h_min, r_temperature_st_2h_max, r_temperature_st_2h_rms, r_umidity_st_2h_rms, r_temperature_st_2h_median, r_umidity_st_2h_mean, r_temperature_st_2h_mean, r_umidity_st_2h_median ; **cluster 21:** r_temperature_st_2h_std ; **cluster 22:** r_temperature_st_2h_diff_last ; **cluster 23:** r_temperature_st_2h_slope, r_umidity_st_2h_slope ; **cluster 24:** r_umidity_st_2h_std ; **cluster 25:** r_umidity_st_2h_diff_last ; **cluster 26:** r_NH3_st_3h_median, r_NH3_st_3h_rms, r_NH3_st_3h_mean, r_NH3_st_3h_max, r_NH3_st_3h_min ; **cluster 27:** r_NH3_st_3h_std ; **cluster 28:** r_NH3_st_3h_diff_last ; **cluster 29:** r_NH3_st_3h_slope ; **cluster 30:** r_umidity_st_3h_min, r_umidity_st_3h_mean, r_umidity_st_3h_max, r_temperature_st_3h_median, r_umidity_st_3h_median, r_temperature_st_3h_min, r_temperature_st_3h_mean, r_umidity_st_3h_rms, r_temperature_st_3h_max, r_temperature_st_3h_rms ; **cluster 31:** r_temperature_st_3h_std, r_umidity_st_3h_std ; **cluster 32:** r_temperature_st_3h_diff_last ; **cluster 33:** r_umidity_st_3h_slope, r_temperature_st_3h_slope ; **cluster 34:** r_umidity_st_3h_diff_last ;

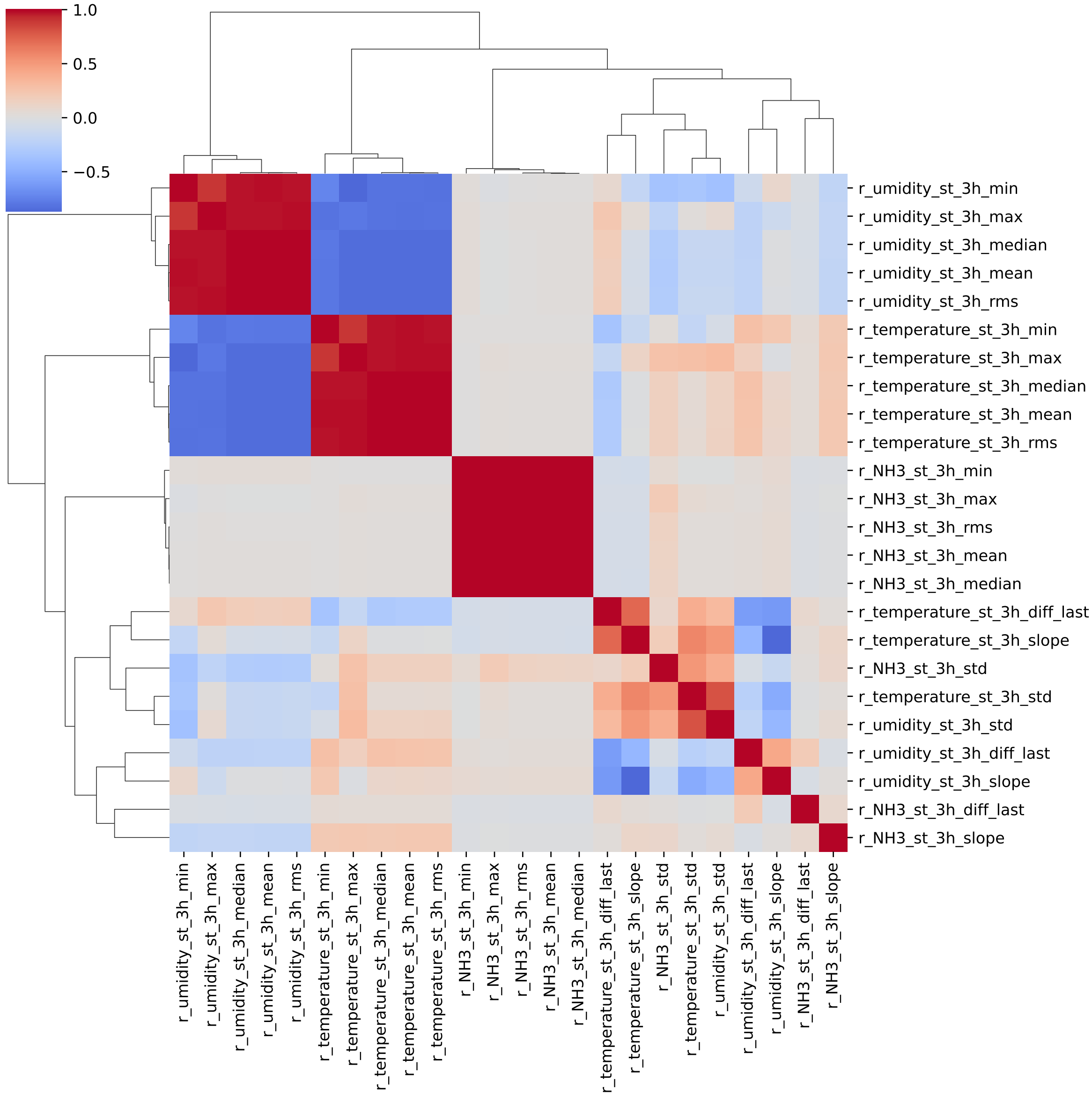
2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_1h_mean, r_NH3_st_1h_std, r_NH3_st_1h_diff_last, r_NH3_st_1h_slope, r_temperature_st_1h_median, r_temperature_st_1h_std, r_temperature_st_1h_slope, r_umidity_st_1h_std, r_umidity_st_1h_diff_last, r_umidity_st_1h_slope, r_NH3_st_2h_min, r_NH3_st_2h_std, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_umidity_st_2h_min, r_temperature_st_2h_std, r_temperature_st_2h_diff_last, r_temperature_st_2h_slope, r_umidity_st_2h_std, r_umidity_st_2h_diff_last, r_NH3_st_3h_median, r_NH3_st_3h_std, r_NH3_st_3h_diff_last, r_NH3_st_3h_slope, r_umidity_st_3h_min, r_temperature_st_3h_std, r_temperature_st_3h_diff_last, r_umidity_st_3h_slope, r_umidity_st_3h_diff_last




1.0
0.5
0.0
−0.5
r_umidity_st_2h_min
r_umidity_st_2h_max
r_umidity_st_2h_median
r_umidity_st_2h_mean
r_umidity_st_2h_rms
r_temperature_st_2h_min
r_temperature_st_2h_max
r_temperature_st_2h_median
r_temperature_st_2h_mean
r_temperature_st_2h_rms
r_NH3_st_2h_diff_last
r_NH3_st_2h_slope
r_NH3_st_2h_std
r_temperature_st_2h_std
r_umidity_st_2h_std
r_temperature_st_2h_diff_last
r_temperature_st_2h_slope
r_NH3_st_2h_min
r_NH3_st_2h_max
r_NH3_st_2h_rms
r_NH3_st_2h_mean
r_NH3_st_2h_median
r_umidity_st_2h_diff_last
r_umidity_st_2h_slope



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_1h_mean, r_NH3_st_1h_std, r_NH3_st_1h_diff_last, r_NH3_st_1h_slope, r_temperature_st_1h_median, r_temperature_st_1h_std, r_temperature_st_1h_slope, r_umidity_st_1h_std, r_umidity_st_1h_diff_last, r_umidity_st_1h_slope, r_NH3_st_2h_min, r_NH3_st_2h_std, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_umidity_st_2h_min, r_temperature_st_2h_std, r_temperature_st_2h_diff_last, r_temperature_st_2h_slope, r_umidity_st_2h_std, r_umidity_st_2h_diff_last, r_NH3_st_3h_median, r_NH3_st_3h_std, r_NH3_st_3h_diff_last, r_NH3_st_3h_slope, r_umidity_st_3h_min, r_temperature_st_3h_std, r_temperature_st_3h_diff_last, r_umidity_st_3h_slope, r_umidity_st_3h_diff_last

Len(Features_X): 34

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model': None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

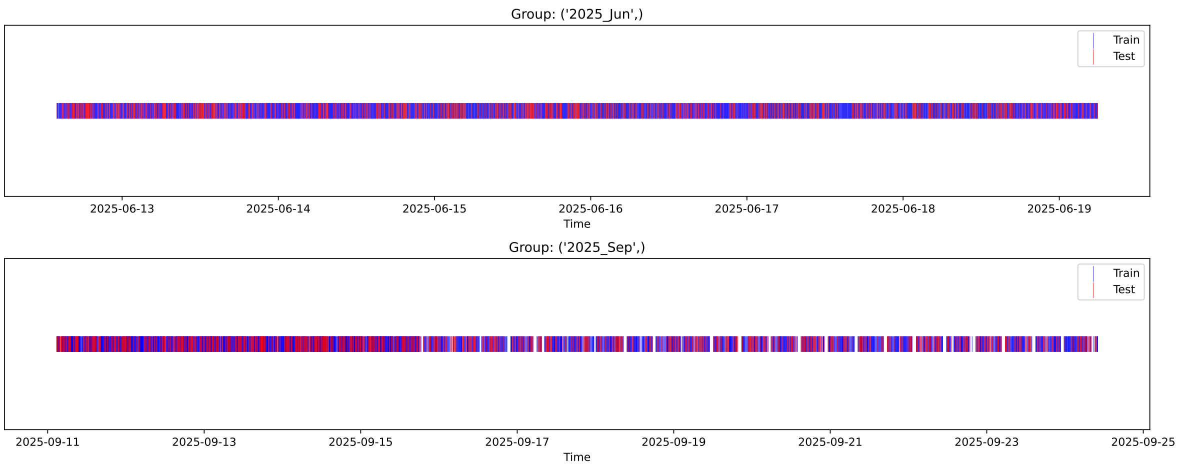
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

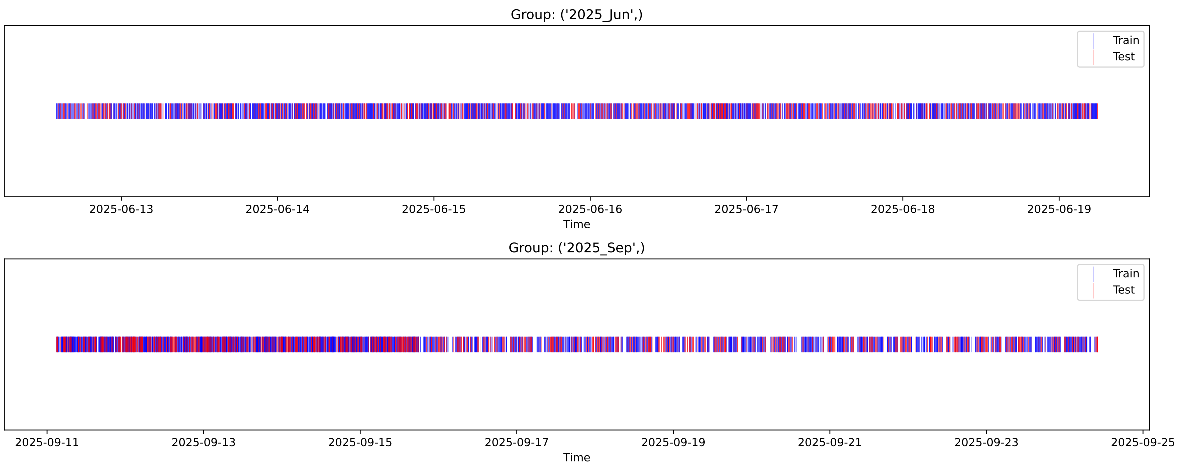
External Split: Train: 5138 samples Test: 2202 samples



3.3.2 Internal CV Folds

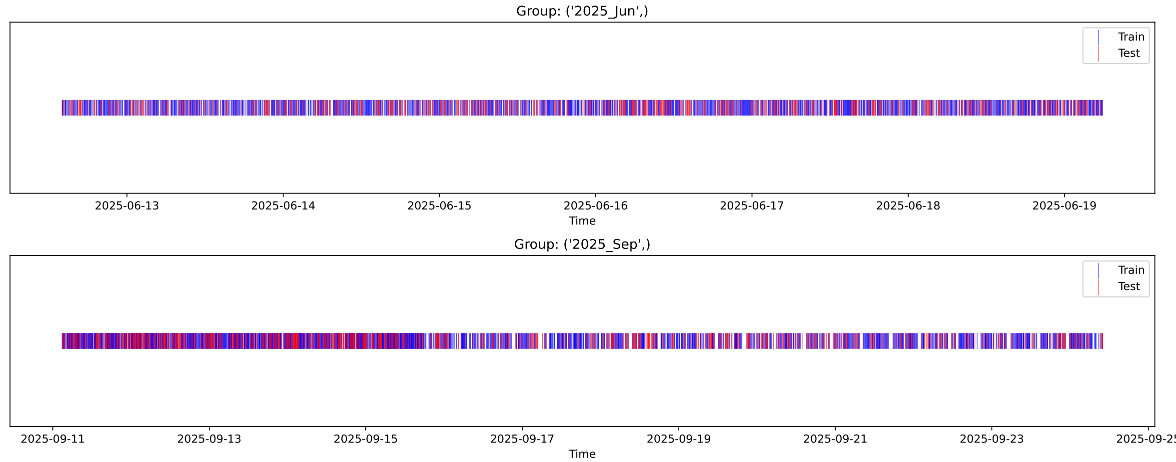
Fold 1/5

Train: 3596 samples Test: 1542 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3596 samples Test: 1542 samples Excluded: 0 samples Window exclusion: 0h

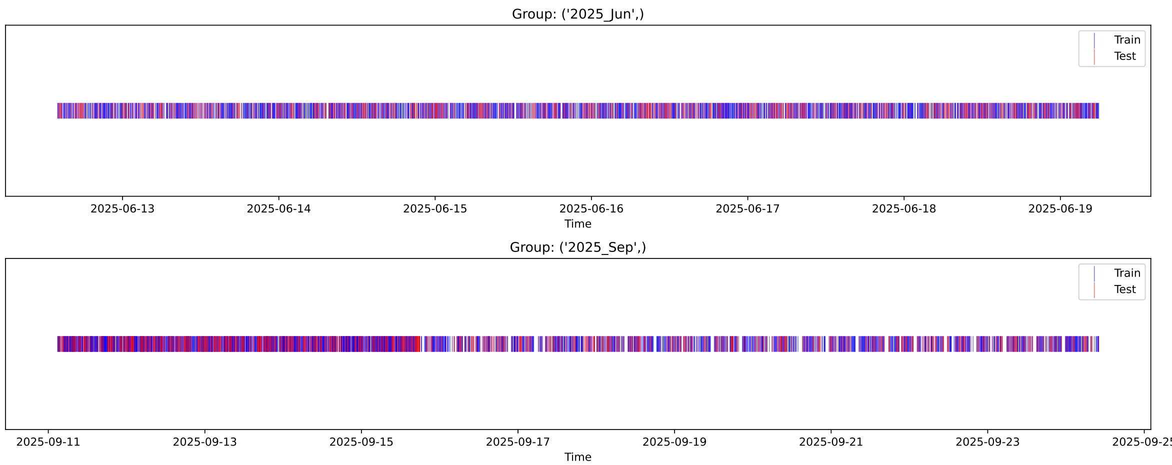


Fold 3/5

Train: 3596 samples Test: 1542 samples Excluded: 0 samples Window exclusion: 0h

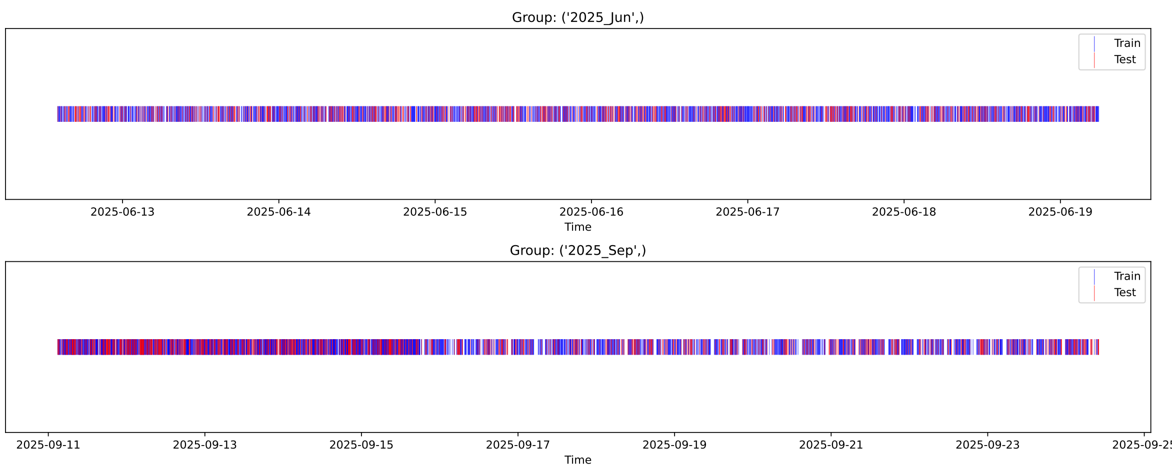
Fold 4/5

Train: 3596 samples Test: 1542 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3596 samples Test: 1542 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.361, std=0.812, min=0.959, max=13.845 y_test: mean=2.360, std=0.767, min=0.973, max=6.594

Fold 0 — y_test: mean=2.35, std=0.80, y_train: mean=2.36, std=0.82

Fold 1 — y_test: mean=2.36, std=0.86, y_train: mean=2.36, std=0.79

Fold 2 — y_test: mean=2.39, std=0.86, y_train: mean=2.35, std=0.79

Fold 3 — y_test: mean=2.37, std=0.82, y_train: mean=2.36, std=0.81

Fold 4 — y_test: mean=2.37, std=0.85, y_train: mean=2.36, std=0.80

train scores CV: [0.52800786 0.56343568 0.56810299 0.52661871 0.56697631]

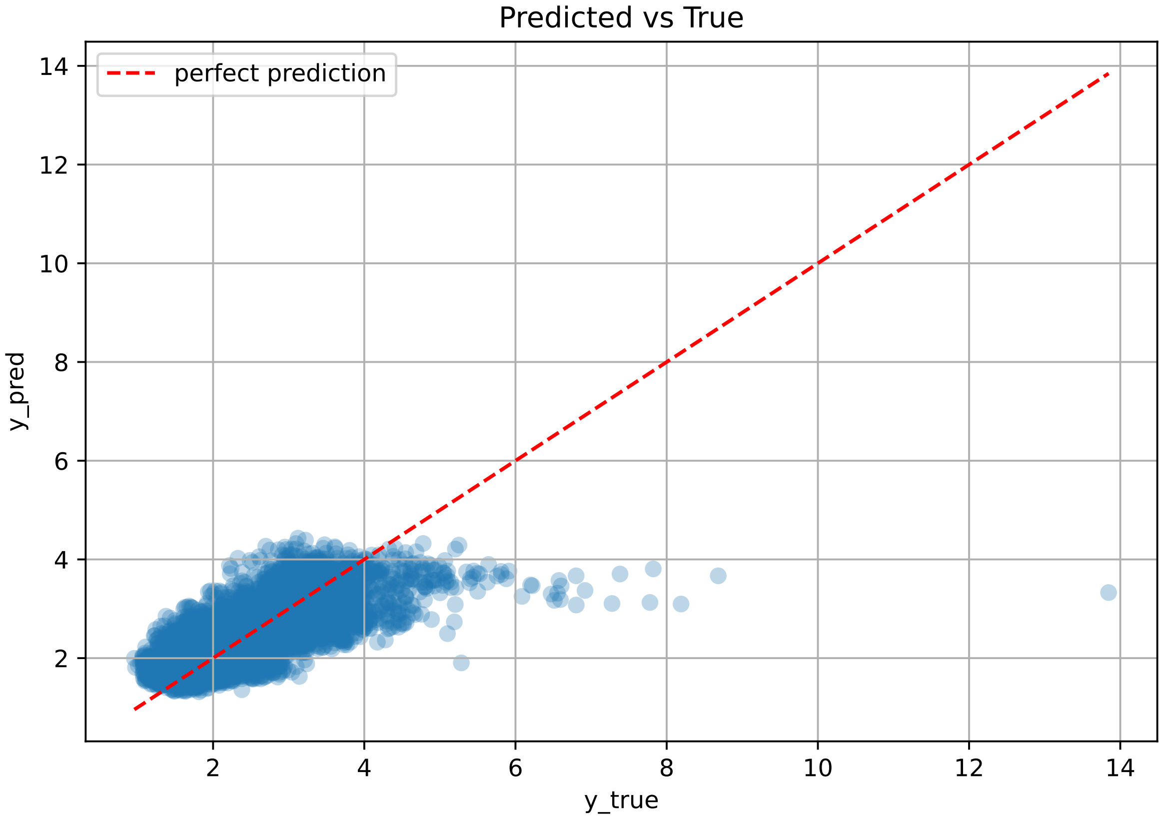
test scores CV: [0.57068975 0.4903515 0.48169112 0.56996698 0.48489147]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5506	0.5195	0.0416	0.5694	0.5688	0.0493	1.0948	1.0599

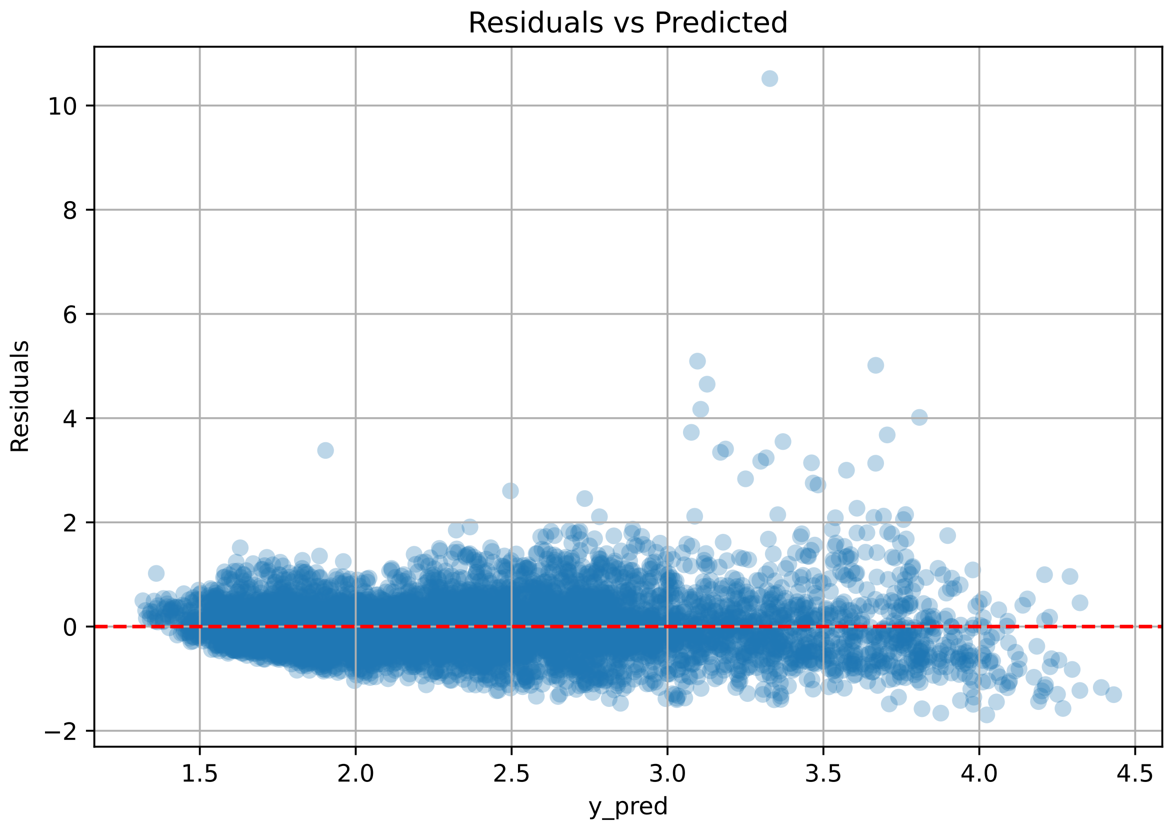
Coefficients:

coef_r_CO2 = -0.0437, coef_r_NH3 = 0.2259, coef_r_NH3_st_1h_diff_last = 0.0010, coef_r_NH3_st_1h_mean = -0.3298, coef_r_NH3_st_1h_slope = 0.0114, coef_r_NH3_st_1h_std = 0.0595, coef_r_NH3_st_2h_diff_last = 0.0010, coef_r_NH3_st_2h_min = -0.3456, coef_r_NH3_st_2h_slope = -0.0248, coef_r_NH3_st_2h_std = -0.0064, coef_r_NH3_st_3h_diff_last = 0.0010, coef_r_NH3_st_3h_median = 0.4119, coef_r_NH3_st_3h_slope = 0.0550, coef_r_NH3_st_3h_std = 0.0384, coef_r_THI = 0.4377, coef_r_temperature = 0.5836, coef_r_temperature_st_1h_median = -1.0384, coef_r_temperature_st_1h_slope = -0.0585, coef_r_temperature_st_1h_std = -0.0104, coef_r_temperature_st_2h_diff_last = -0.0205, coef_r_temperature_st_2h_slope = 0.0444, coef_r_temperature_st_2h_std = -0.0092, coef_r_temperature_st_3h_diff_last = -0.0205, coef_r_temperature_st_3h_std = 0.1565, coef_r_umidity = -0.6309, coef_r_umidity_st_1h_diff_last = -0.0073, coef_r_umidity_st_1h_slope = -0.0353, coef_r_umidity_st_1h_std = 0.0563, coef_r_umidity_st_2h_diff_last = -0.0073, coef_r_umidity_st_2h_min = -0.1201, coef_r_umidity_st_2h_std = 0.0209, coef_r_umidity_st_3h_diff_last = -0.0073, coef_r_umidity_st_3h_min = 0.3717, coef_r_umidity_st_3h_slope = 0.0002, intercept = 2.3612,

3.4.2 Scatter


Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

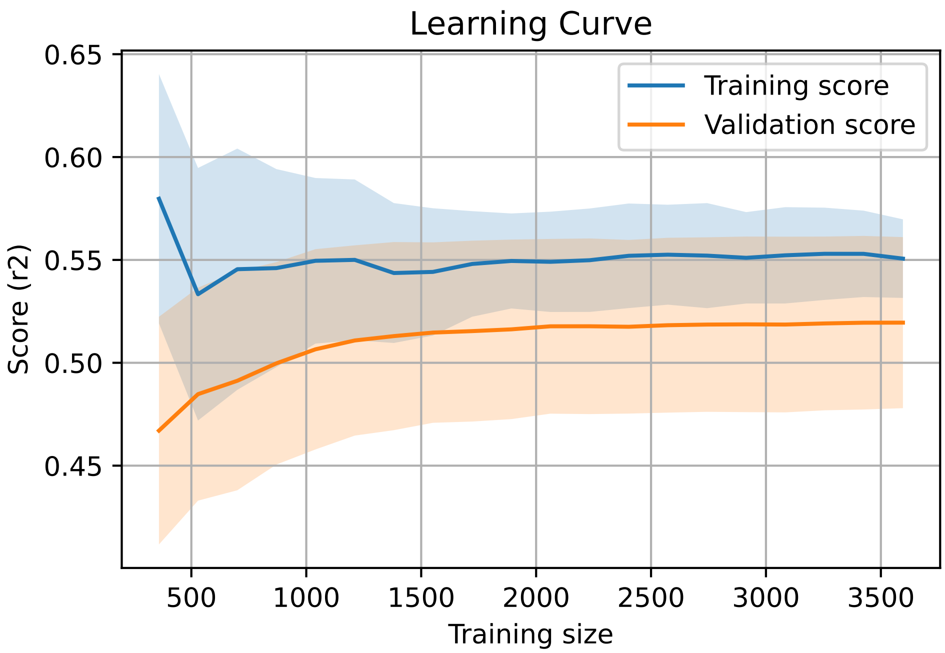
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 7340, **Len(training)** : 5138,

Len(training_real) : 3597, **Len(test_of_training)** : 1541, **Len(test)** : 2202

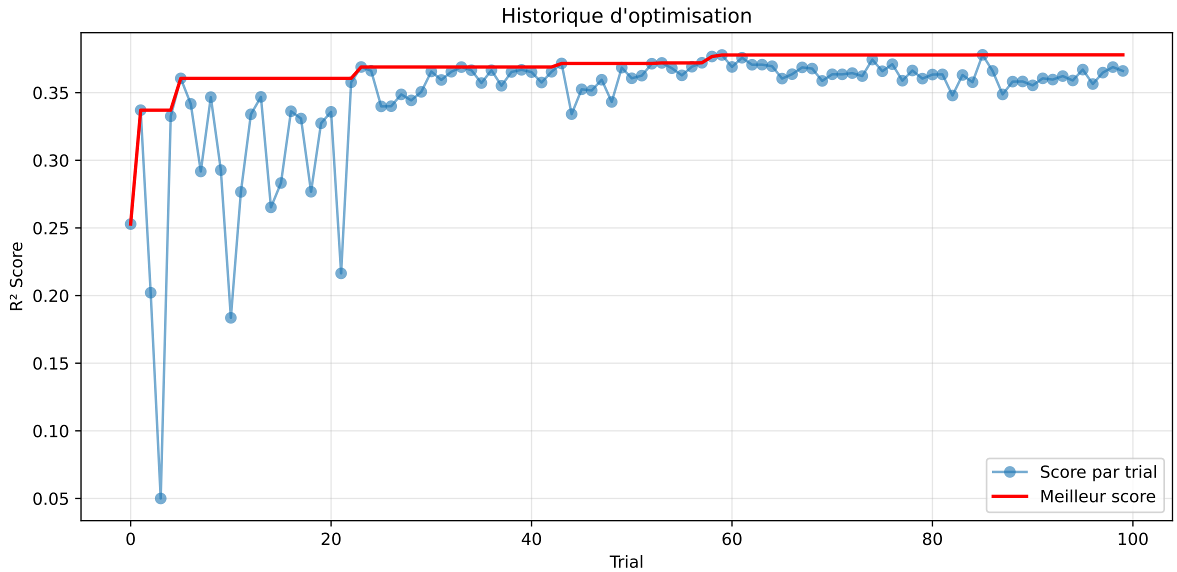


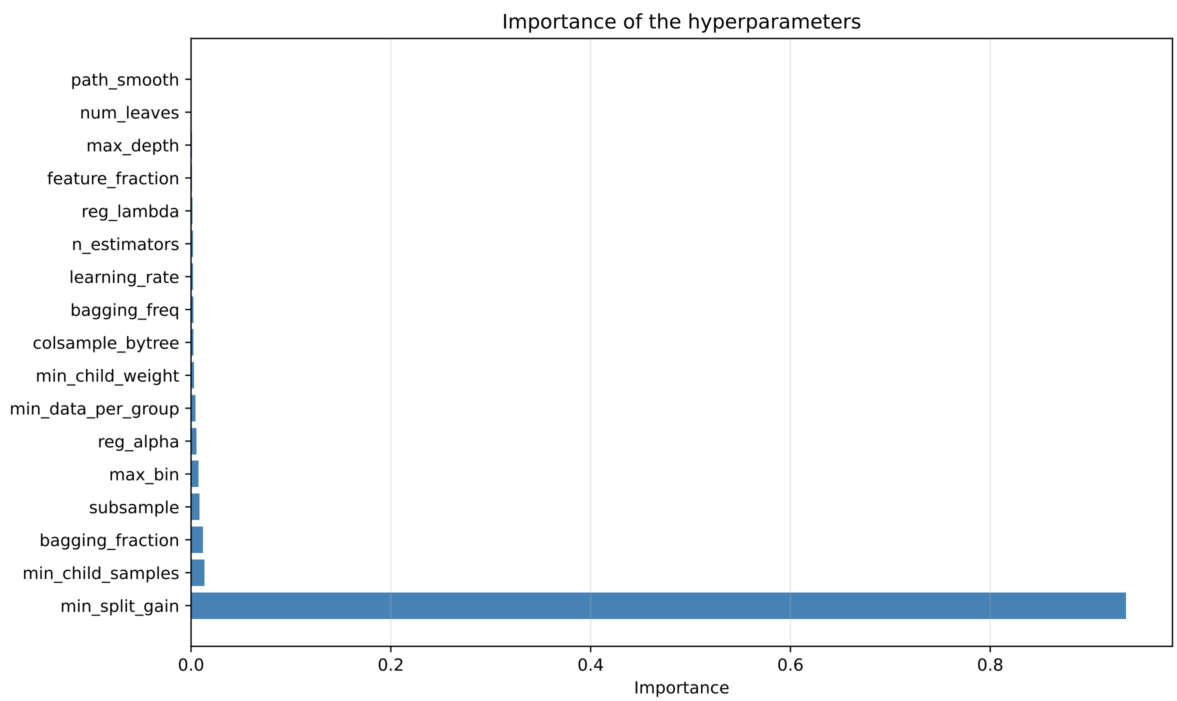
3.5 Model : LGBM

3.5.1 Gridsearch

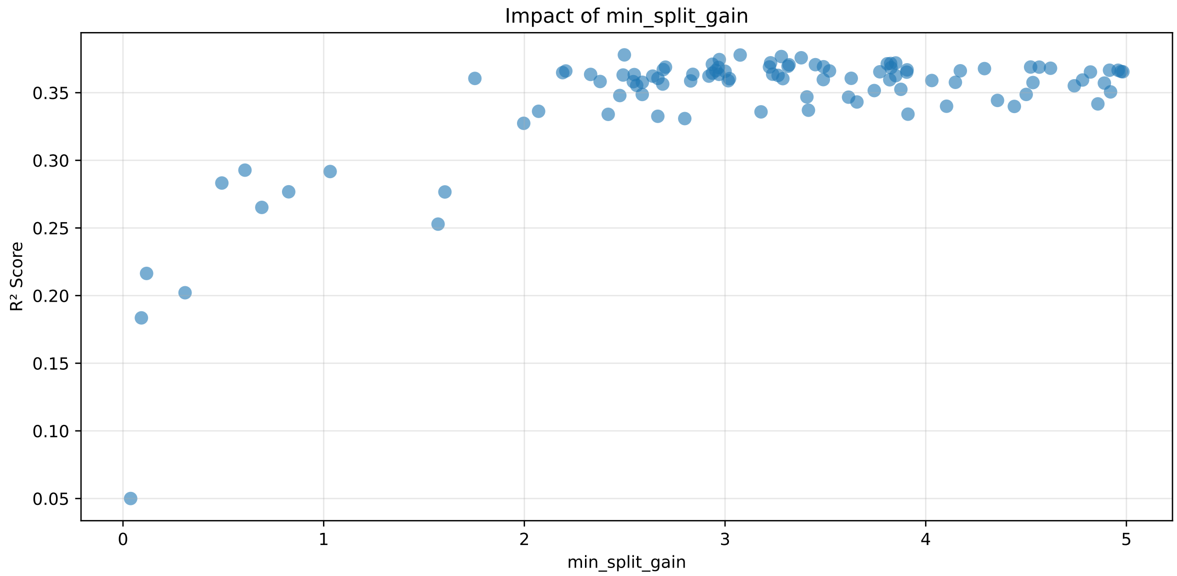
Distribution y_train vs y_test: y_train: mean=2.361, std=0.812, min=0.959, max=13.845 y_test: mean=2.360, std=0.767, min=0.973, max=6.594

==================== TOP Importance FEATURES ==================== feature importance r_umidity 78 r_CO2 63 r_temperature_st_2h_slope 38 r_NH3_st_2h_std 31 r_NH3_st_3h_std 28 r_temperature_st_3h_std 23 r_temperature_st_2h_diff_last 22 r_NH3_st_3h_median 20 r_temperature_st_1h_slope 20 r_NH3_st_1h_std 16 r_NH3_st_3h_slope 15 r_NH3 14 r_temperature_st_2h_std 13 r_NH3_st_1h_diff_last 12 r_umidity_st_2h_min 11 r_umidity_st_3h_min 11 r_temperature 7 r_NH3_st_2h_slope 7 r_umidity_st_1h_std 7 r_umidity_st_3h_slope 7 r_NH3_st_1h_slope 6 r_umidity_st_1h_slope 5 r_umidity_st_1h_diff_last 4 r_THI 4 r_temperature_st_1h_median 3 r_umidity_st_2h_std 3 r_temperature_st_3h_diff_last 2 r_temperature_st_1h_std 2 r_NH3_st_2h_diff_last 1 r_NH3_st_3h_diff_last 1 r_umidity_st_2h_diff_last 0 r_NH3_st_2h_min 0 r_NH3_st_1h_mean 0 r_umidity_st_3h_diff_last 0





Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6301	0.5881	0.6397	0.0412	0.3779	0.6531	0.065	1.1106	1.0715
2	0.6224	0.5816	0.634	0.0376	0.3778	0.6477	0.0661	1.1137	1.0701
3	0.6224	0.5815	0.6331	0.0408	0.3767	0.6466	0.0652	1.1121	1.0704
4	0.6196	0.5789	0.6305	0.0396	0.3758	0.6474	0.0685	1.1183	1.0702
5	0.6288	0.5864	0.6382	0.0449	0.3745	0.6563	0.07	1.1193	1.0723
6	0.6227	0.581	0.6341	0.0404	0.3721	0.6396	0.0587	1.101	1.0719
7	0.6112	0.5713	0.6239	0.0361	0.372	0.6423	0.071	1.1243	1.0698
8	0.6133	0.573	0.6262	0.0411	0.3716	0.6464	0.0734	1.128	1.0703
9	0.6189	0.5776	0.6311	0.0397	0.3714	0.6373	0.0597	1.1034	1.0714
10	0.6283	0.5854	0.6379	0.045	0.371	0.6661	0.0807	1.1379	1.0732
---	---	---	---	---	---	---	---	---	---
100	0.8236	0.6946	0.7555	0.0411	0.05	0.778	0.0834	1.12	1.1856

Hyperparameters:

Rank 1: {'reg___n_estimators': 800, 'reg___max_depth': 3, 'reg___learning_rate': 0.04798998489717568, 'reg___num_leaves': 28, 'reg___subsample': 0.6720887784739754, 'reg___colsample_bytree': 0.6846508722466956, 'reg___min_child_samples': 48, 'reg___reg_alpha': 3.3991843160647677, 'reg___reg_lambda': 46.037912079126684, 'reg___min_split_gain': 2.4984117467546887, 'reg___path_smooth': 1.4829462862775014, 'reg___min_child_weight': 0.002320911221828884, 'reg___feature_fraction': 0.8565560645700439, 'reg___bagging_fraction': 0.5113519892060552, 'reg___bagging_freq': 6, 'reg___max_bin': 72, 'reg___min_data_per_group': 97}

Rank 2: {'reg___n_estimators': 800, 'reg___max_depth': 3, 'reg___learning_rate': 0.04010738501555048, 'reg___num_leaves': 28, 'reg___subsample': 0.533230433884448, 'reg___colsample_bytree': 0.5646560994028968, 'reg___min_child_samples': 43, 'reg___reg_alpha': 3.9657158799652605, 'reg___reg_lambda': 36.917031787591604, 'reg___min_split_gain': 3.075377402732183, 'reg___path_smooth': 2.043767475943296, 'reg___min_child_weight': 0.0020116305312193688, 'reg___feature_fraction': 0.7601874413443673, 'reg___bagging_fraction': 0.5254518191056874, 'reg___bagging_freq': 6, 'reg___max_bin': 80, 'reg___min_data_per_group': 80}

Rank 3: {'reg___n_estimators': 800, 'reg___max_depth': 3, 'reg___learning_rate': 0.039210556637652896, 'reg___num_leaves': 34, 'reg___subsample': 0.5860028298163911, 'reg___colsample_bytree': 0.5645539313597541, 'reg___min_child_samples': 43, 'reg___reg_alpha': 3.5235040757608784, 'reg___reg_lambda': 35.82964529996346, 'reg___min_split_gain': 3.280530557612467, 'reg___path_smooth': 2.1372677048624698, 'reg___min_child_weight': 0.0020746137836782975, 'reg___feature_fraction': 0.8700426583762974, 'reg___bagging_fraction': 0.5237203324409208, 'reg___bagging_freq': 6, 'reg___max_bin': 80, 'reg___min_data_per_group': 77}

Rank 4: {'reg___n_estimators': 800, 'reg___max_depth': 3, 'reg___learning_rate': 0.039471597661673574, 'reg___num_leaves': 34, 'reg___subsample': 0.5293211031385233, 'reg___colsample_bytree': 0.5637099594588451, 'reg___min_child_samples': 42, 'reg___reg_alpha': 3.7351950801549663, 'reg___reg_lambda': 35.25853052581645, 'reg___min_split_gain': 3.3795931967044623, 'reg___path_smooth': 2.169676361222587, 'reg___min_child_weight': 0.007503513322818184, 'reg___feature_fraction': 0.7604233312813069, 'reg___bagging_fraction': 0.5279219153651236, 'reg___bagging_freq': 6, 'reg___max_bin': 80, 'reg___min_data_per_group': 78}

Rank 5: {'reg___n_estimators': 800, 'reg___max_depth': 3, 'reg___learning_rate': 0.02977354235227434, 'reg___num_leaves': 40, 'reg___subsample': 0.6162220620626511, 'reg___colsample_bytree': 0.5878069248998304, 'reg___min_child_samples': 37, 'reg___reg_alpha': 2.498979772440777, 'reg___reg_lambda': 30.46617554286565, 'reg___min_split_gain': 2.971593074886549, 'reg___path_smooth': 2.7187014894123545, 'reg___min_child_weight': 0.01659730393666145, 'reg___feature_fraction': 0.7165972757347748, 'reg___bagging_fraction': 0.5184300367488867, 'reg___bagging_freq': 7, 'reg___max_bin': 64, 'reg___min_data_per_group': 84}

Rank 6: {'reg___n_estimators': 800, 'reg___max_depth': 3, 'reg___learning_rate': 0.039915847852919246, 'reg___num_leaves': 34, 'reg___subsample': 0.5870932983836267, 'reg___colsample_bytree': 0.560579647572455, 'reg___min_child_samples': 43, 'reg___reg_alpha': 3.9456125720120667, 'reg___reg_lambda': 36.95066463248976, 'reg___min_split_gain': 3.2269914300115965, 'reg___path_smooth': 1.9888604506137615, 'reg___min_child_weight': 0.006350054458162612, 'reg___feature_fraction': 0.7454214105863431, 'reg___bagging_fraction': 0.5277020745139868, 'reg___bagging_freq': 6, 'reg___max_bin': 83, 'reg___min_data_per_group': 76}

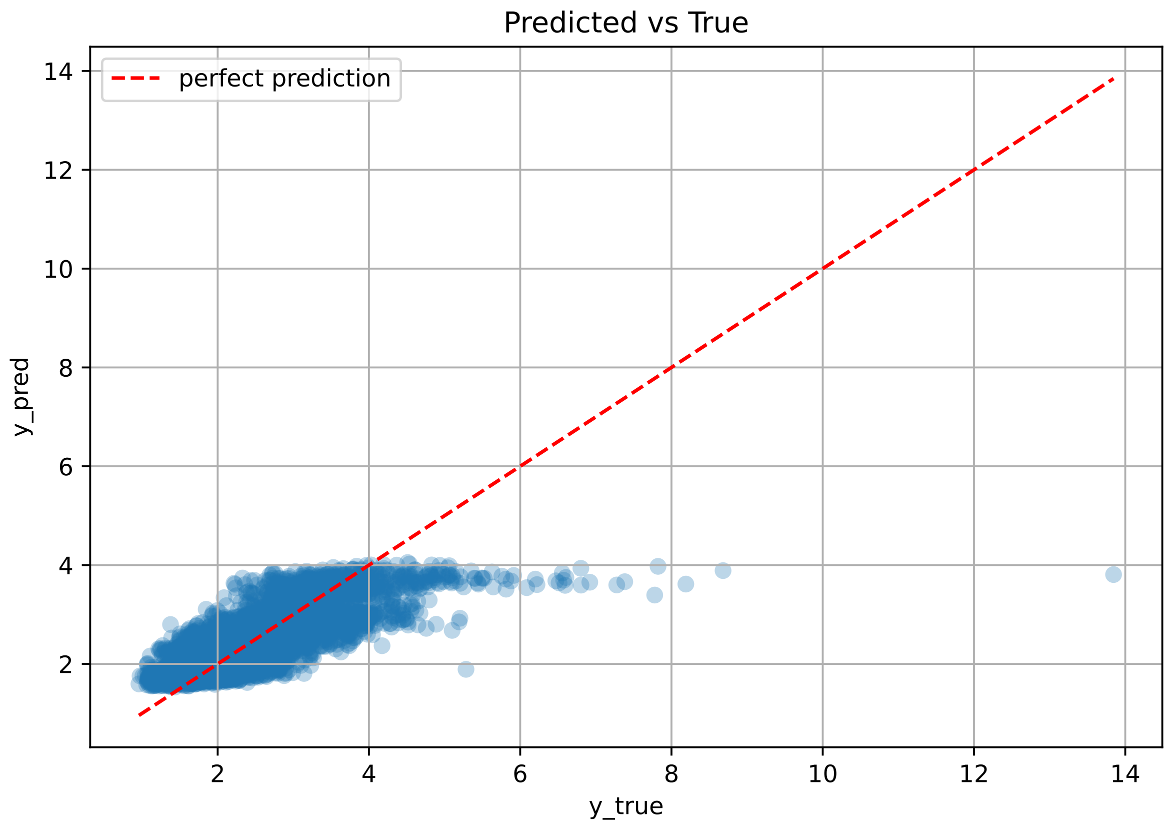
Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.03711461892691812, 'reg__num_leaves': 34, 'reg__subsample': 0.5863115322323055, 'reg__colsample_bytree': 0.5195109867348991, 'reg__min_child_samples': 43, 'reg__reg_alpha': 5.325952211843006, 'reg__reg_lambda': 36.899649339887155, 'reg__min_split_gain': 3.8505696981180955, 'reg__path_smooth': 1.230321259100263, 'reg__min_child_weight': 0.006427363297455944, 'reg__feature_fraction': 0.7448712538522859, 'reg__bagging_fraction': 0.5800851537250952, 'reg__bagging_freq': 6, 'reg__max_bin': 79, 'reg__min_data_per_group': 73}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.05784840372853279, 'reg__num_leaves': 31, 'reg__subsample': 0.550161028159075, 'reg__colsample_bytree': 0.5001651305871753, 'reg__min_child_samples': 46, 'reg__reg_alpha': 4.660795428203054, 'reg__reg_lambda': 40.49041350076189, 'reg__min_split_gain': 3.824988699251272, 'reg__path_smooth': 3.0642222065835036, 'reg__min_child_weight': 0.0032091746005480583, 'reg__feature_fraction': 0.8225289103756163, 'reg__bagging_fraction': 0.536650233334736, 'reg__bagging_freq': 5, 'reg__max_bin': 109, 'reg__min_data_per_group': 51}

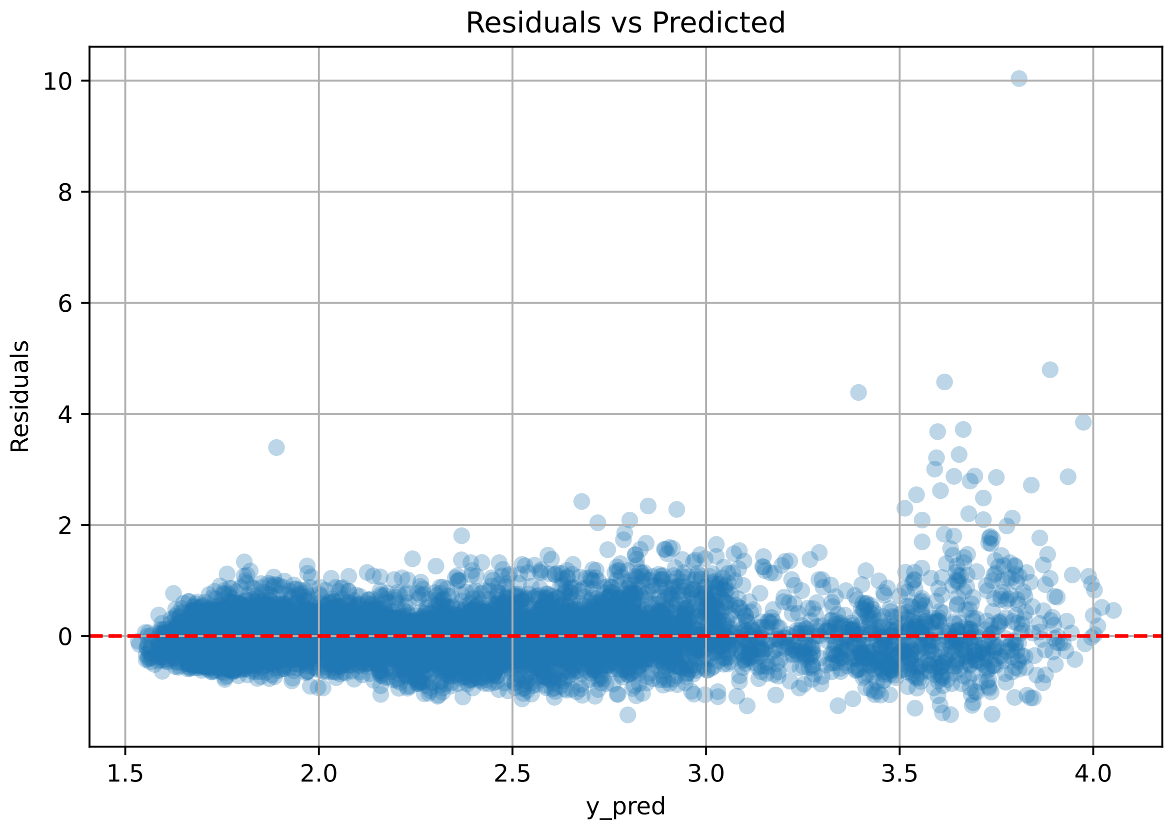
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.03888799133877456, 'reg__num_leaves': 32, 'reg__subsample': 0.5940504952284122, 'reg__colsample_bytree': 0.5233005570962016, 'reg__min_child_samples': 44, 'reg__reg_alpha': 4.094807604788956, 'reg__reg_lambda': 39.39058042475599, 'reg__min_split_gain': 3.80840309816695, 'reg__path_smooth': 1.318586419719677, 'reg__min_child_weight': 0.007612960076419037, 'reg__feature_fraction': 0.7419307672163254, 'reg__bagging_fraction': 0.5930310090286601, 'reg__bagging_freq': 6, 'reg__max_bin': 81, 'reg__min_data_per_group': 50}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.030324343902549396, 'reg__num_leaves': 44, 'reg__subsample': 0.6172736104794345, 'reg__colsample_bytree': 0.5868717260908294, 'reg__min_child_samples': 38, 'reg__reg_alpha': 2.859504285023492, 'reg__reg_lambda': 30.23691944151112, 'reg__min_split_gain': 2.93672255589593, 'reg__path_smooth': 2.6494724659394477, 'reg__min_child_weight': 0.01694286698141702, 'reg__feature_fraction': 0.7226816598691961, 'reg__bagging_fraction': 0.5165852047767903, 'reg__bagging_freq': 7, 'reg__max_bin': 65, 'reg__min_data_per_group': 90}

3.5.2 Scatter


Predicted vs True
perfect prediction
14
12
10
8
6
4
2
y_pred
2
4
6
8
10
12
14
y_true

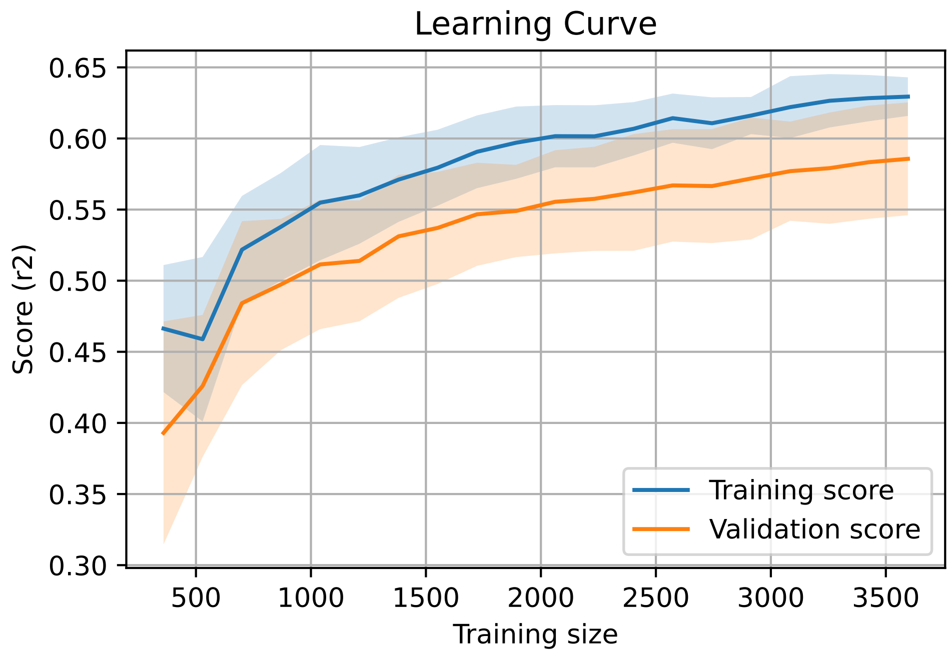
3.5.3 Scatter residuals



3.5.4 Learning curve

$\mathbf{Len(group)} : 7340$, $\mathbf{Len(training)} : 5138$,

$\mathbf{Len(training\_real)} : 3597$, $\mathbf{Len(test\_of\_training)} : 1541$, $\mathbf{Len(test)} : 2202$

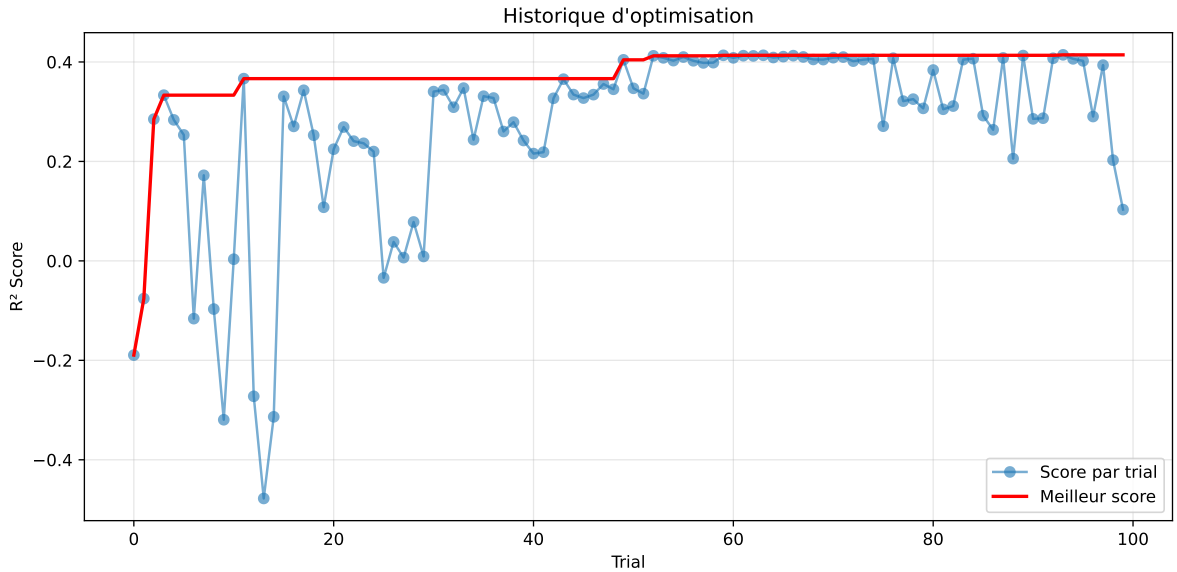


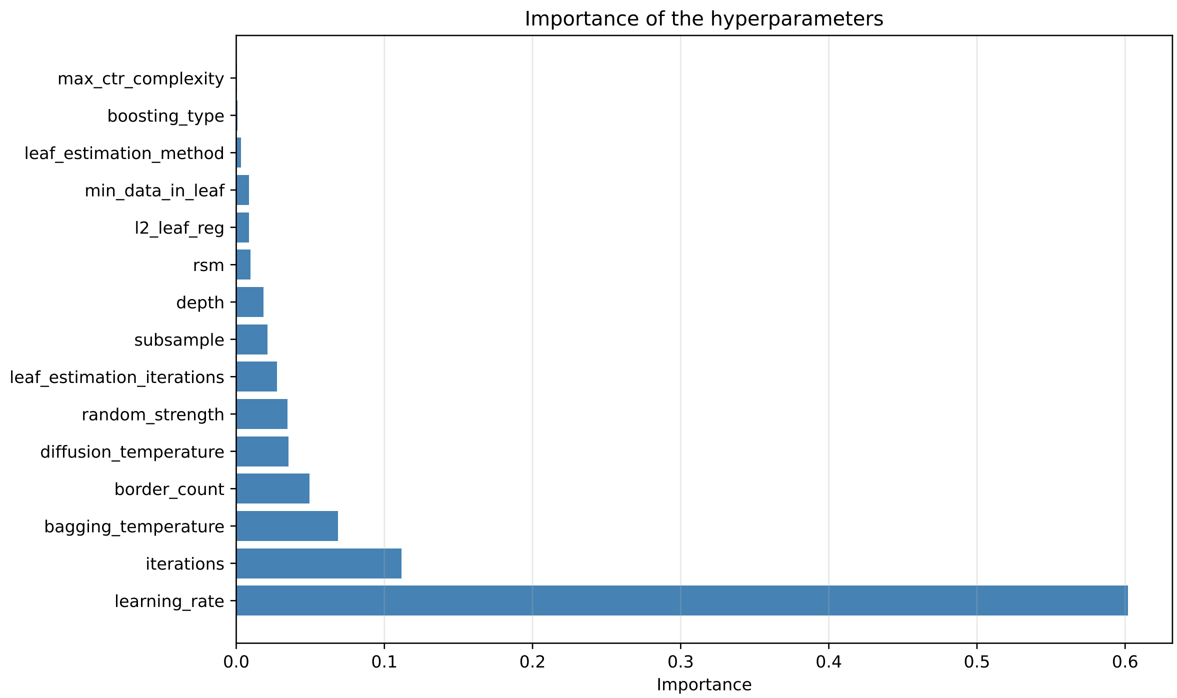
3.6 Model : CatBoost

3.6.1 Gridsearch

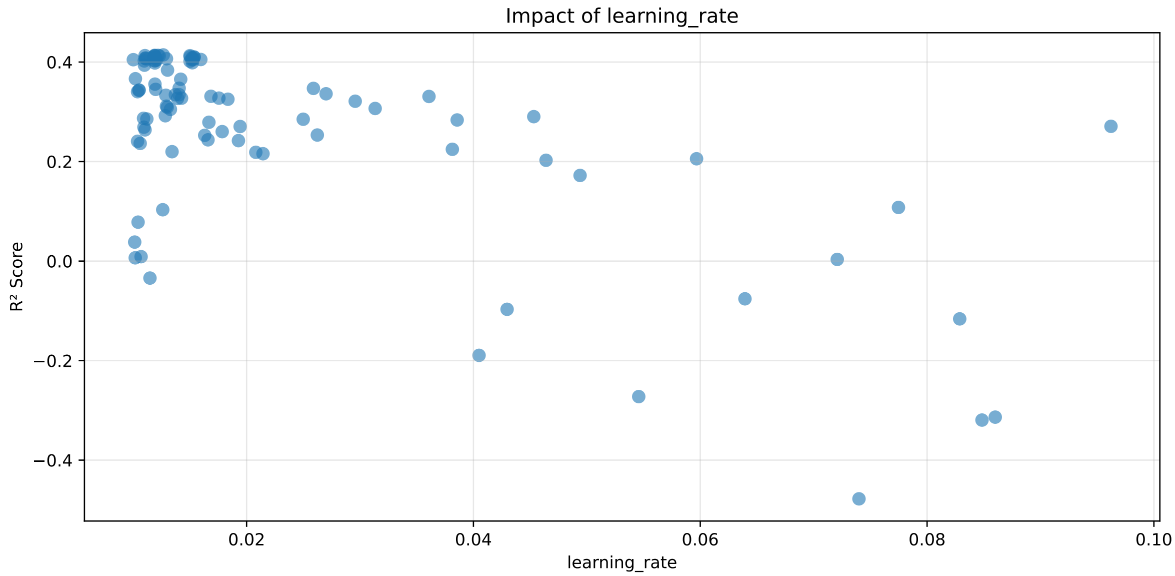
Distribution y_train vs y_test: y_train: mean=2.361, std=0.812, min=0.959, max=13.845 y_test: mean=2.360, std=0.767, min=0.973, max=6.594

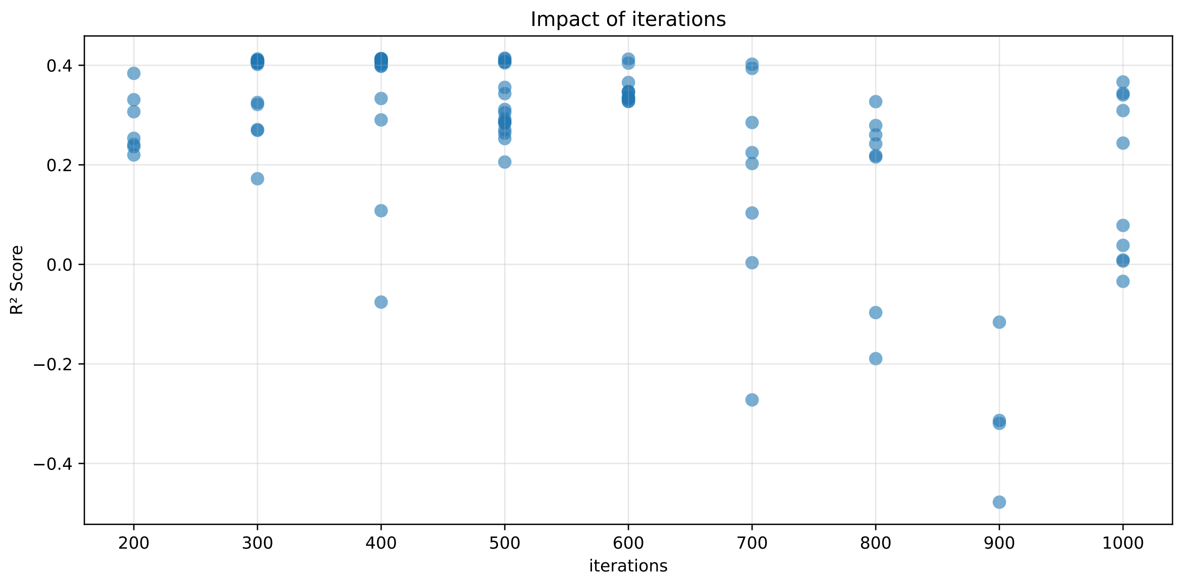
===================== TOP Importance FEATURES ===================== feature importance r_umidity 18.271615 r_temperature_st_2h_std 9.105213 r_temperature_st_2h_slope 8.788011 r_temperature_st_1h_slope 7.564700 r_temperature_st_3h_std 6.579137 r_temperature 5.529071 r_umidity_st_2h_min 5.414313 r_umidity_st_3h_slope 5.069770 r_umidity_st_3h_min 4.199204 r_NH3_st_3h_std 3.661763 r_temperature_st_2h_diff_last 2.616264 r_NH3_st_2h_std 2.558452 r_CO2 2.517226 r_temperature_st_3h_diff_last 1.820671 r_umidity_st_1h_slope 1.793135 r_temperature_st_1h_std 1.151775 r_THI 1.067620 r_NH3 1.060236 r_NH3_st_1h_std 0.989607 r_temperature_st_1h_median 0.911850 r_NH3_st_1h_slope 0.894529 r_umidity_st_2h_diff_last 0.882272 r_NH3_st_2h_min 0.869949 r_NH3_st_3h_slope 0.857560 r_NH3_st_2h_slope 0.832847 r_NH3_st_3h_median 0.806431 r_NH3_st_1h_mean 0.760382 r_umidity_st_3h_diff_last 0.595754 r_umidity_st_1h_diff_last 0.588232 r_NH3_st_3h_diff_last 0.553130 r_umidity_st_2h_std 0.490374 r_NH3_st_2h_diff_last 0.436220 r_umidity_st_1h_std 0.427434 r_NH3_st_1h_diff_last 0.335252

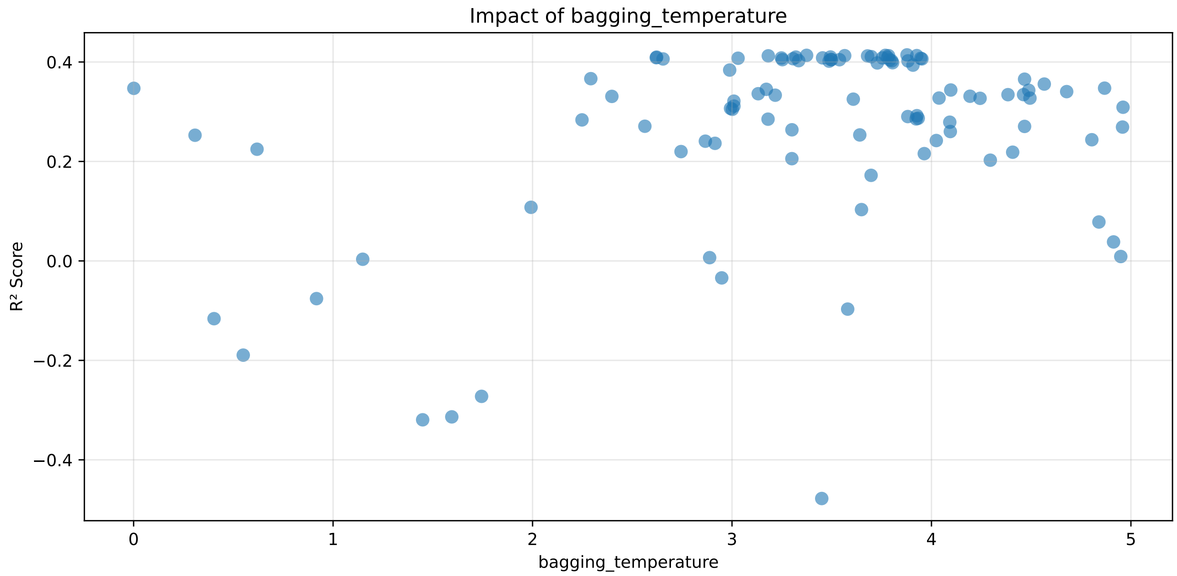




Paramètres avec >5% d'importance : ['learning_rate', 'iterations', 'bagging_temperature']

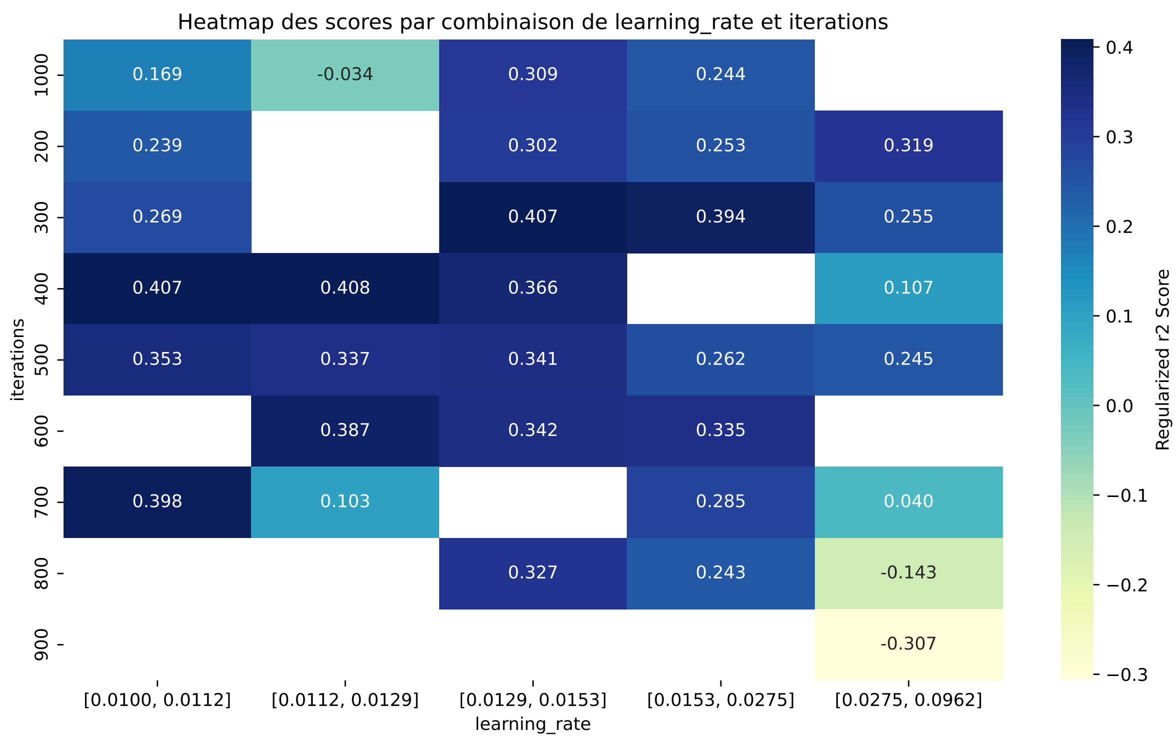

Impact of learning_rate
R² Score
0.4
0.2
0.0
−0.2
−0.4
0.02
0.04
0.06
0.08
0.10
learning_rate


Impact of iterations
R² Score
0.4
0.2
0.0
−0.2
−0.4
200
300
400
500
600
700
800
900
1000
iterations



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6154	0.5819	0.6271	0.0382	0.4143	0.6316	0.0497	1.0854	1.0576
2	0.5979	0.5671	0.6115	0.0377	0.4134	0.6154	0.0483	1.0852	1.0542
3	0.5963	0.5658	0.6103	0.0379	0.4134	0.6143	0.0485	1.0858	1.0539
4	0.6061	0.5739	0.6175	0.038	0.4129	0.6221	0.0481	1.0839	1.0561
5	0.5921	0.5622	0.6077	0.0383	0.4127	0.6116	0.0494	1.0878	1.0532
6	0.5973	0.5665	0.6099	0.0385	0.4127	0.6145	0.048	1.0848	1.0543
7	0.6216	0.5867	0.6324	0.0378	0.4124	0.6399	0.0532	1.0906	1.0594
8	0.5958	0.5652	0.608	0.0382	0.4123	0.6152	0.0501	1.0886	1.0541
9	0.5897	0.5599	0.6045	0.037	0.4107	0.6081	0.0483	1.0862	1.0533
10	0.5901	0.5601	0.6032	0.038	0.4101	0.609	0.0488	1.0872	1.0536
---	---	---	---	---	---	---	---	---	---
100	0.9942	0.7488	0.8091	0.0427	-0.4778	0.8248	0.0759	1.1014	1.3276

Hyperparameters:

Rank 1: {'reg___iterations': 500, 'reg___depth': 3, 'reg___learning_rate': 0.01265050747664389, 'reg___l2_leaf_reg': 31.608484327989476, 'reg___bagging_temperature': 3.8773530552095705, 'reg___random_strength': 4.163305597505727, 'reg___subsample': 0.5618170385271666, 'reg___min_data_in_leaf': 37, 'reg___leaf_estimation_iterations': 2, 'reg___rsm': 0.5820348227235445, 'reg___border_count': 32, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 9094.860967095861}

Rank 2: {'reg___iterations': 400, 'reg___depth': 3, 'reg___learning_rate': 0.011894091421549107, 'reg___l2_leaf_reg': 34.54410982687221, 'reg___bagging_temperature': 3.374015635585759, 'reg___random_strength': 4.367961943786849, 'reg___subsample': 0.6869003584388288, 'reg___min_data_in_leaf': 38, 'reg___leaf_estimation_iterations': 6, 'reg___rsm': 0.5569574548411165, 'reg___border_count': 34, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 8845.149679932594}

Rank 3: {'reg___iterations': 400, 'reg___depth': 3, 'reg___learning_rate': 0.012188761747659408, 'reg___l2_leaf_reg': 35.19407500696547, 'reg___bagging_temperature': 3.7687294903513155, 'reg___random_strength': 4.332106814220087, 'reg___subsample': 0.6757725890354034, 'reg___min_data_in_leaf': 38, 'reg___leaf_estimation_iterations': 3, 'reg___rsm': 0.5592852218179296, 'reg___border_count': 32, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 8701.498780408734}

Rank 4: {'reg___iterations': 500, 'reg___depth': 3, 'reg___learning_rate': 0.011038440121664612, 'reg___l2_leaf_reg': 32.030295350065956, 'reg___bagging_temperature': 3.9262864118414202, 'reg___random_strength': 4.477445369634452, 'reg___subsample': 0.5399990633525253, 'reg___min_data_in_leaf': 37, 'reg___leaf_estimation_iterations': 2, 'reg___rsm':

0.5847053916242931, 'reg___border_count': 32, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 9250.928117811764}

Rank 5: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.015002131131632129, 'reg___l2_leaf_reg': 38.41657504847278, 'reg___bagging_temperature': 3.5649896541855615, 'reg___random_strength': 4.823720677405369, 'reg___subsample': 0.5011089805871274, 'reg___min_data_in_leaf': 38, 'reg___leaf_estimation_iterations': 2, 'reg___rsm': 0.6246707465998904, 'reg___border_count': 32, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 9950.193442961849}

Rank 6: {'reg___iterations': 400, 'reg___depth': 3, 'reg___learning_rate': 0.011890150485118248, 'reg___l2_leaf_reg': 33.891619095492075, 'reg___bagging_temperature': 3.7850596886719154, 'reg___random_strength': 4.302286392688825, 'reg___subsample': 0.5694674903360359, 'reg___min_data_in_leaf': 36, 'reg___leaf_estimation_iterations': 2, 'reg___rsm': 0.566073340206648, 'reg___border_count': 32, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 9973.083009796239}

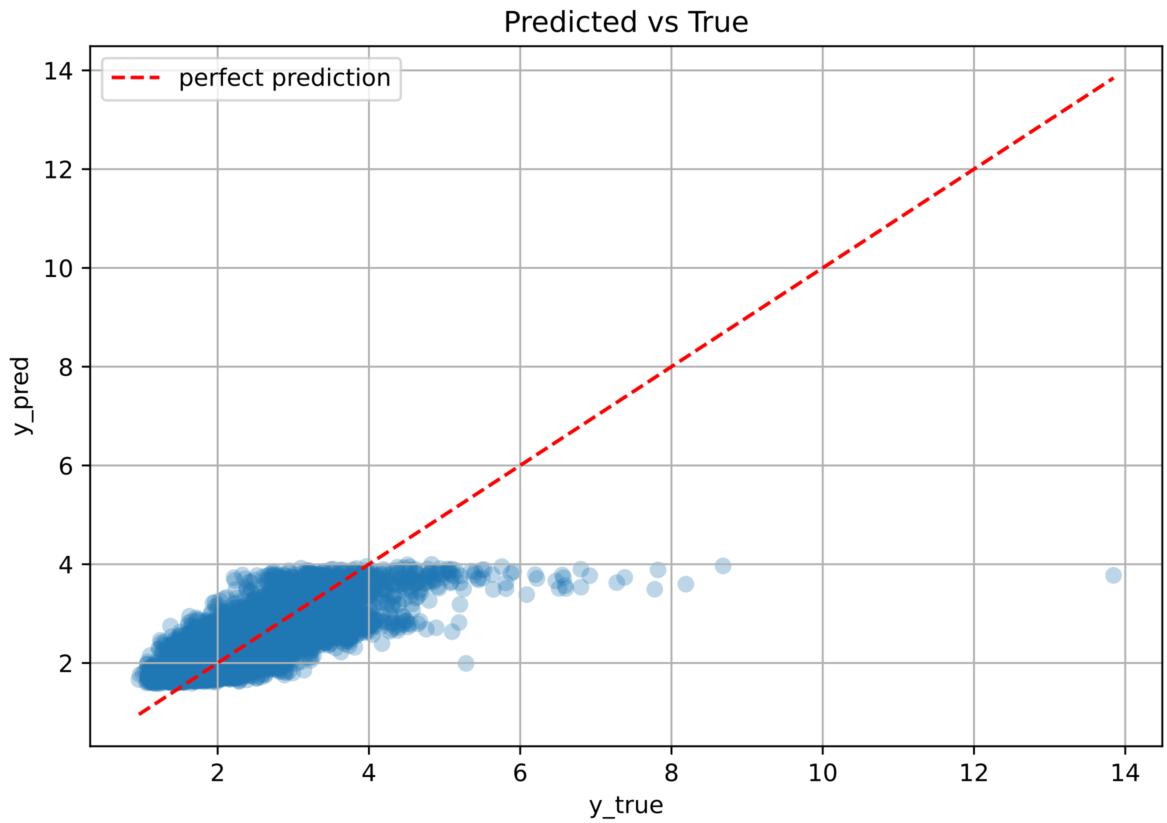
Rank 7: {'reg___iterations': 600, 'reg___depth': 3, 'reg___learning_rate': 0.012300869554038248, 'reg___l2_leaf_reg': 30.66173009369534, 'reg___bagging_temperature': 3.181929298126045, 'reg___random_strength': 4.275030535246315, 'reg___subsample': 0.7052529422850015, 'reg___min_data_in_leaf': 32, 'reg___leaf_estimation_iterations': 2, 'reg___rsm': 0.5672816174265803, 'reg___border_count': 44, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 8964.374434643672}

Rank 8: {'reg___iterations': 400, 'reg___depth': 3, 'reg___learning_rate': 0.01191149027206818, 'reg___l2_leaf_reg': 32.9963436231797, 'reg___bagging_temperature': 3.679670294130889, 'reg___random_strength': 4.345149077038294, 'reg___subsample': 0.5034345297422032, 'reg___min_data_in_leaf': 37, 'reg___leaf_estimation_iterations': 6, 'reg___rsm': 0.5596697279883492, 'reg___border_count': 32, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 8920.906017354195}

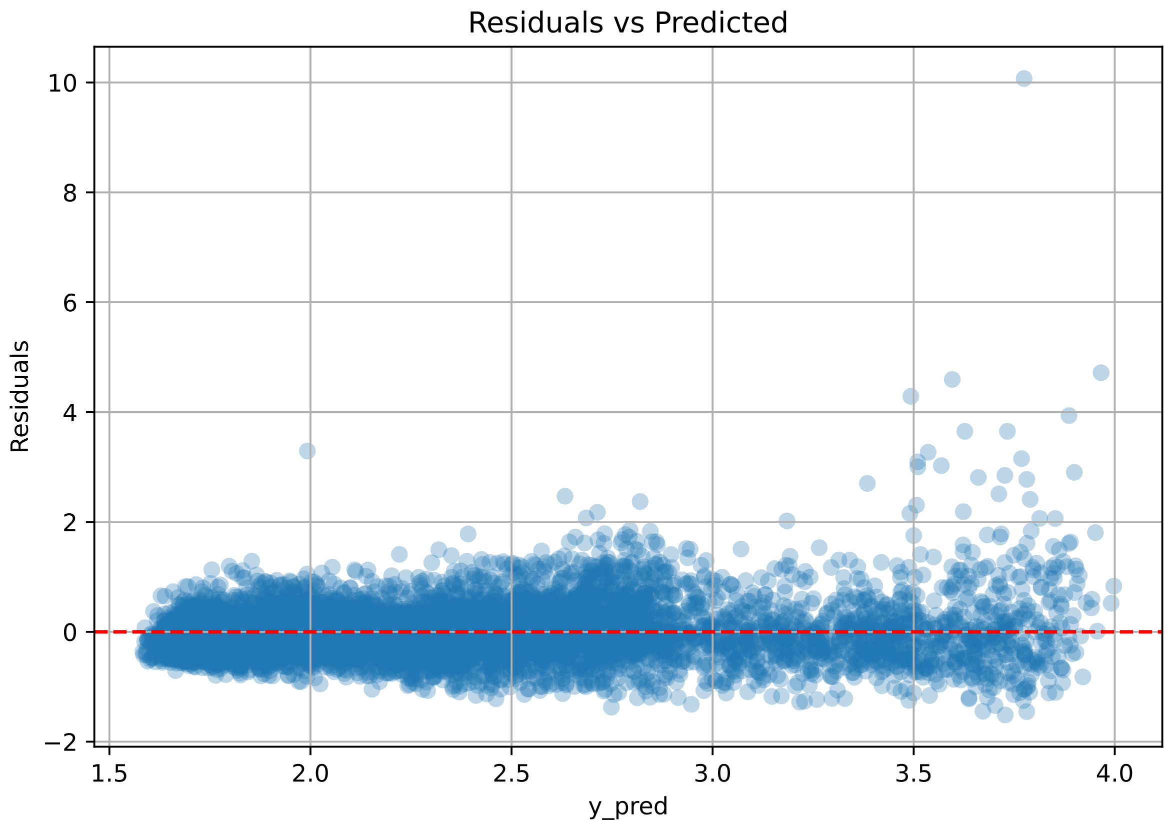
Rank 9: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.015029812345166958, 'reg___l2_leaf_reg': 35.65338171458278, 'reg___bagging_temperature': 3.698612596667453, 'reg___random_strength': 4.903948473287812, 'reg___subsample': 0.5776221556510537, 'reg___min_data_in_leaf': 37, 'reg___leaf_estimation_iterations': 2, 'reg___rsm': 0.5539254106037269, 'reg___border_count': 33, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 9905.232754241659}

Rank 10: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.015377143184168485, 'reg___l2_leaf_reg': 36.69854924767132, 'reg___bagging_temperature': 3.493688003152752, 'reg___random_strength': 4.941329844259265, 'reg___subsample': 0.5085398174555497, 'reg___min_data_in_leaf': 38, 'reg___leaf_estimation_iterations': 2, 'reg___rsm': 0.5516159989680803, 'reg___border_count': 32, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Plain', 'reg___max_ctr_complexity': 2, 'reg___diffusion_temperature': 9891.018110476265}

3.6.2 Scatter



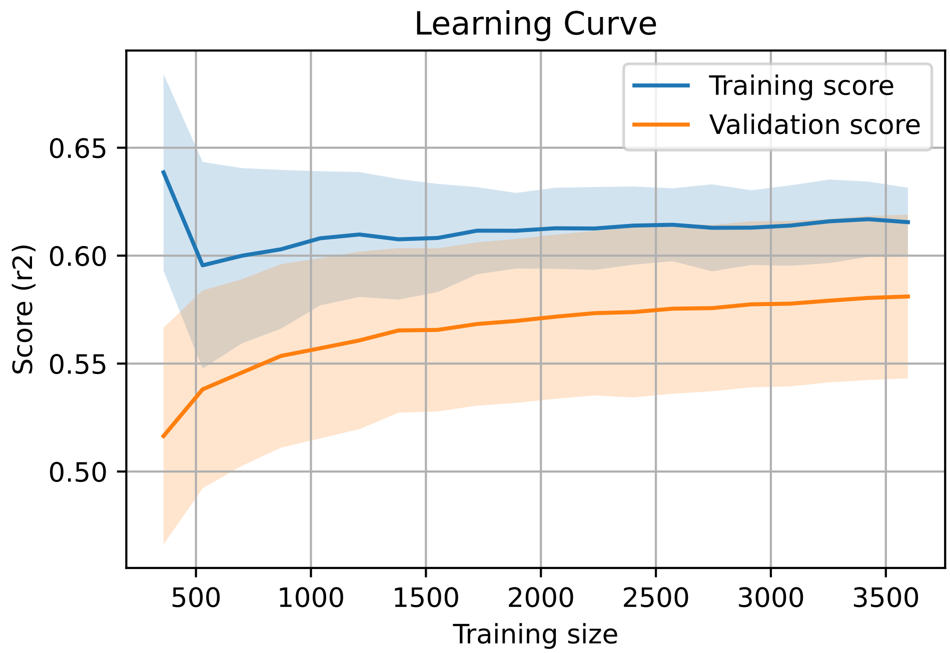
3.6.3 Scatter residuals



3.6.4 Learning curve

$\textbf{Len(group)} : 7340$, $\textbf{Len(training)} : 5138$,

$\textbf{Len(training\_real)} : 3597$, $\textbf{Len(test\_of\_training)} : 1541$, $\textbf{Len(test)} : 2202$



3.7 Sumup-table

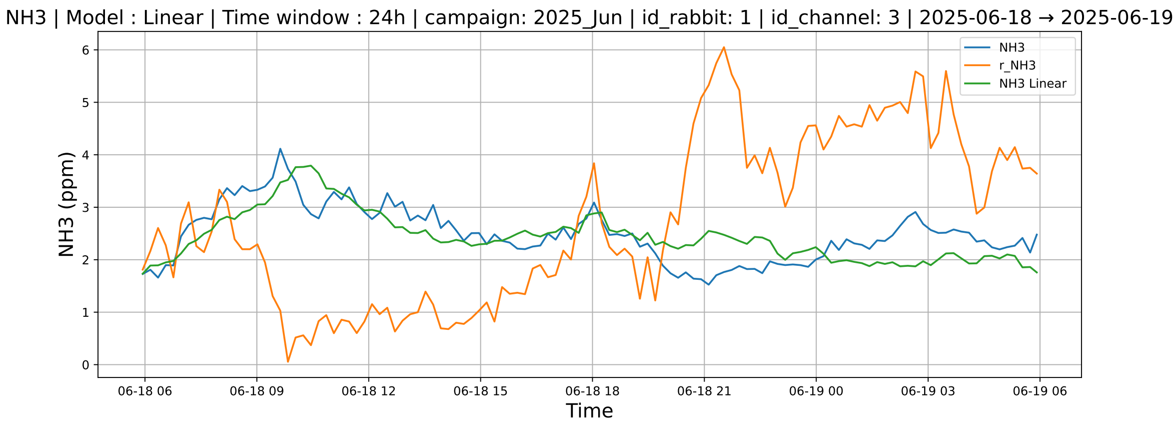
model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.551	0.52	0.042	0.569	0.569	0.049	1.095	1.06	nan
LGBM	0.63	0.588	0.041	0.64	0.653	0.065	1.111	1.072	0.3779
CatBoost	0.615	0.582	0.038	0.627	0.632	0.05	1.085	1.058	0.4143

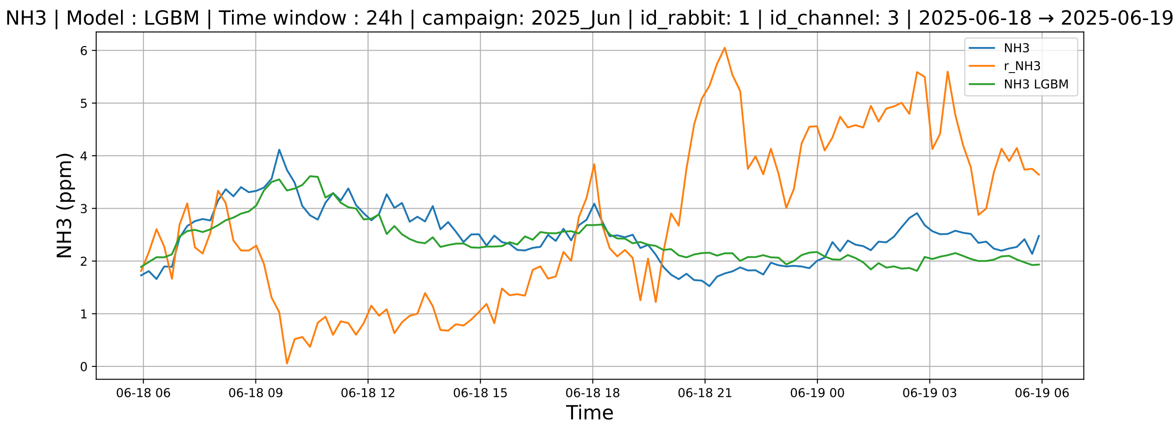
4 Plot

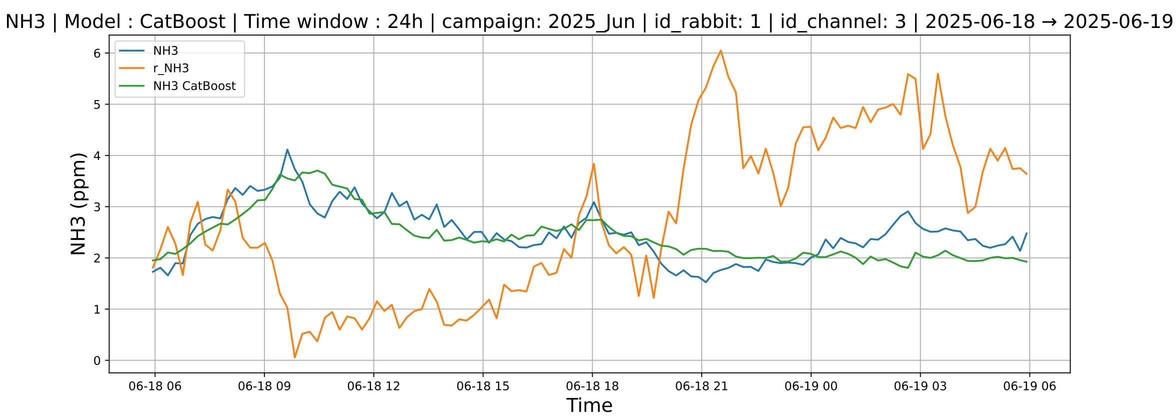
4.1 Standalone plots

4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

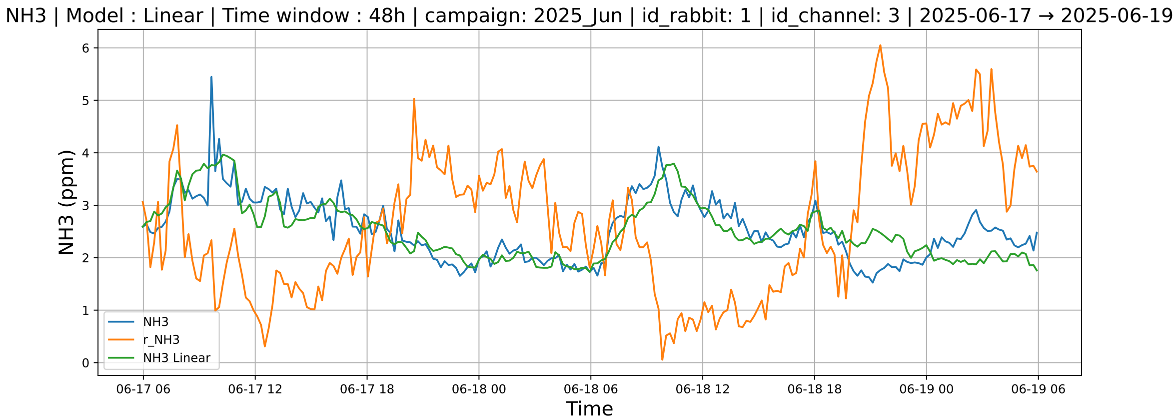
4.1.1.1 Time window : 24

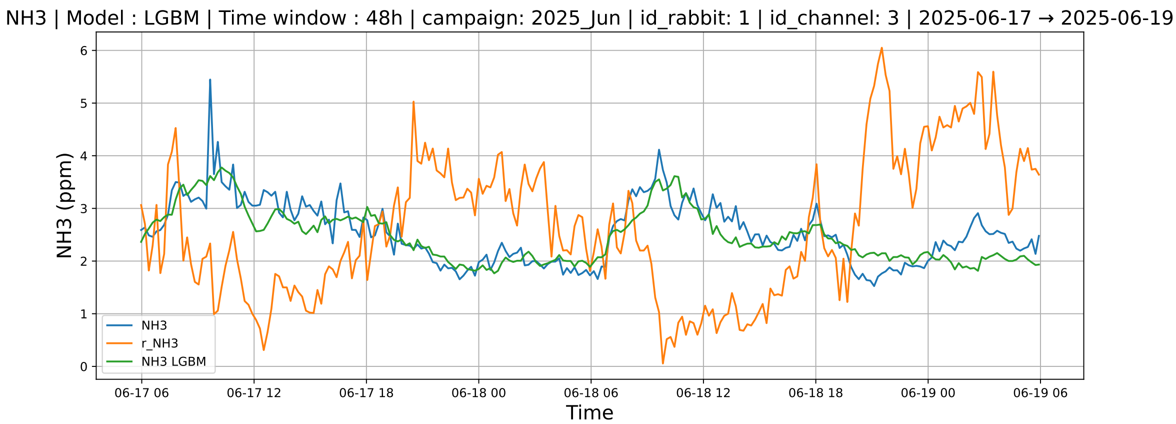


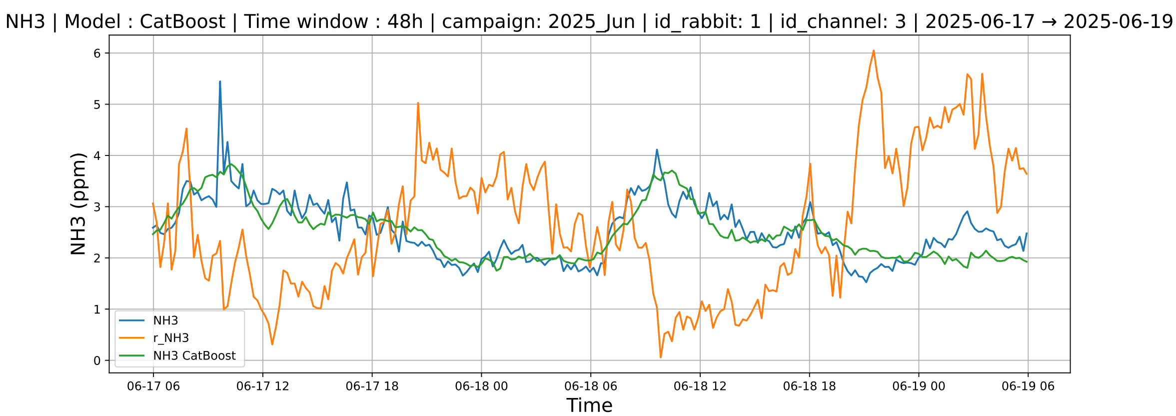




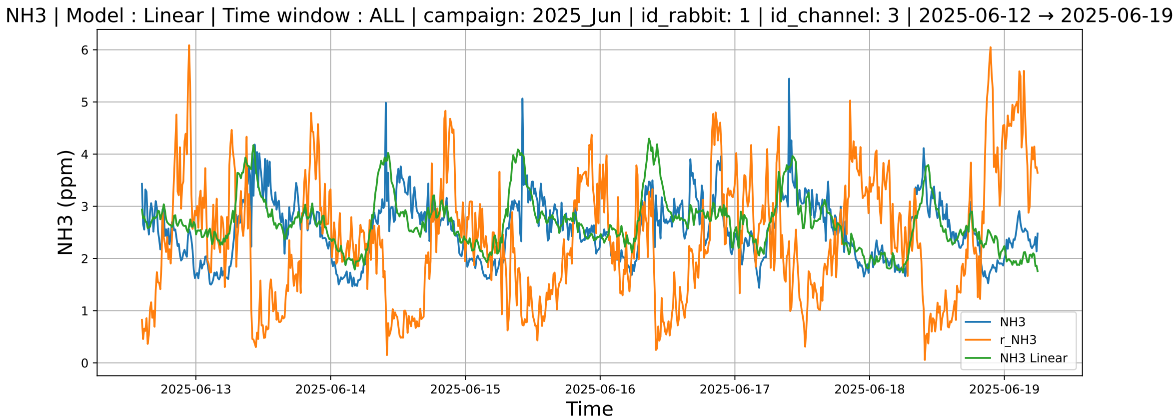
4.1.1.2 Time window : 48

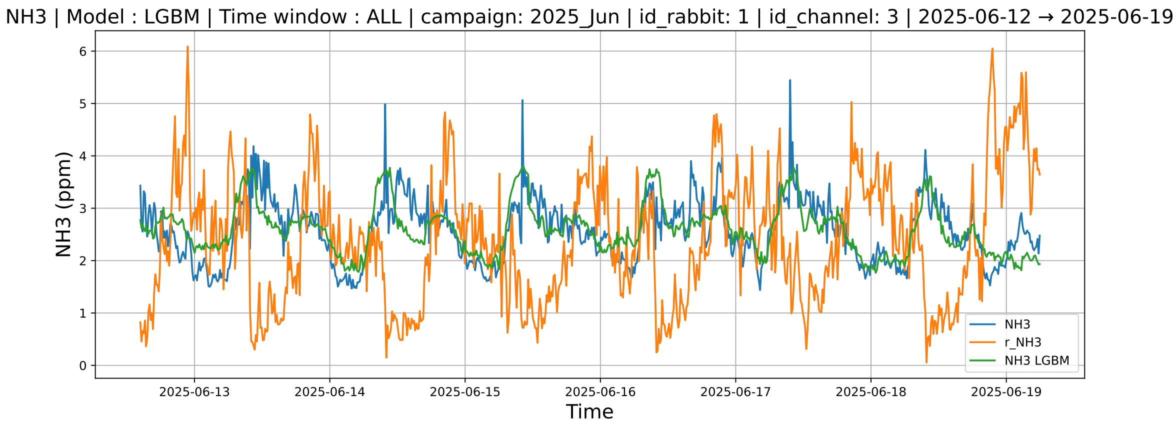


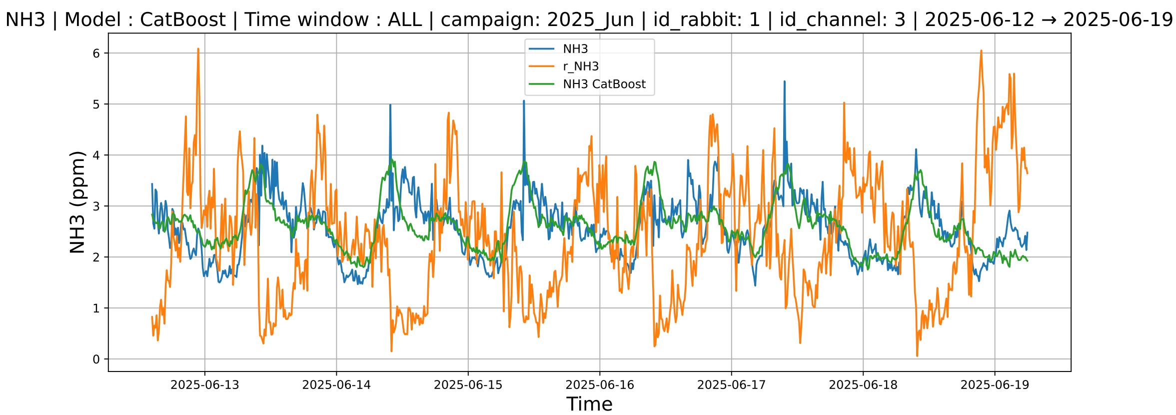




4.1.1.3 Time window : ALL

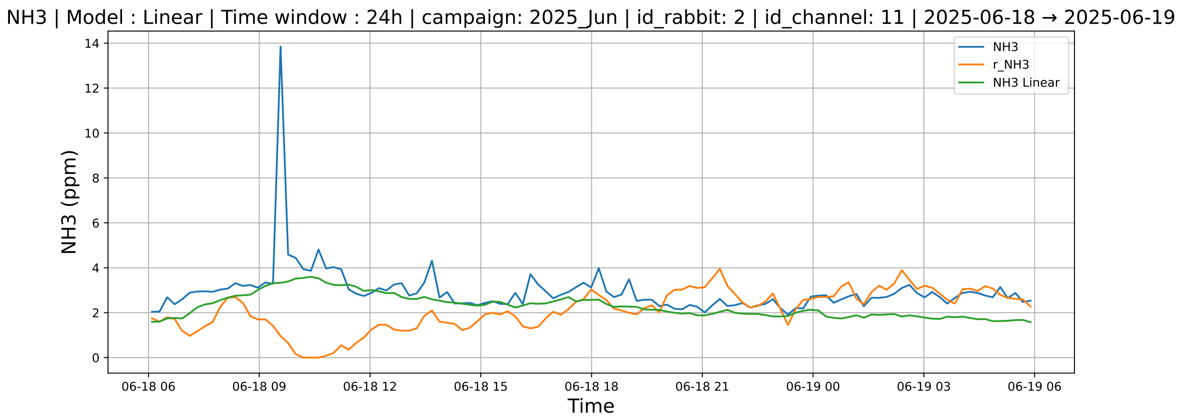


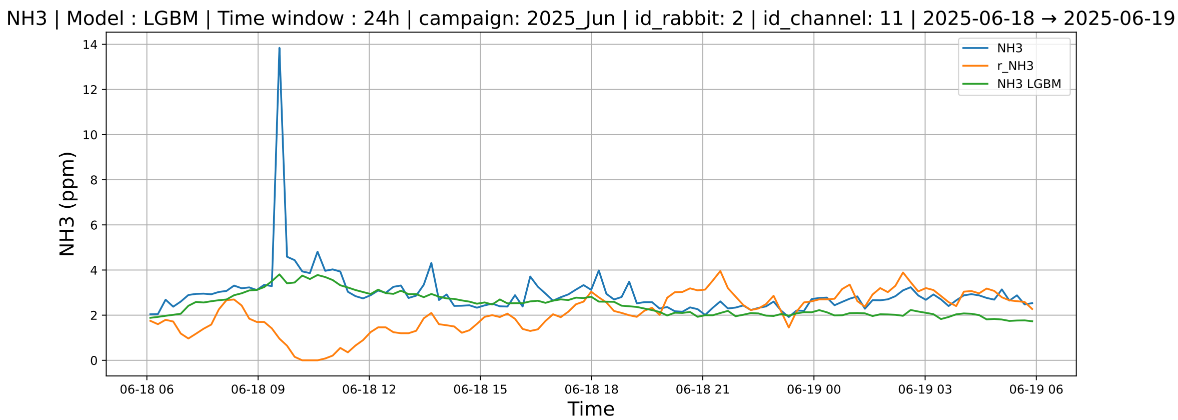


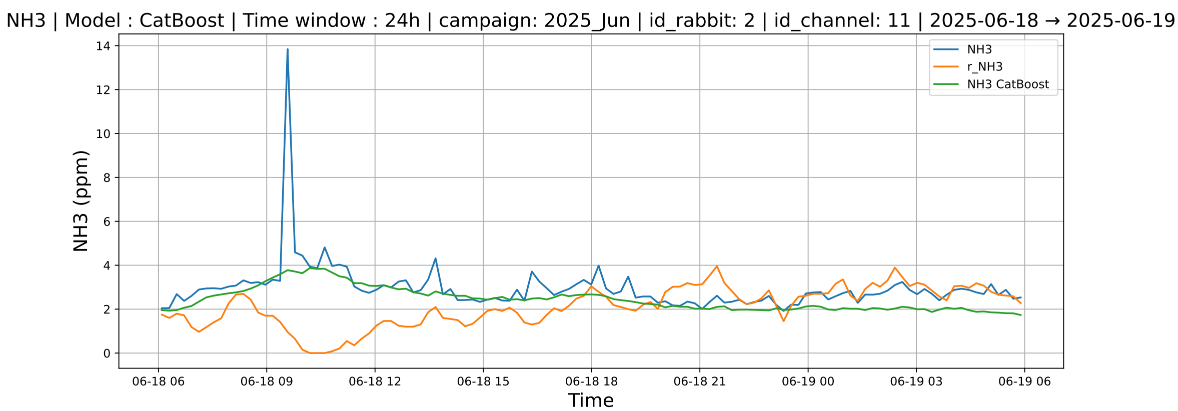


4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

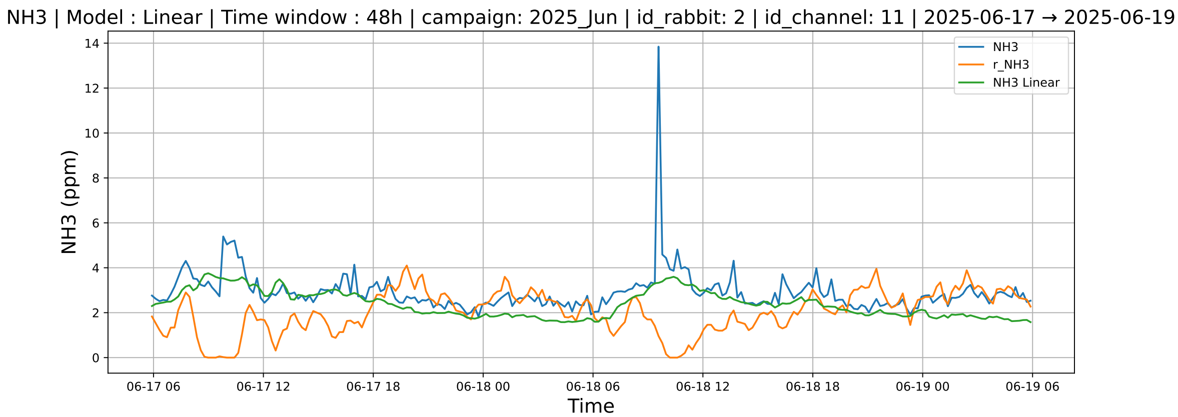
4.1.2.1 Time window : 24

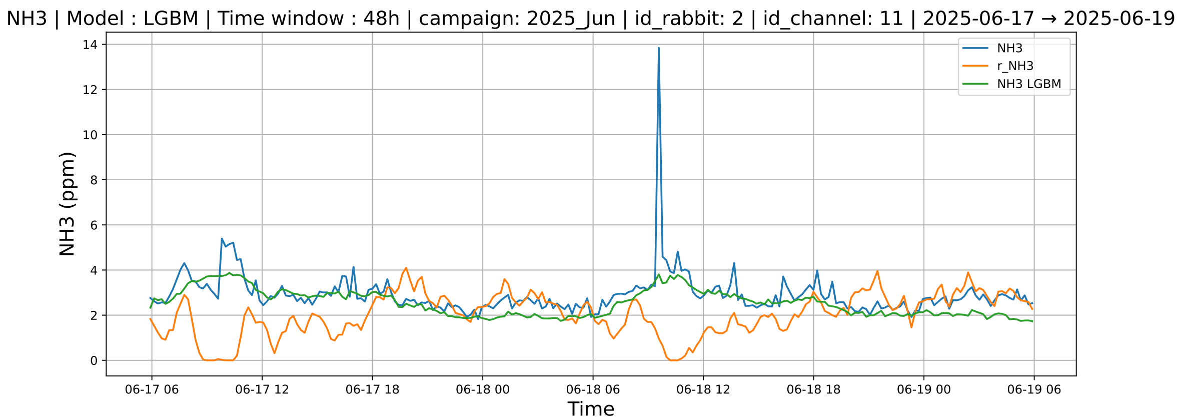


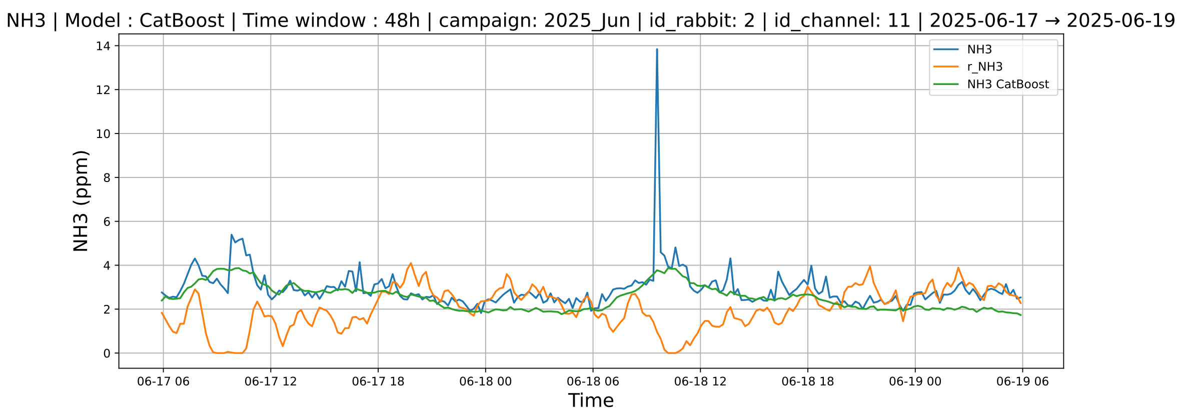




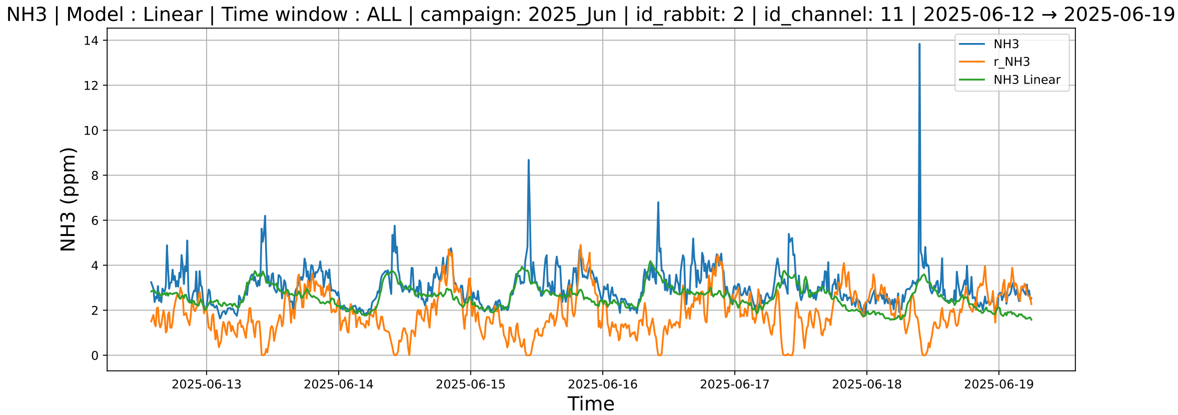
4.1.2.2 Time window : 48

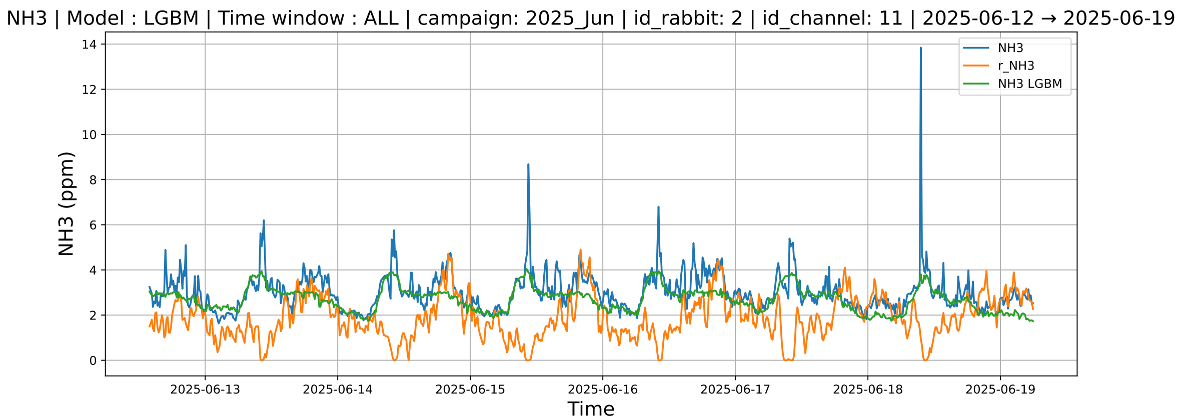


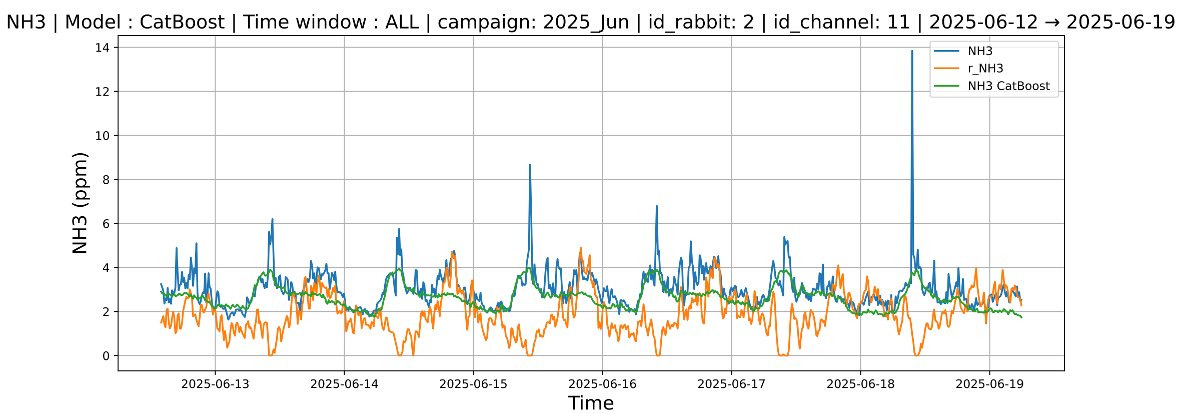




4.1.2.3 Time window : ALL

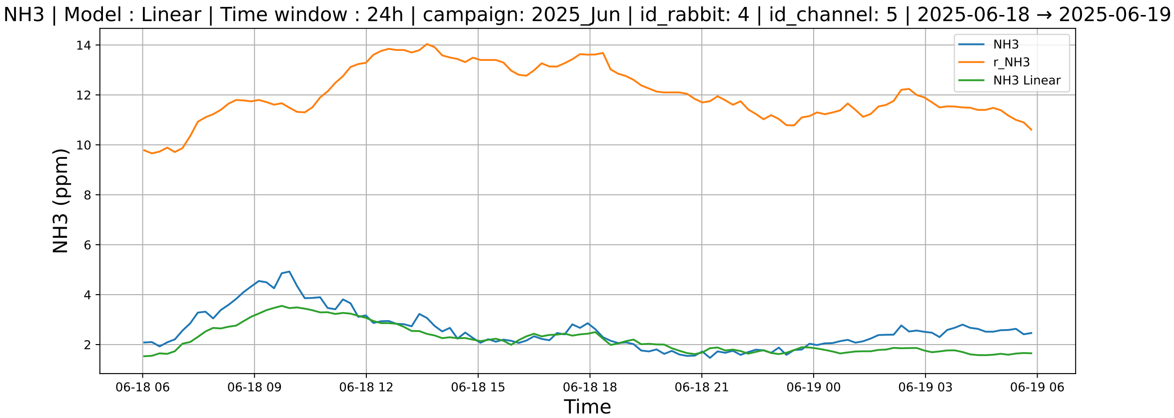


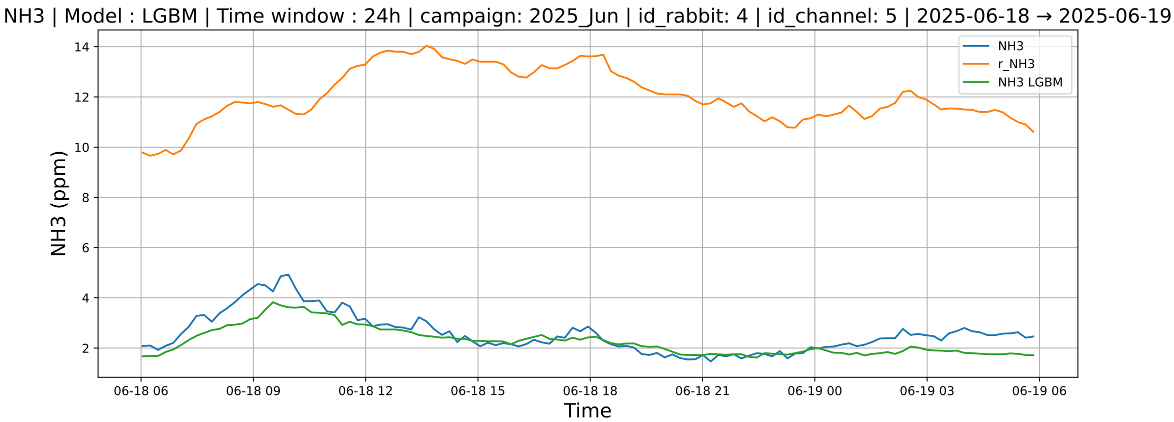


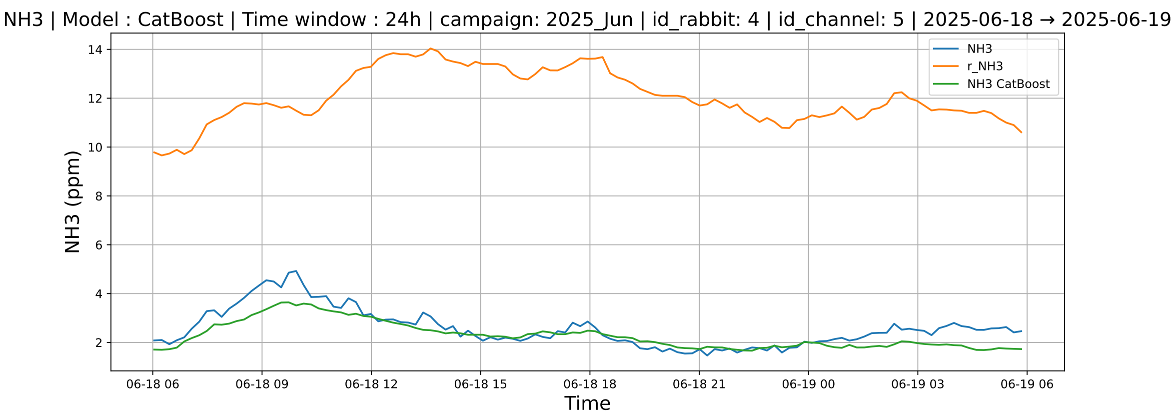


4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

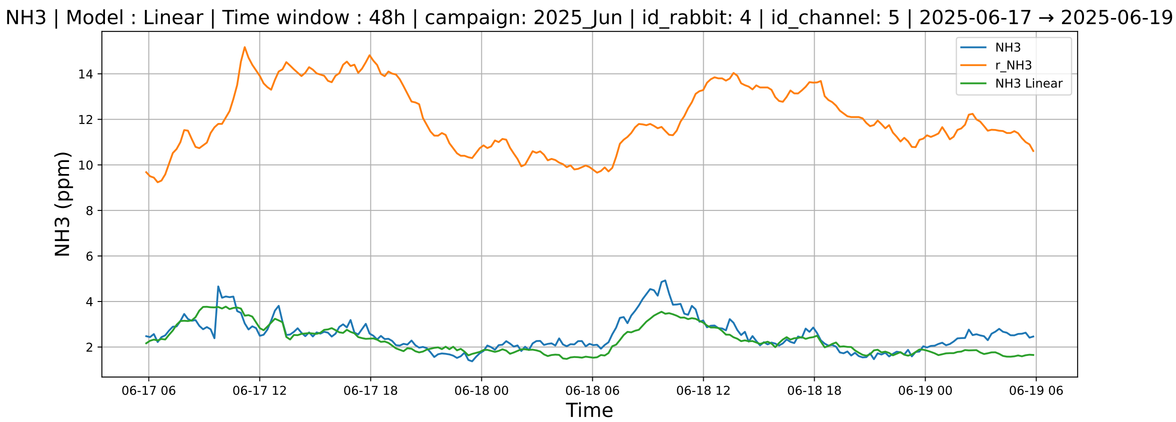
4.1.3.1 Time window : 24

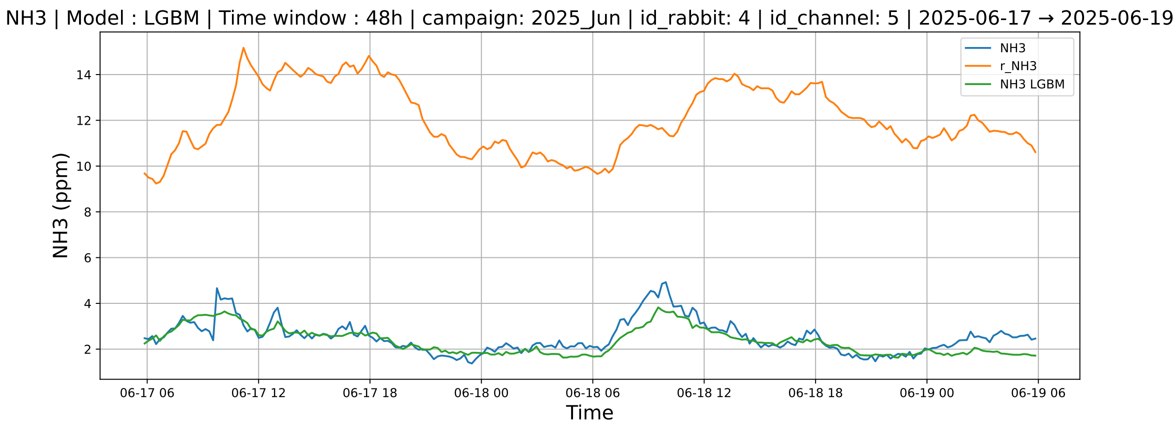


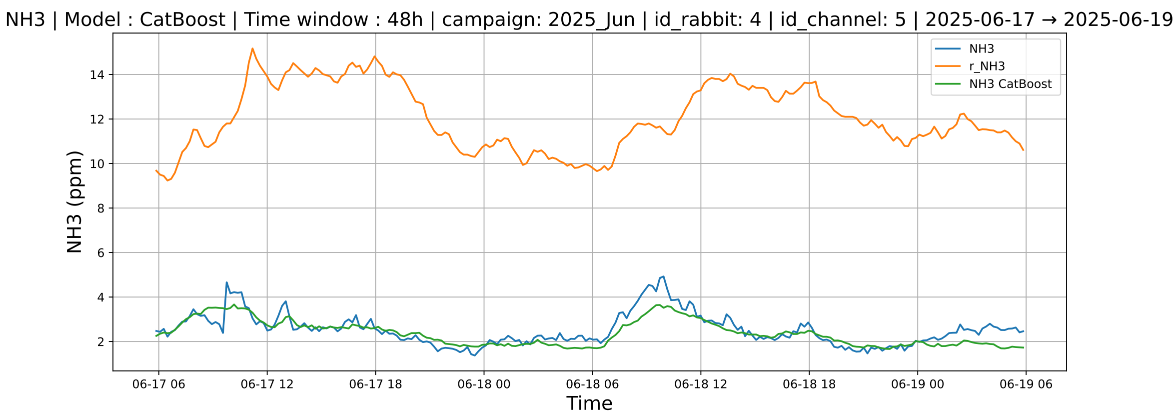




4.1.3.2 Time window : 48

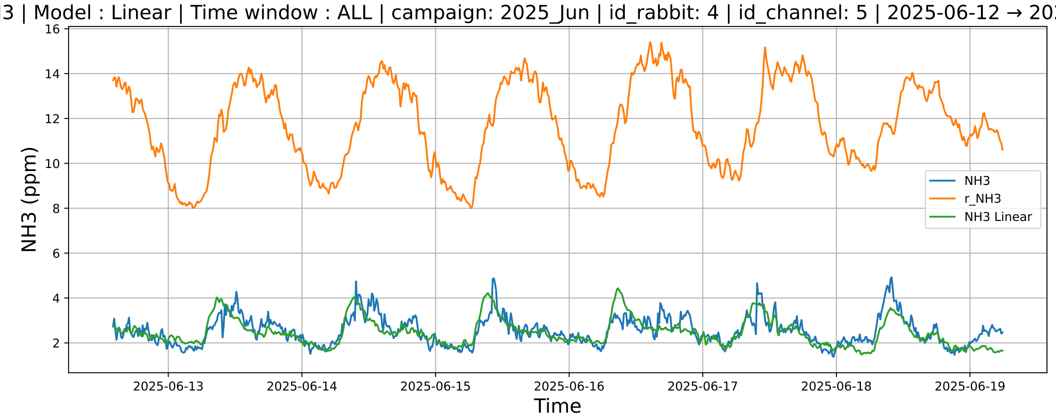




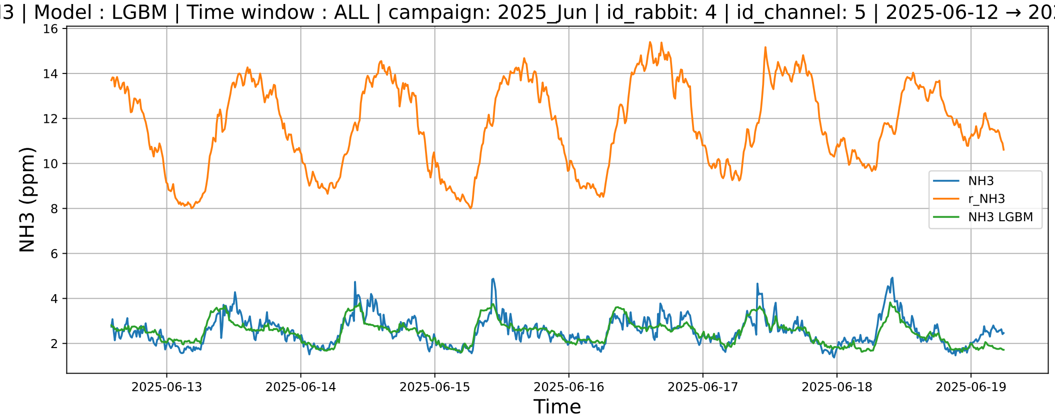


4.1.3.3 Time window : ALL

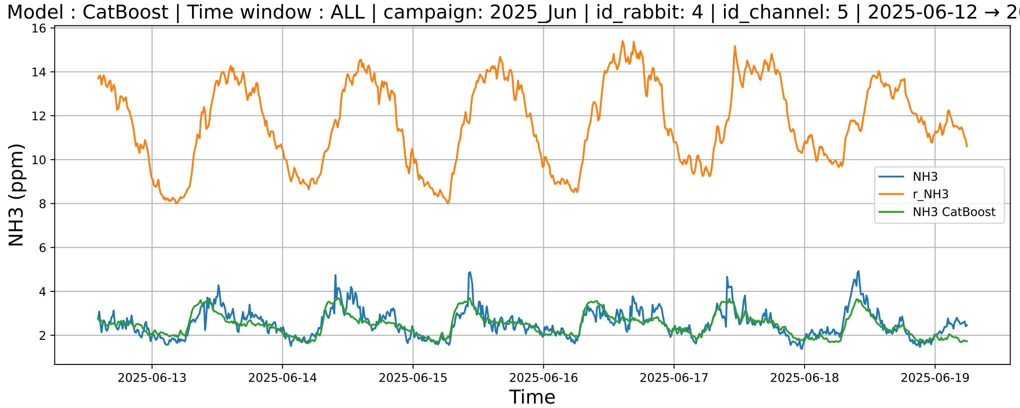
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19

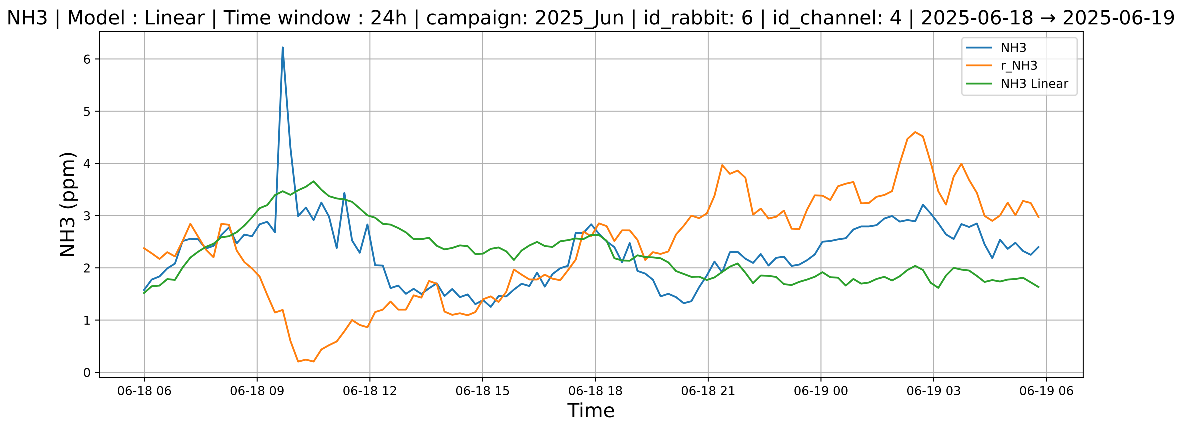


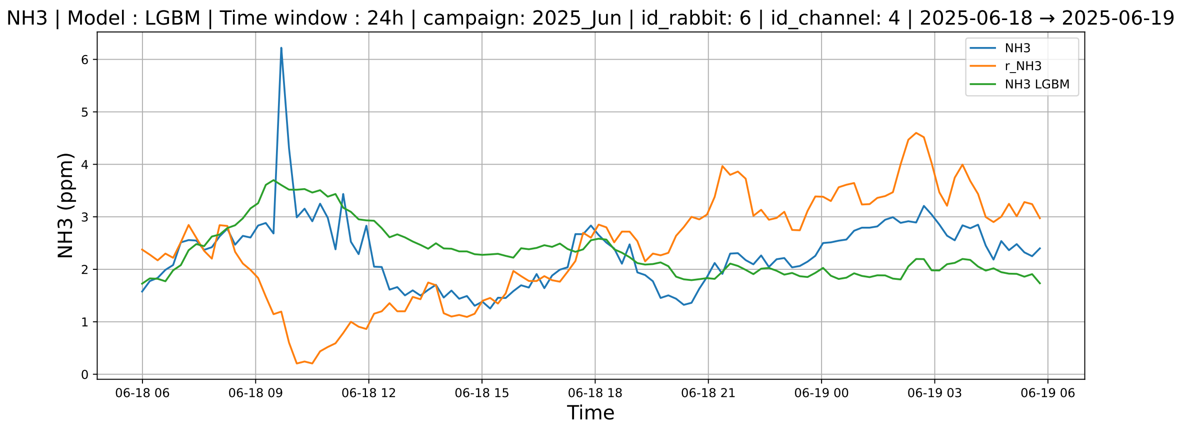
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19

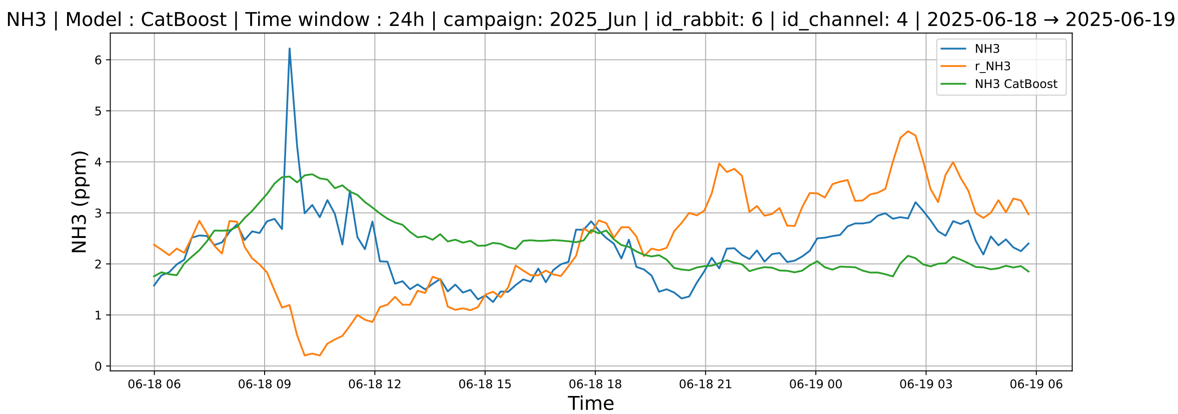


4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

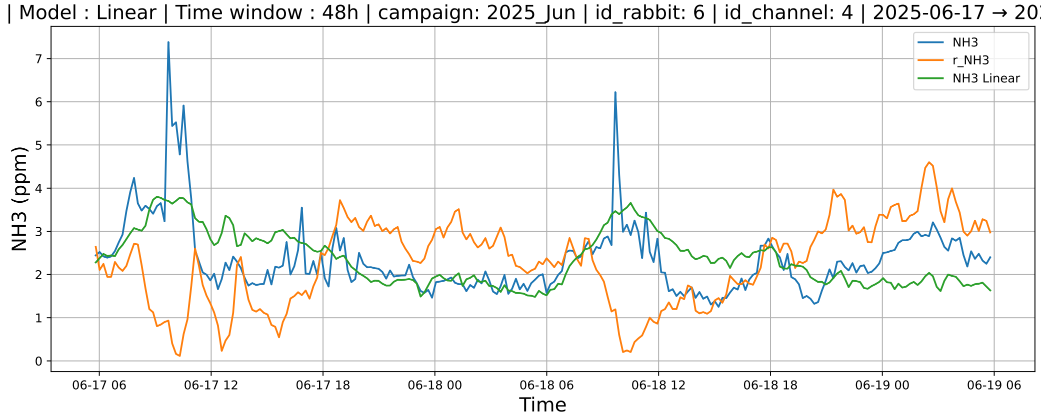




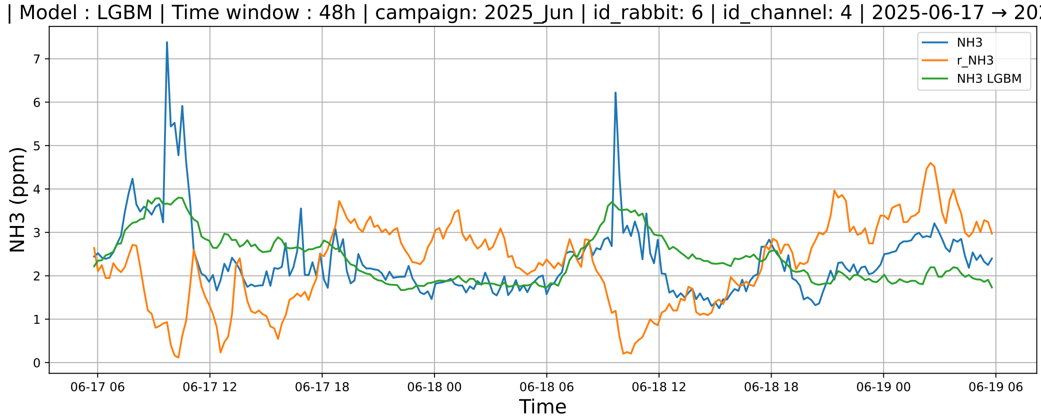


4.1.4.2 Time window : 48

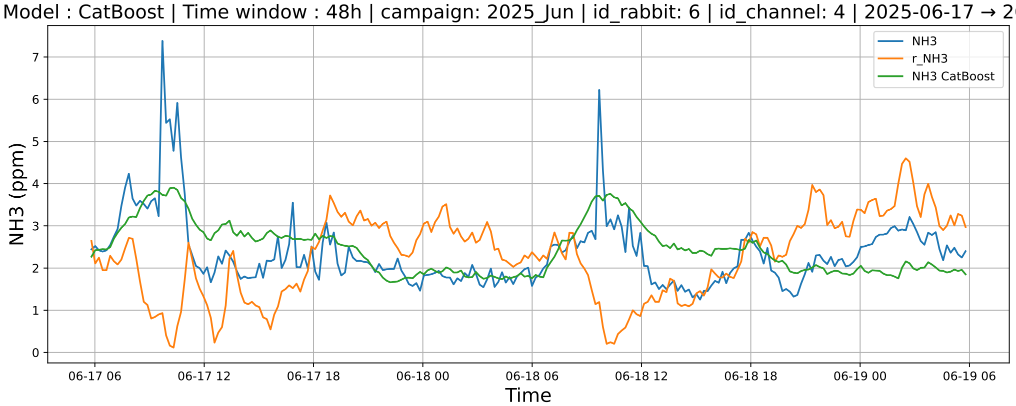
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



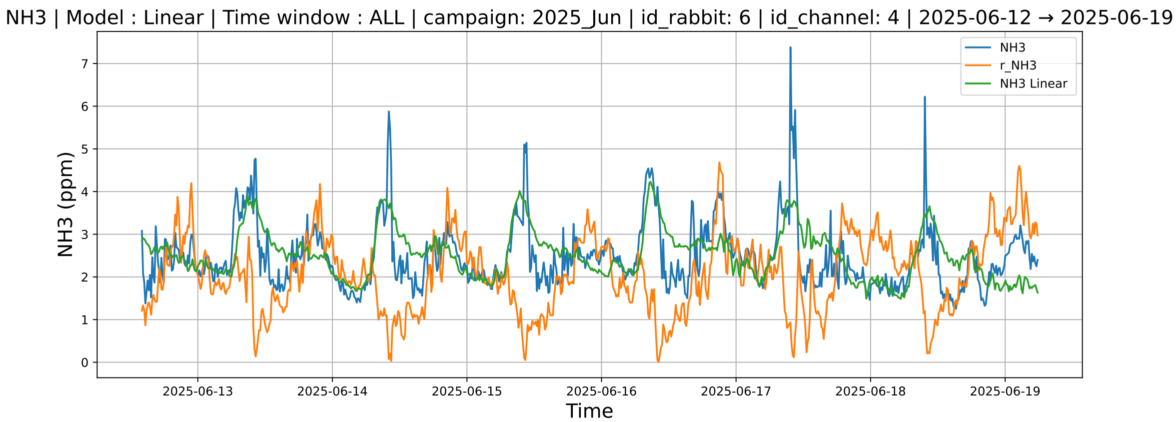
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

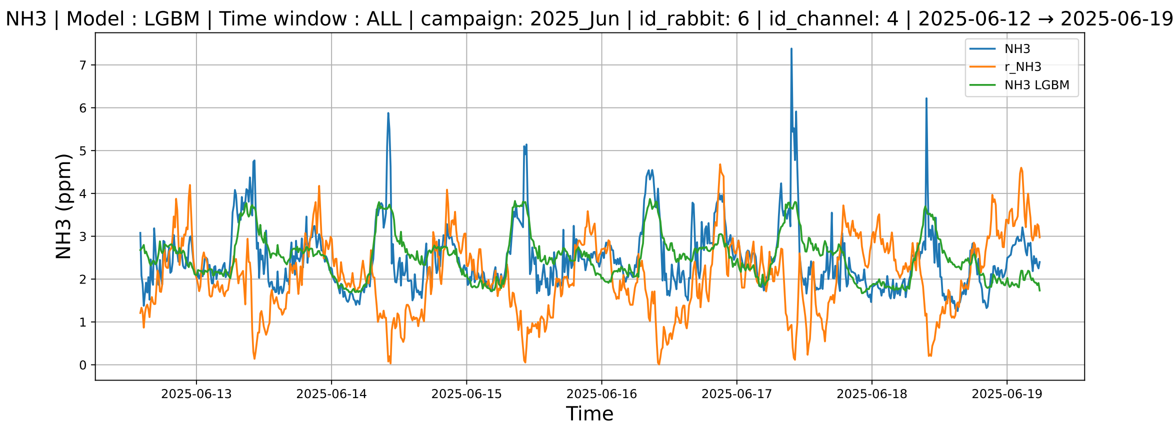


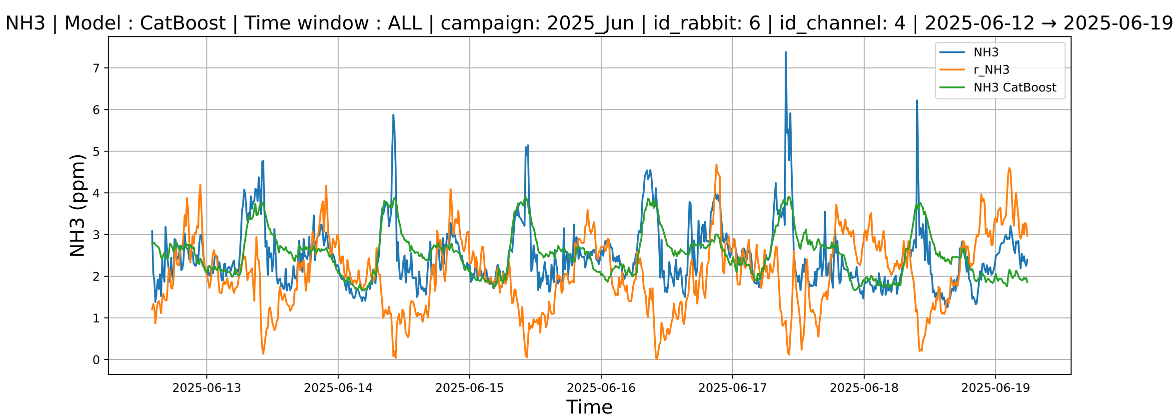
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.1.4.3 Time window : ALL

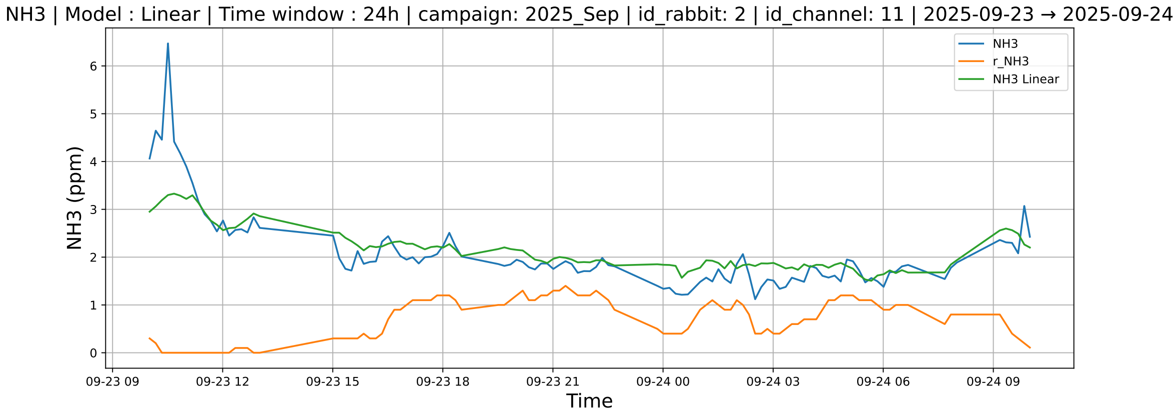


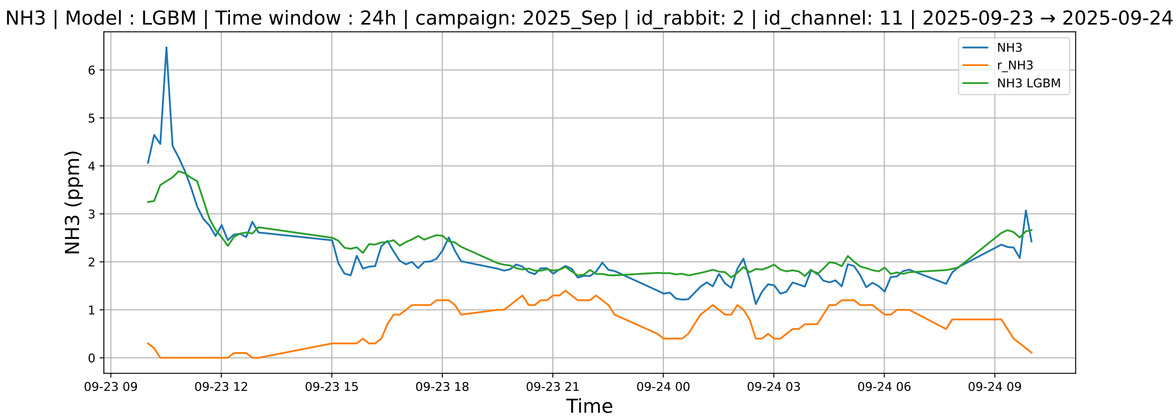


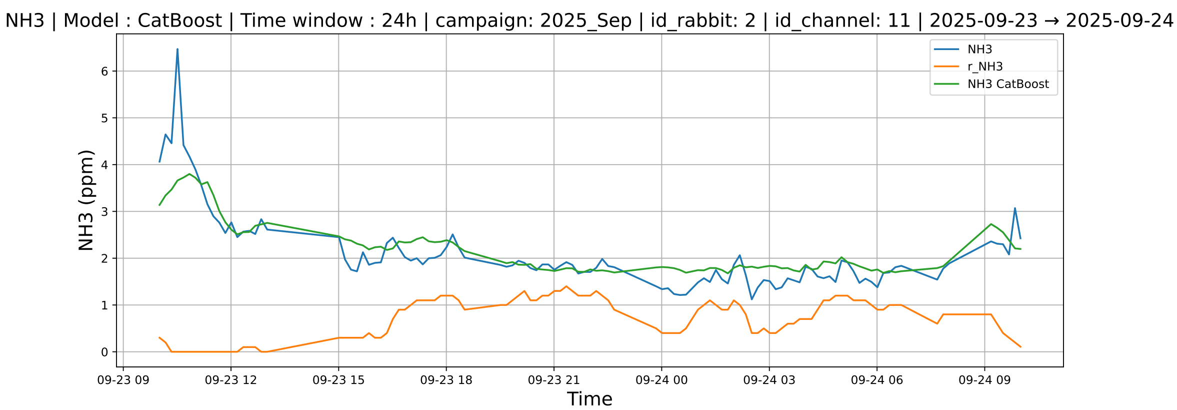


4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

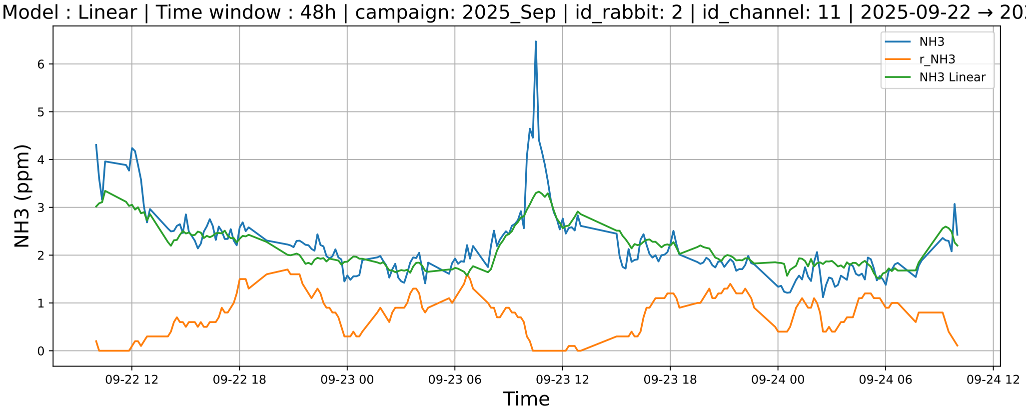




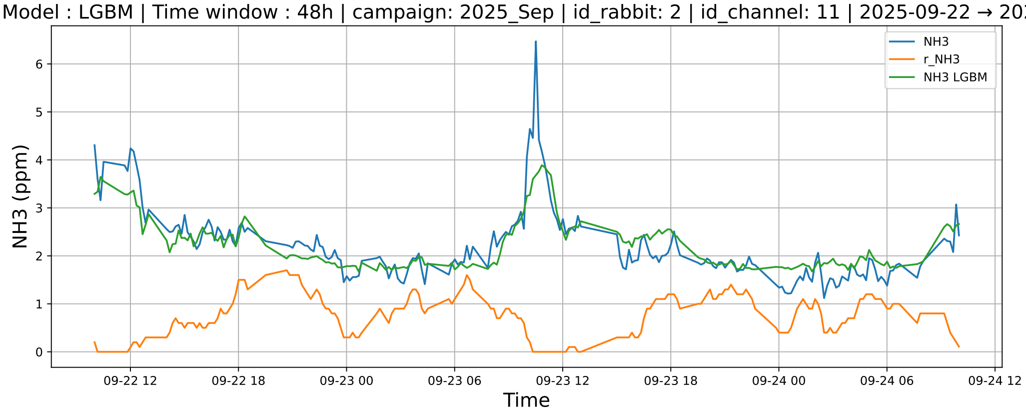


4.1.5.2 Time window : 48

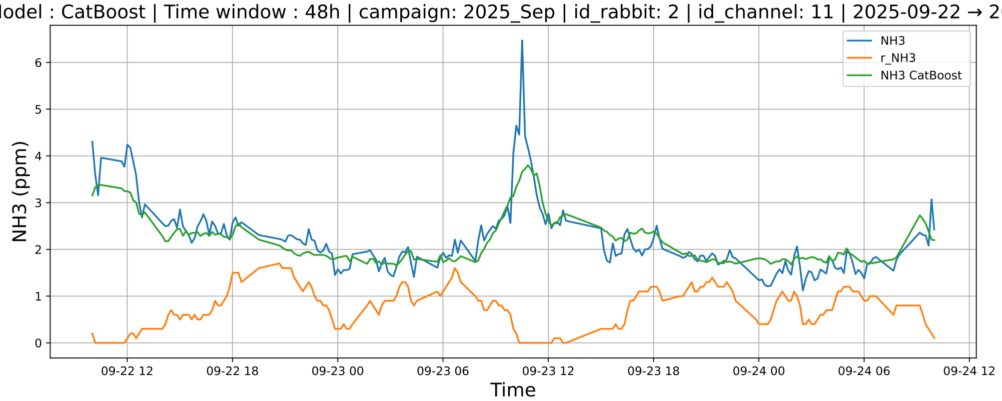
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

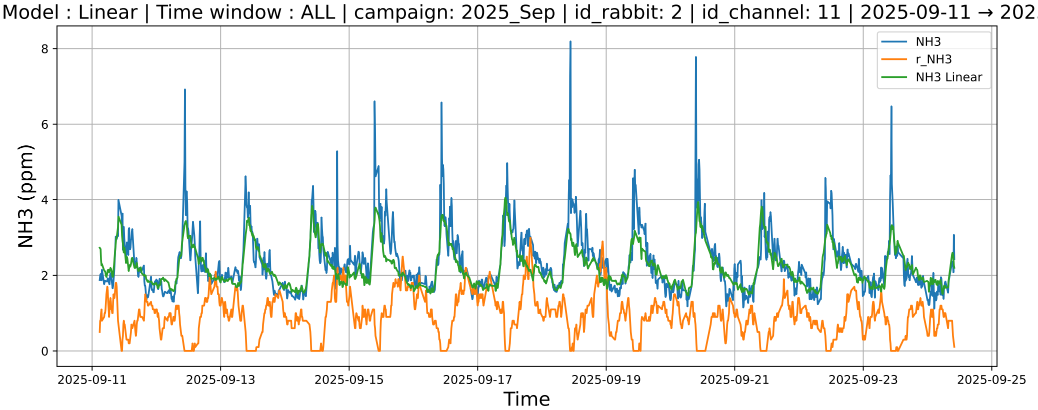


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

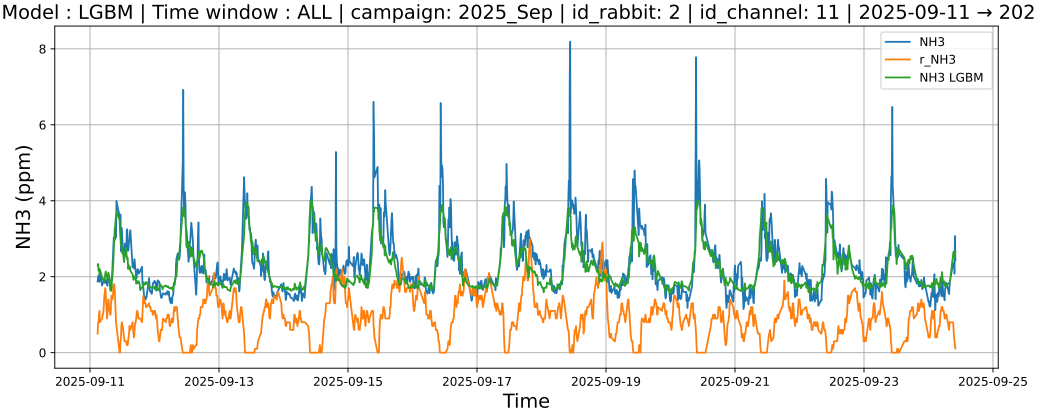


4.1.5.3 Time window : ALL

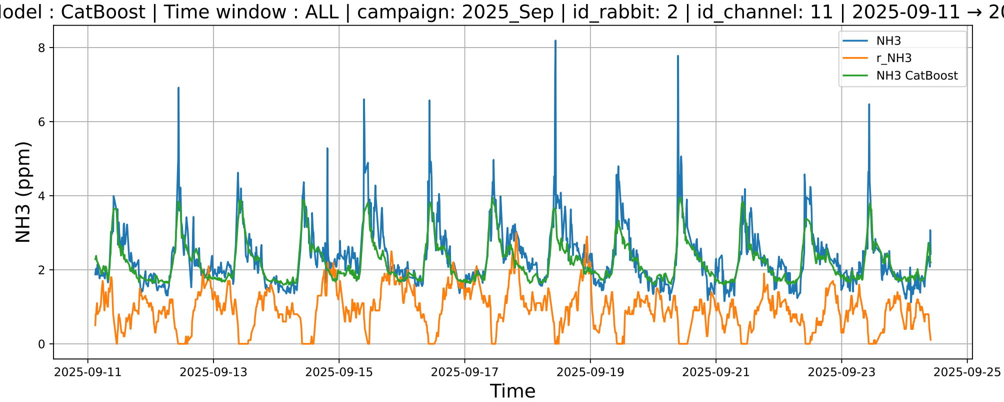
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24

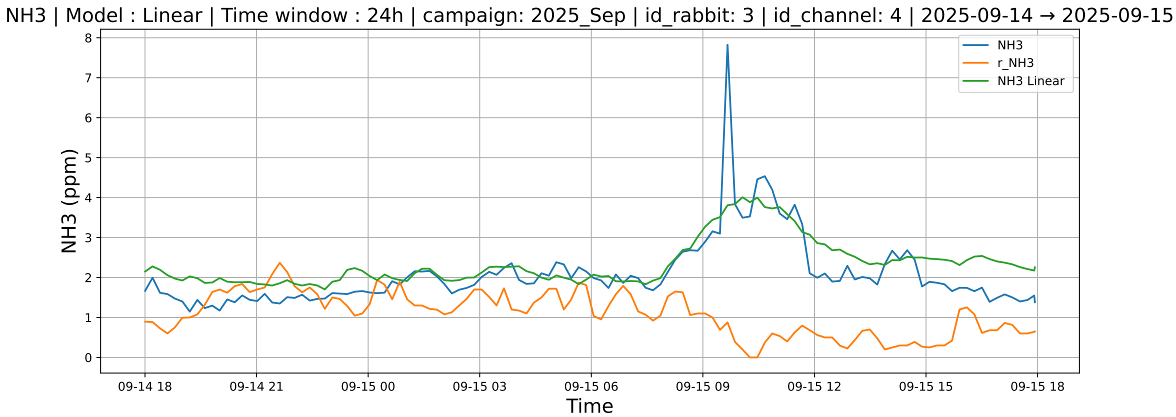


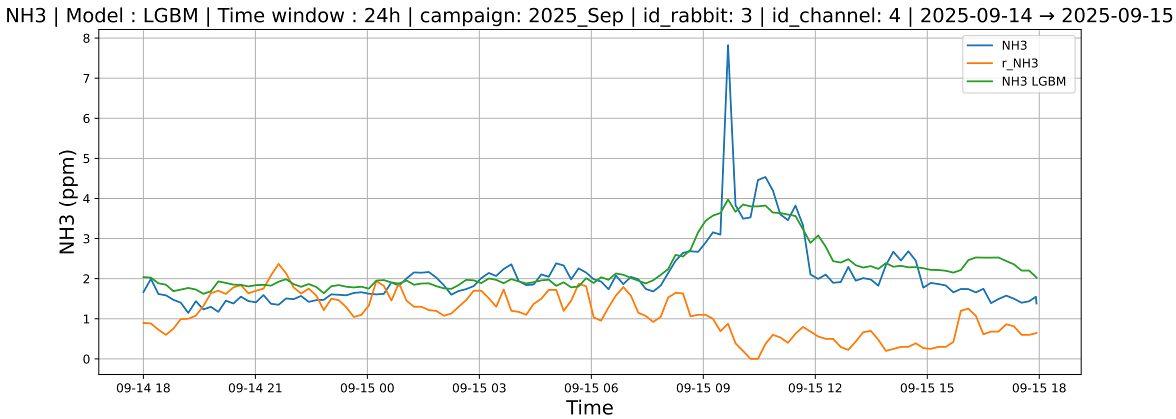
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24

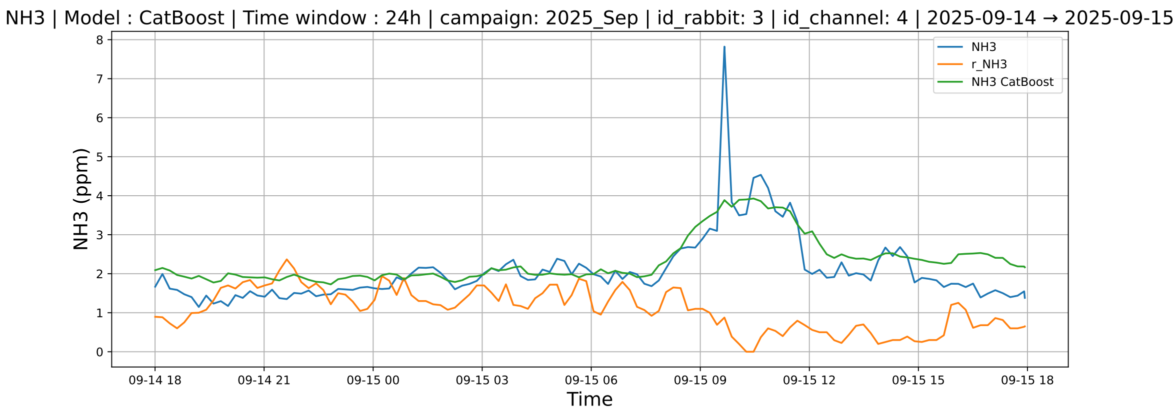


4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

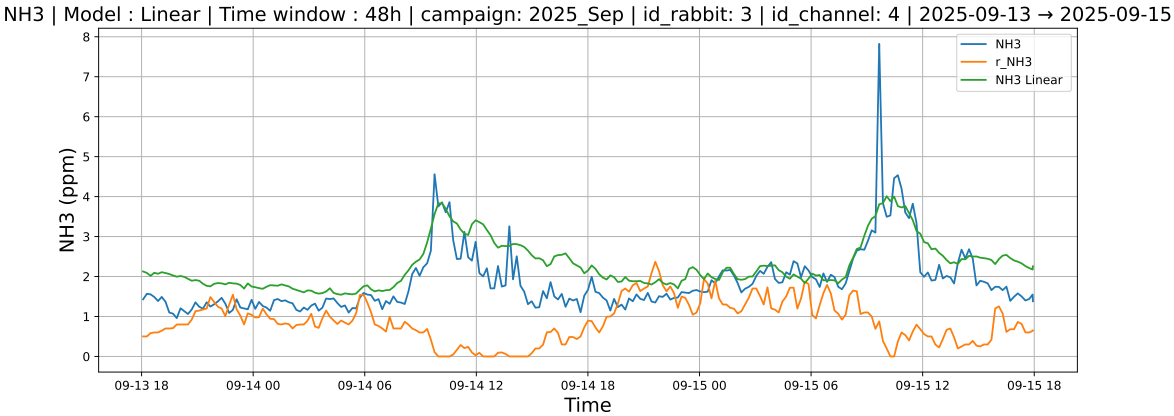
4.1.6.1 Time window : 24

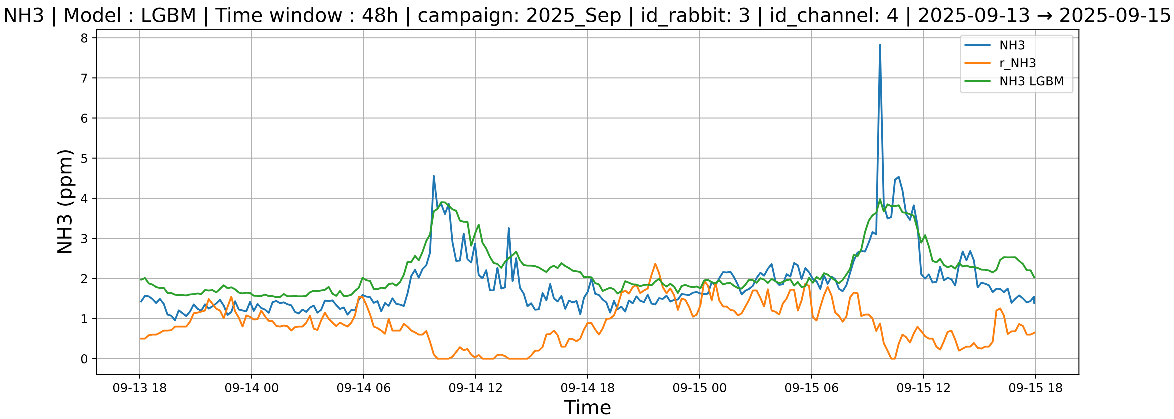


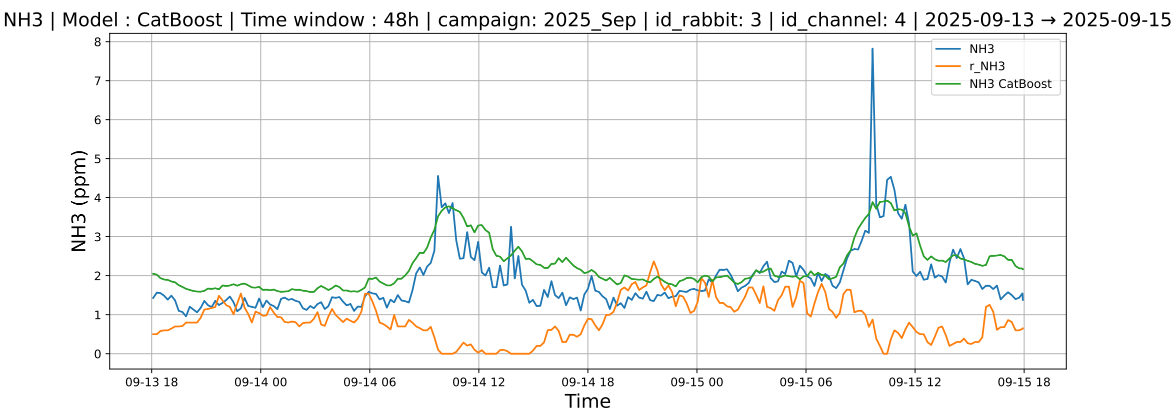




4.1.6.2 Time window : 48

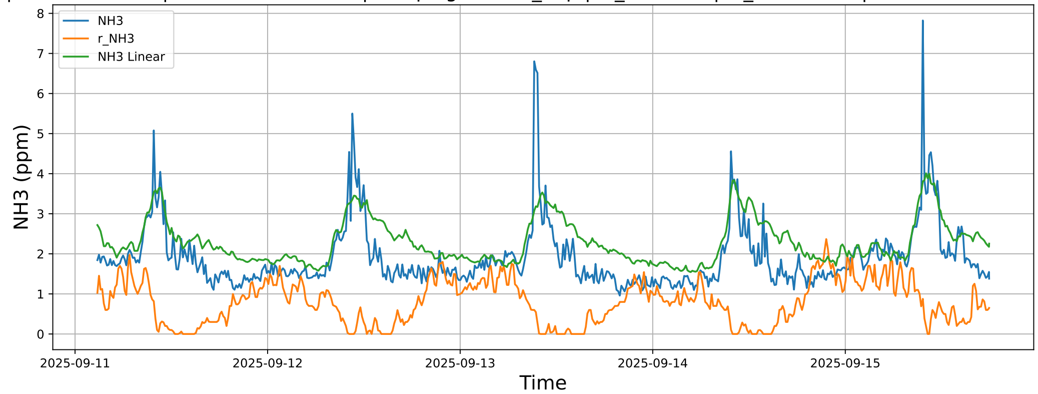




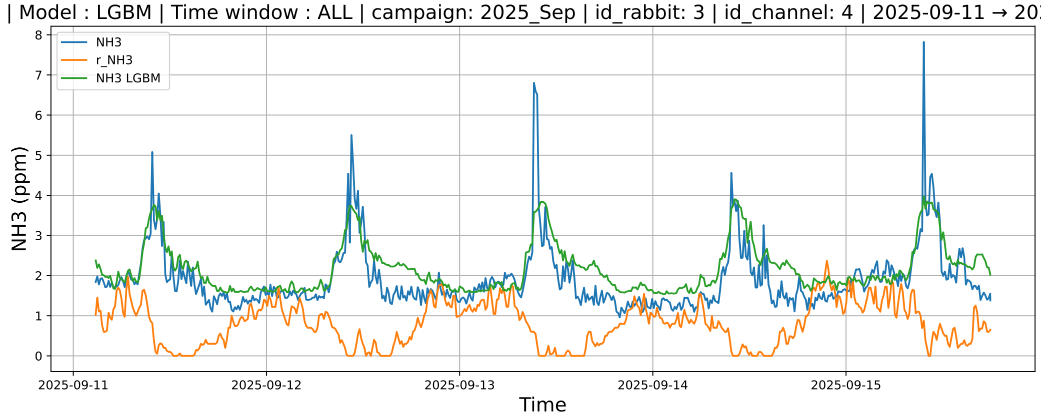


4.1.6.3 Time window : ALL

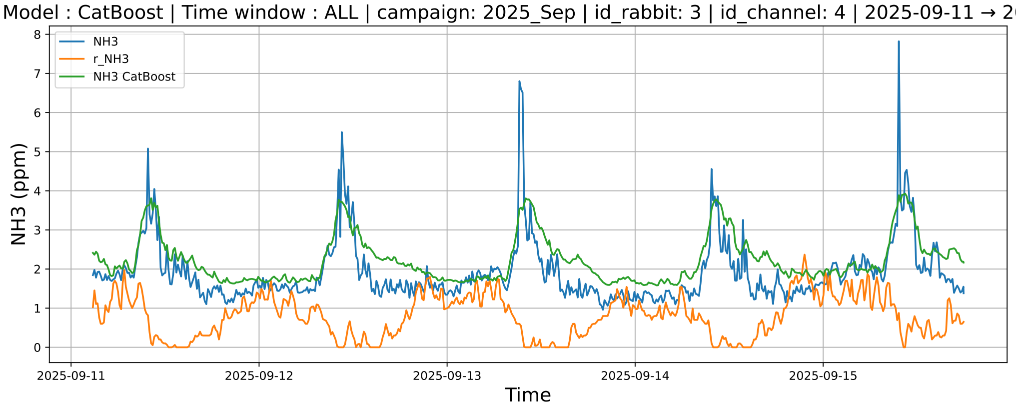
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

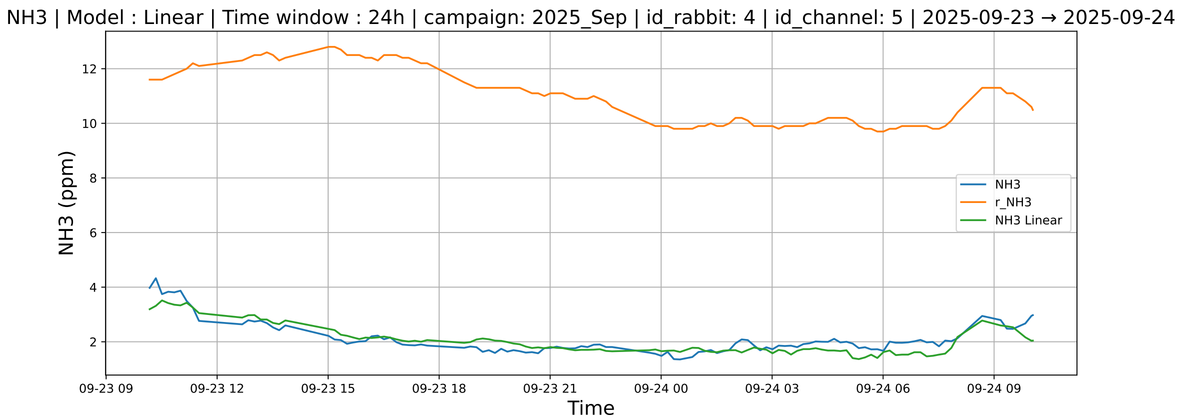


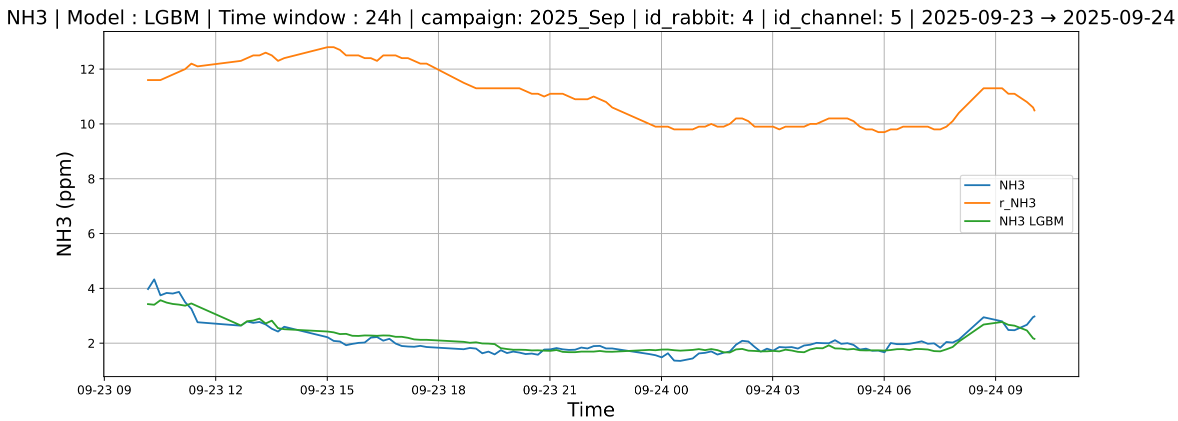
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

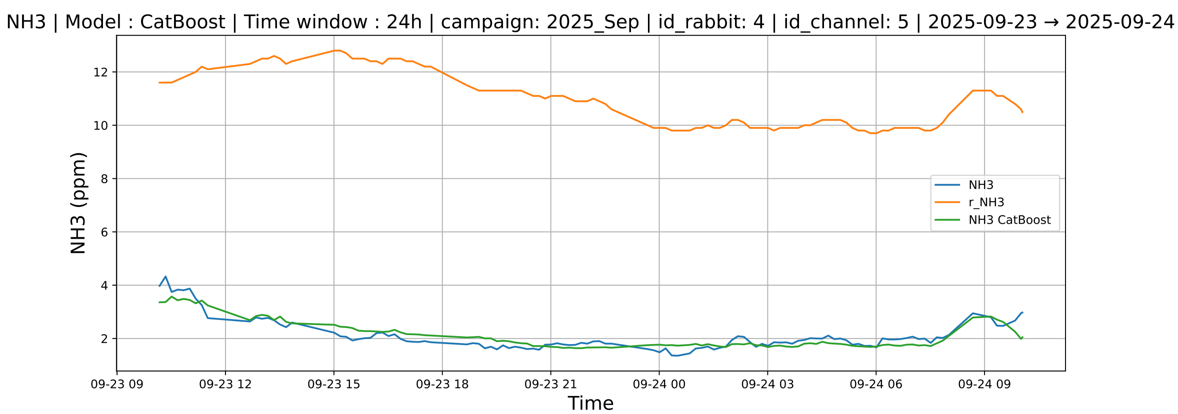


4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

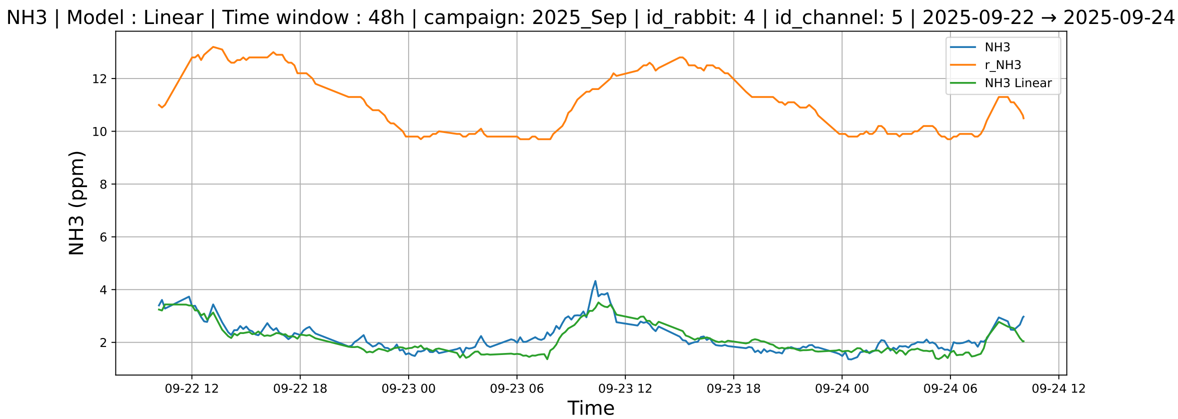
4.1.7.1 Time window : 24

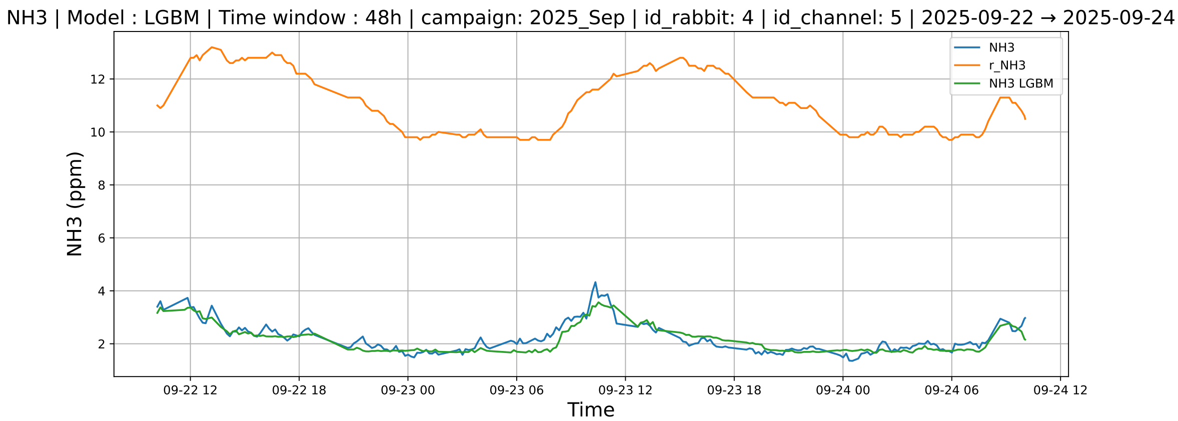


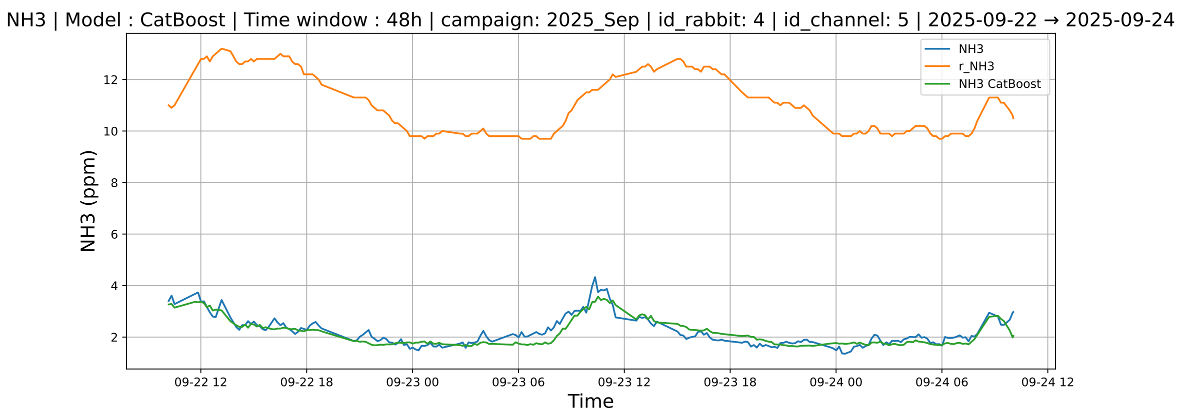




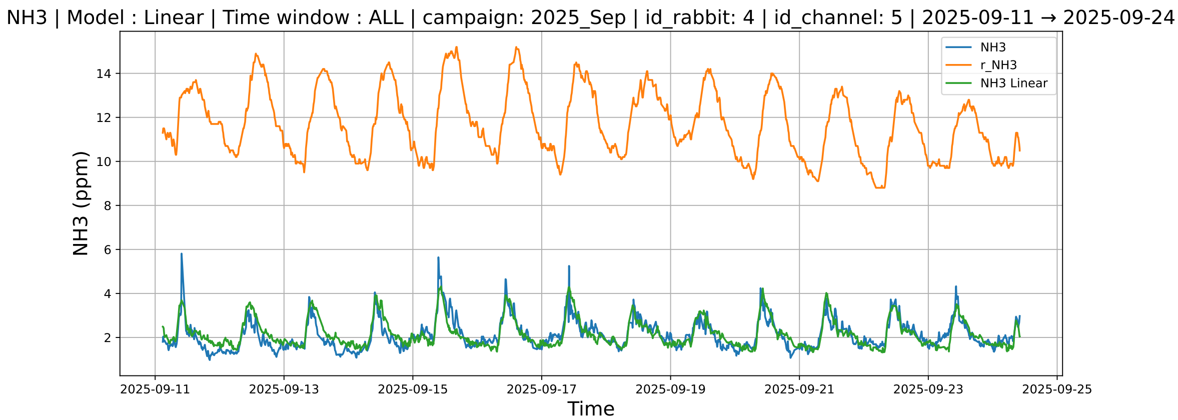
4.1.7.2 Time window : 48

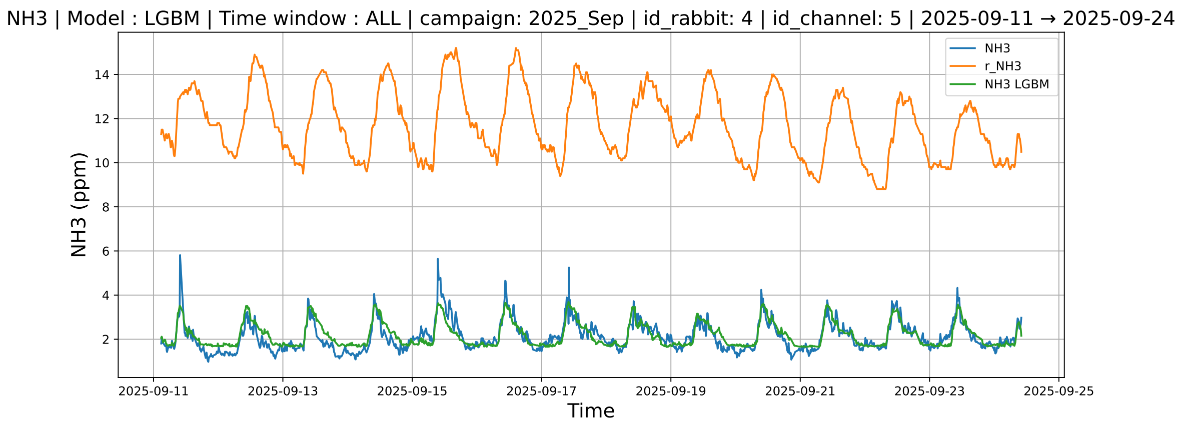


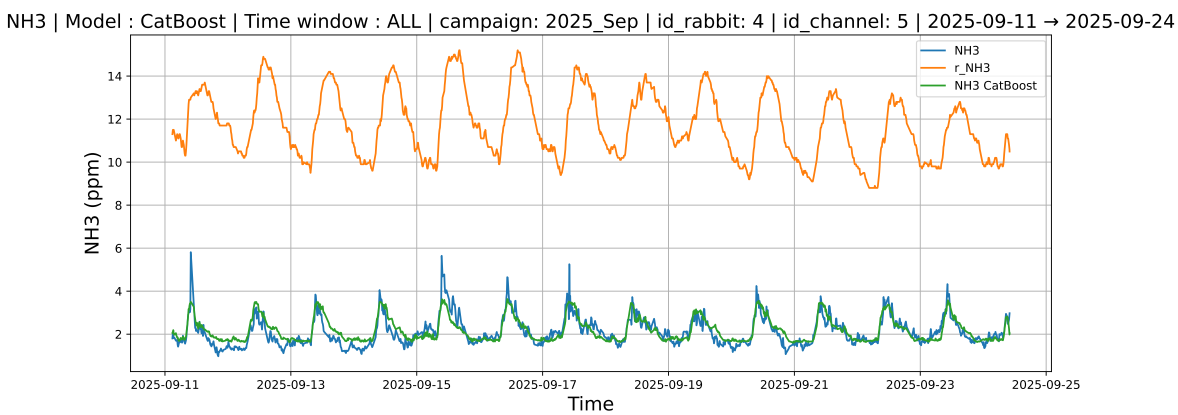




4.1.7.3 Time window : ALL

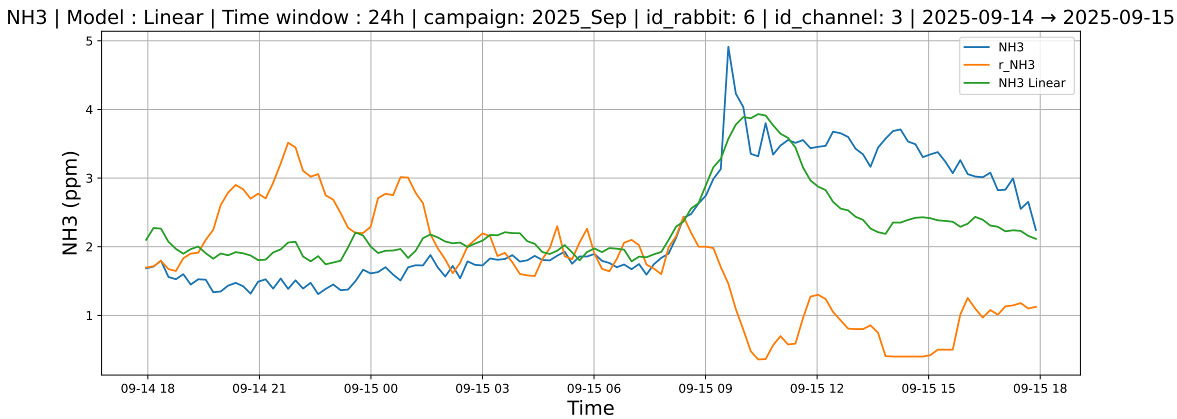


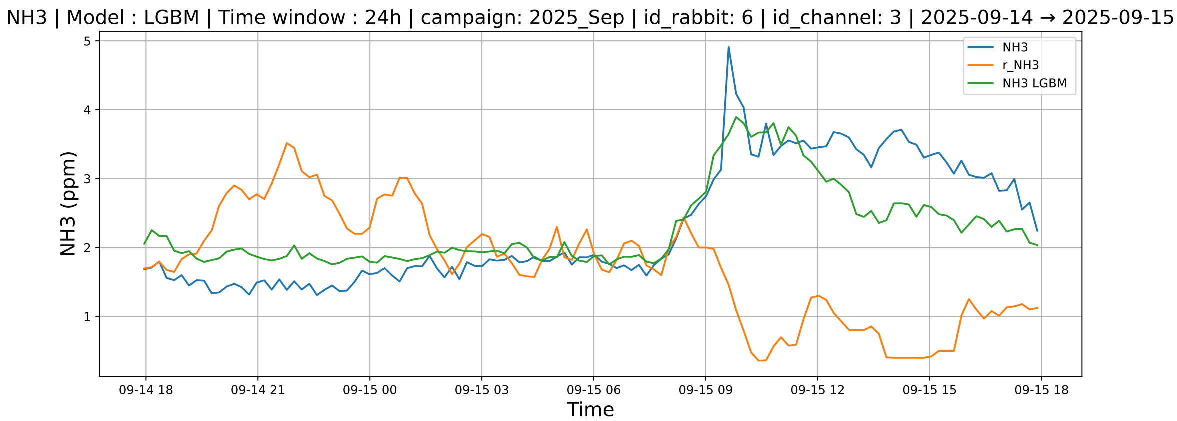


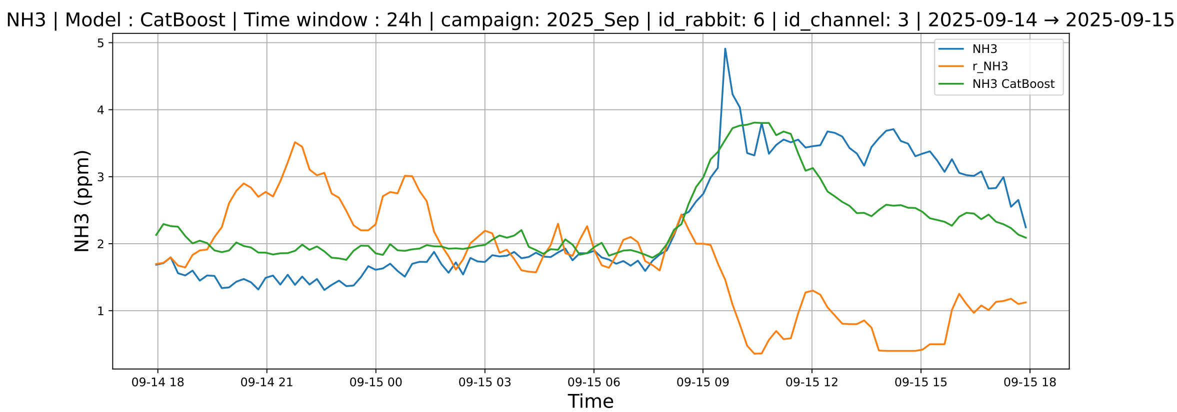


4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

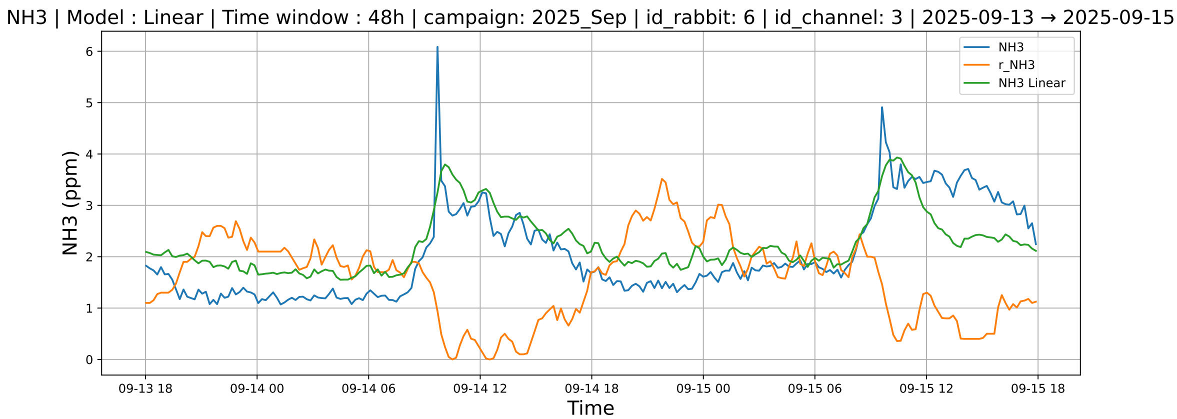
4.1.8.1 Time window : 24

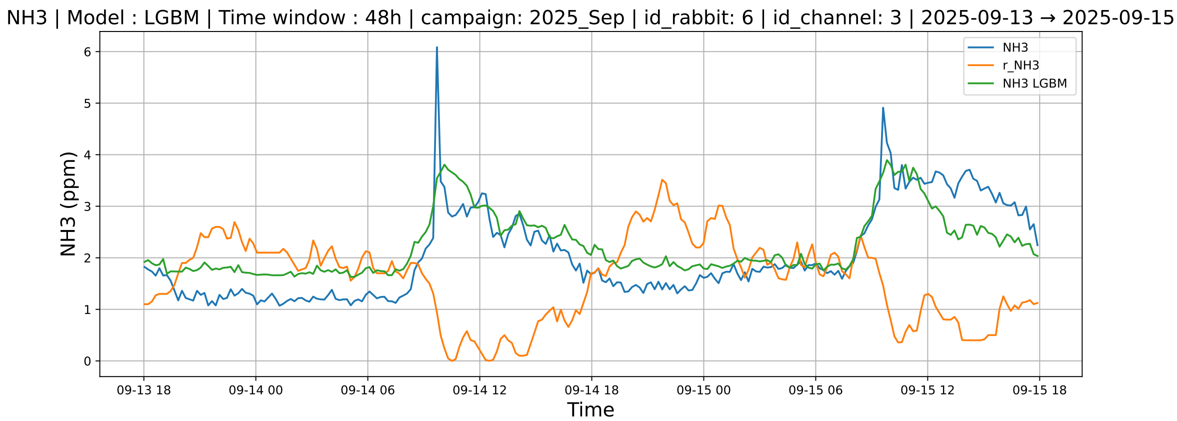


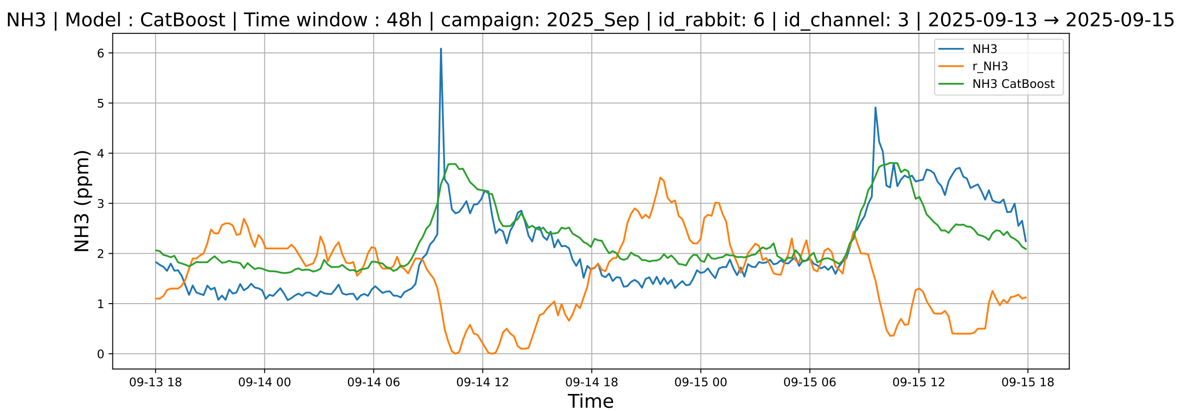




4.1.8.2 Time window : 48

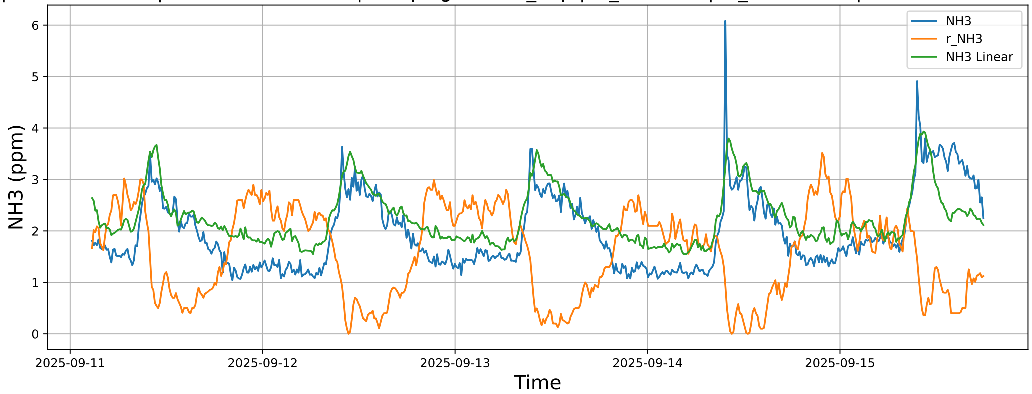




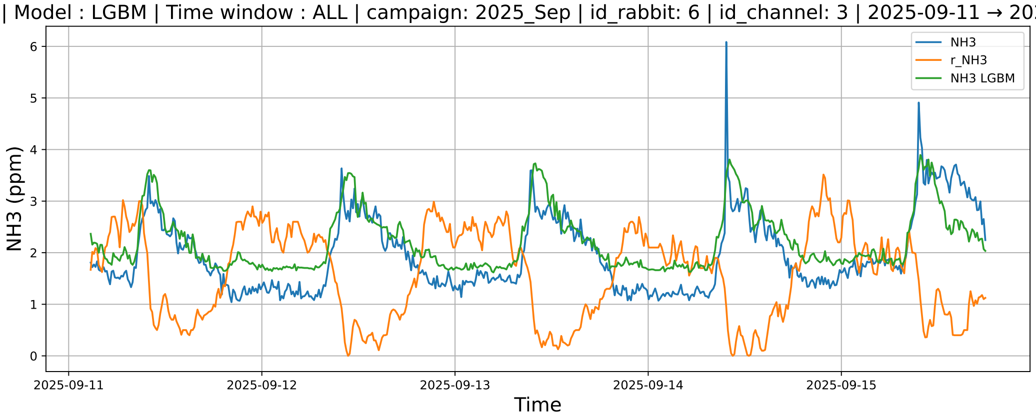


4.1.8.3 Time window : ALL

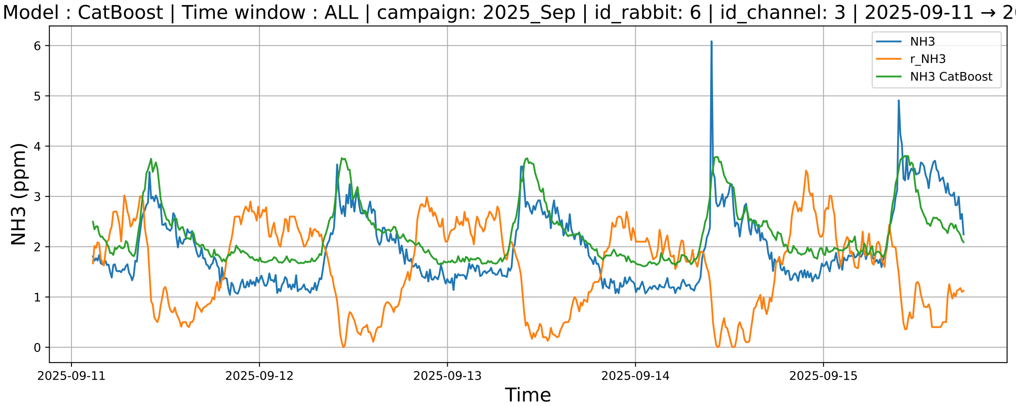
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



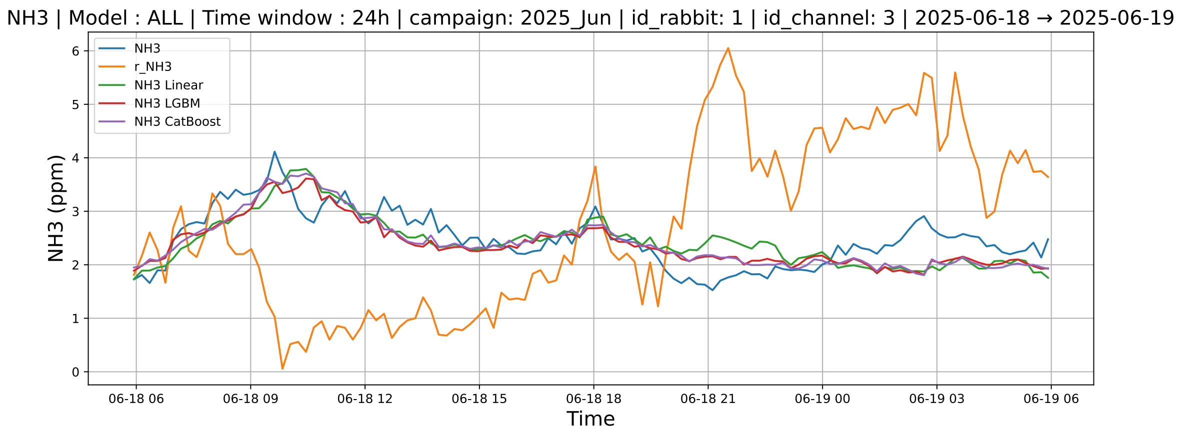
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



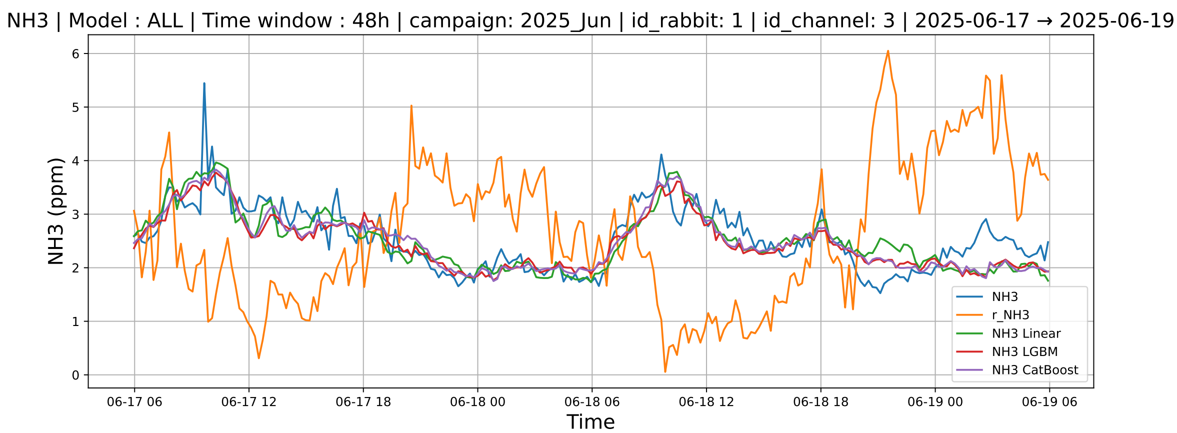
4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

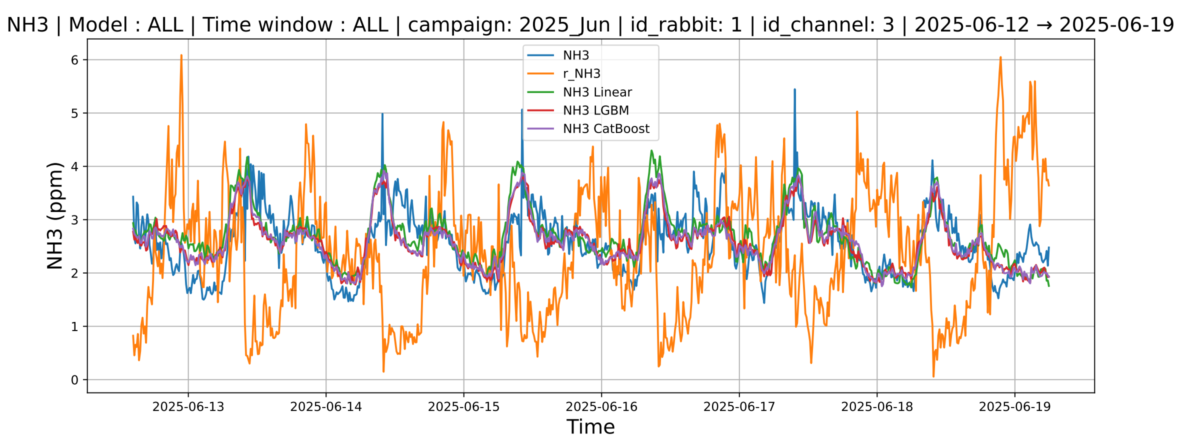
4.2.1.1 Time window : 24



4.2.1.2 Time window : 48



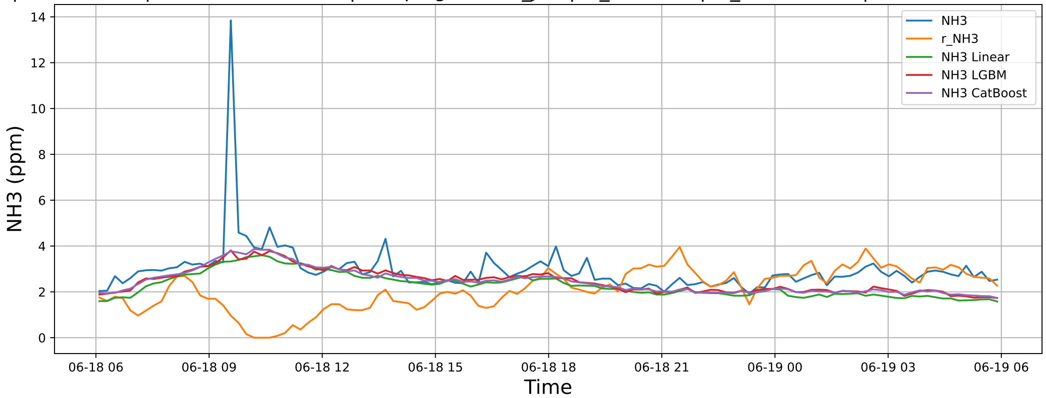
4.2.1.3 Time window : ALL



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

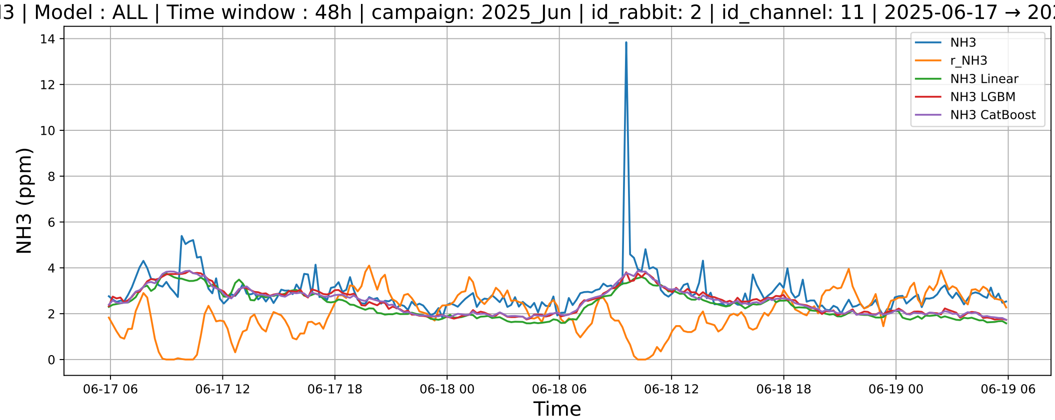
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



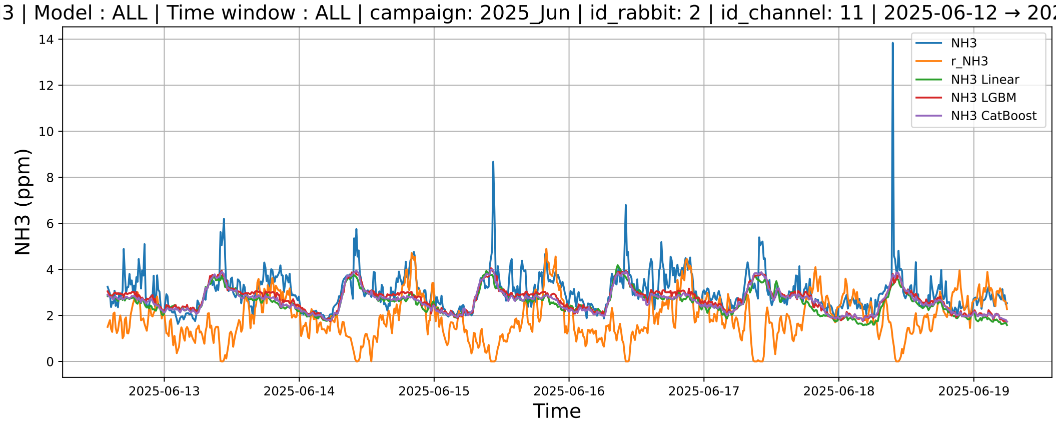
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

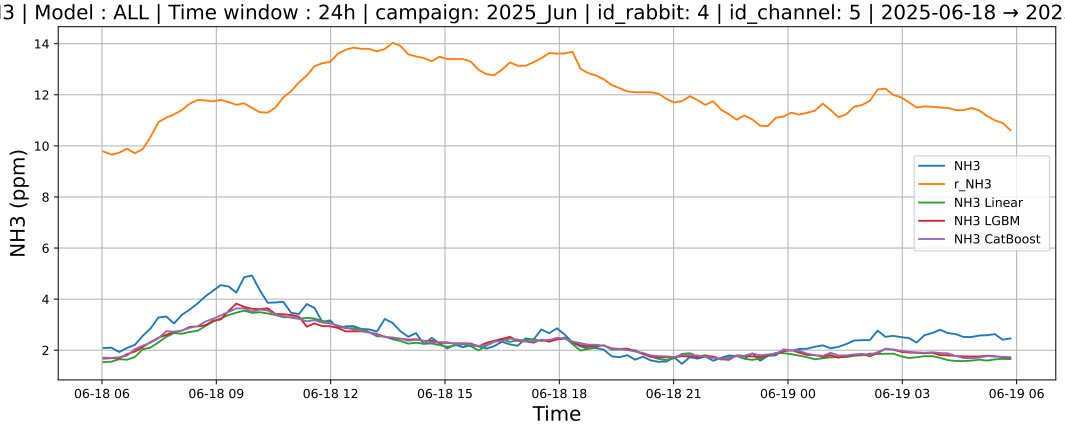
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



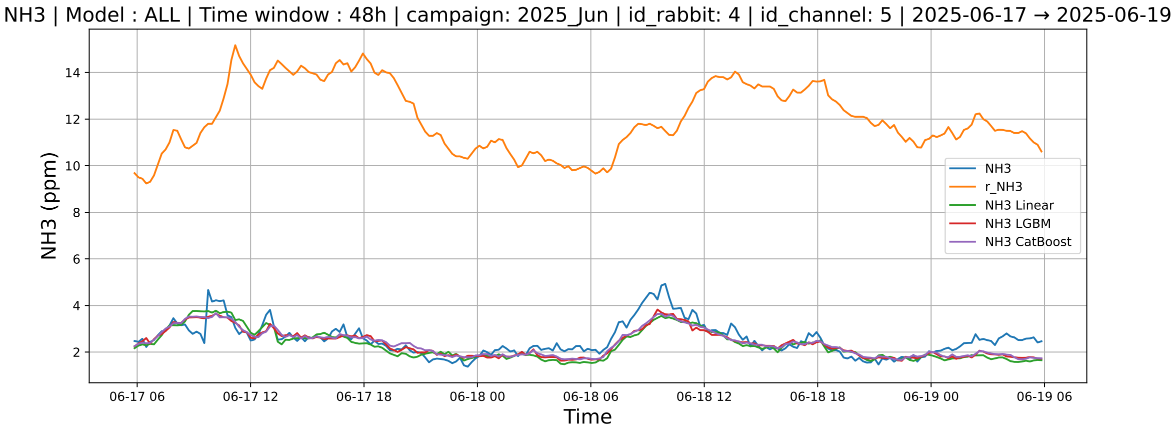
4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.2.3.1 Time window : 24

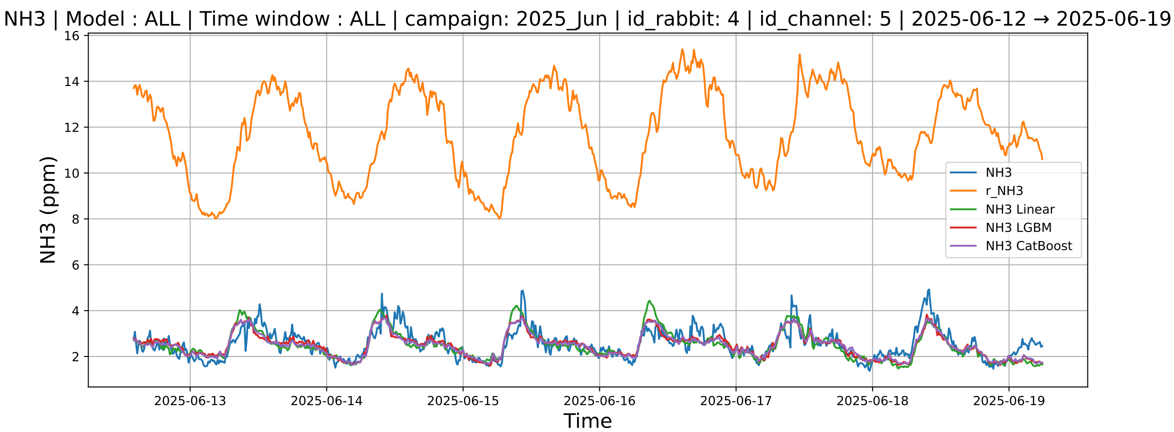
NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



4.2.3.2 Time window : 48

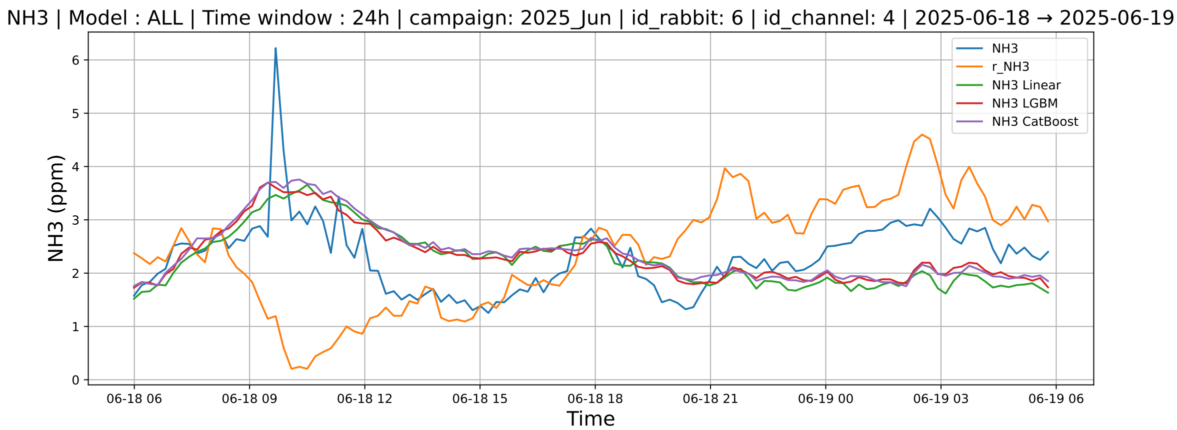


4.2.3.3 Time window : ALL

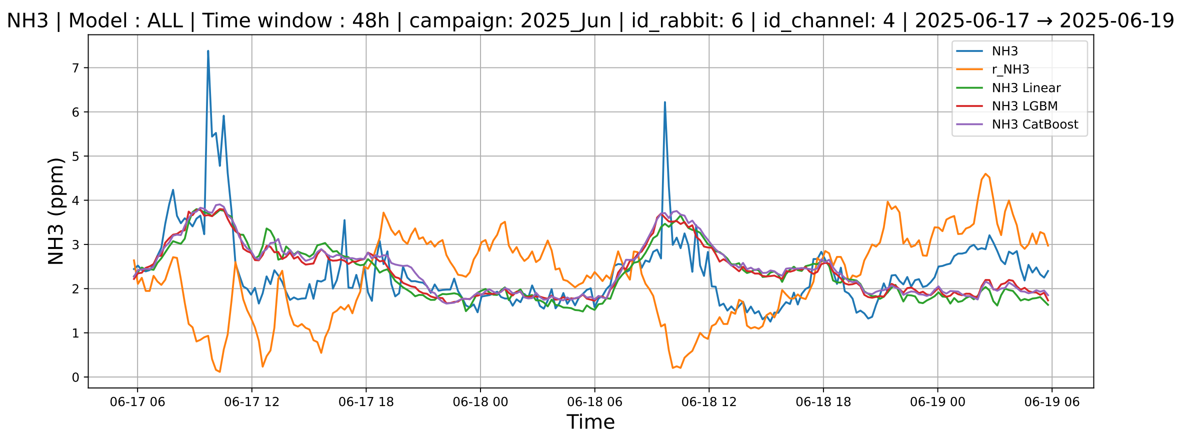


4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

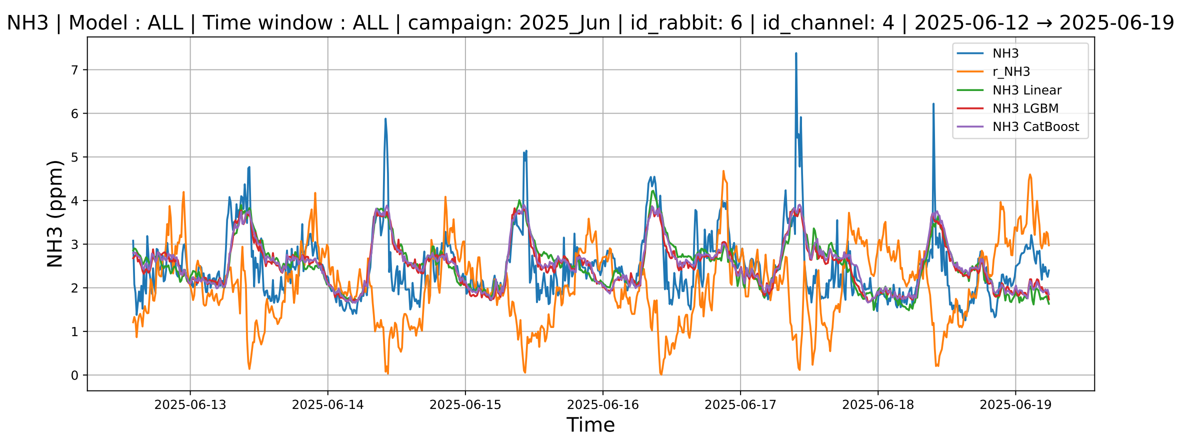
4.2.4.1 Time window : 24



4.2.4.2 Time window : 48



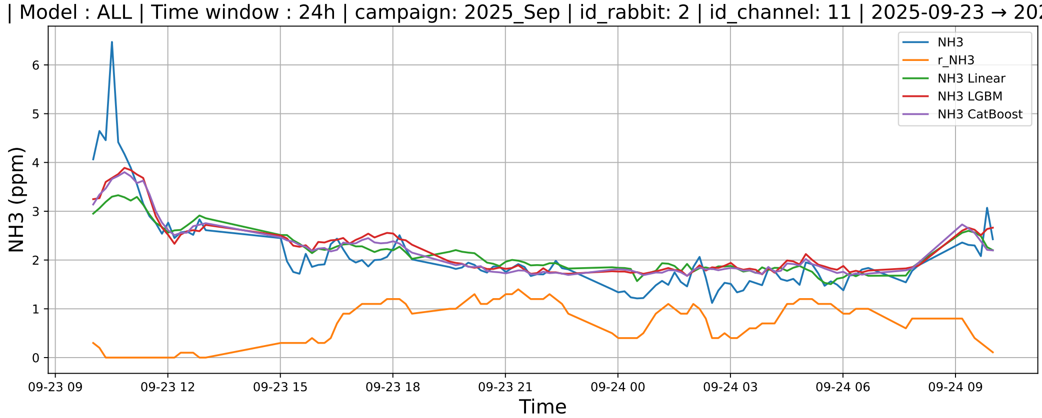
4.2.4.3 Time window : ALL



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

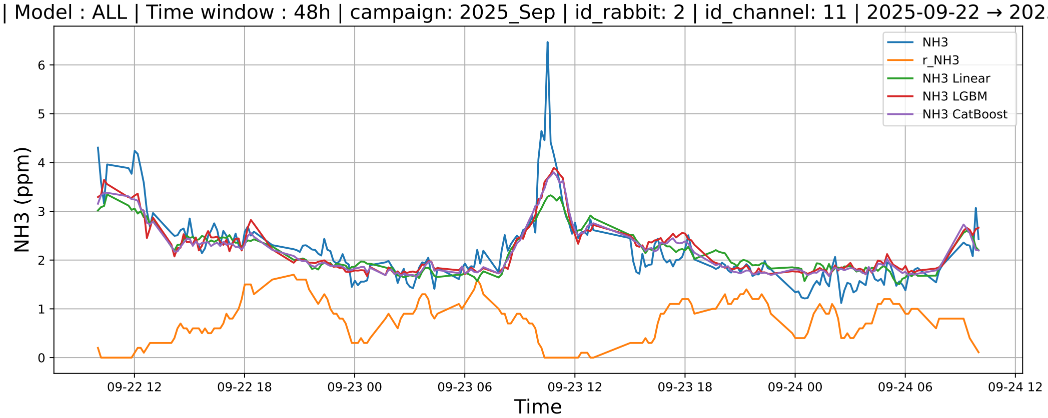
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

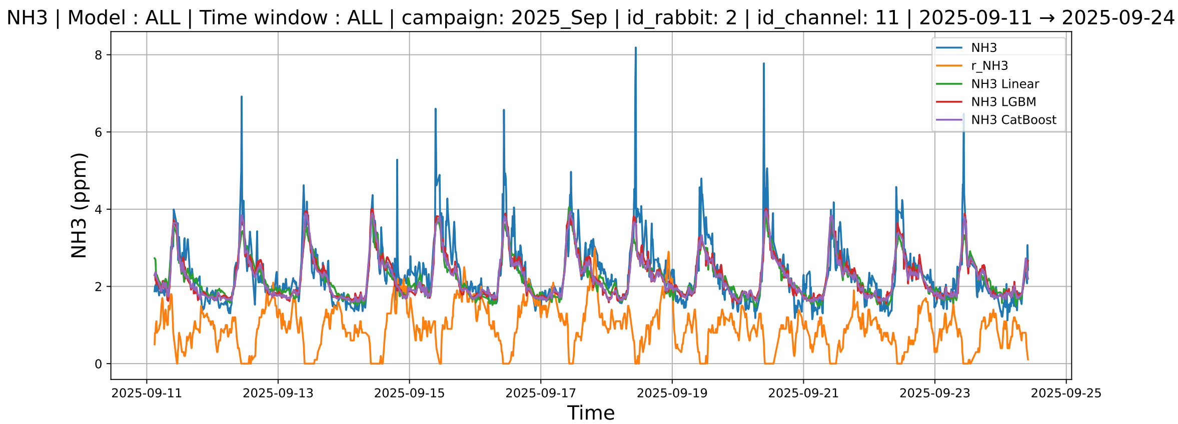


4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

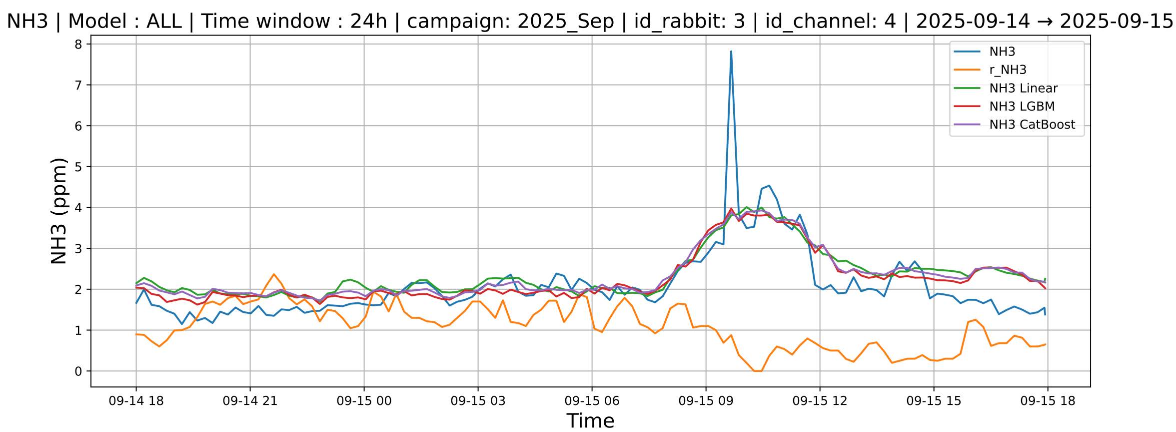


4.2.5.3 Time window : ALL



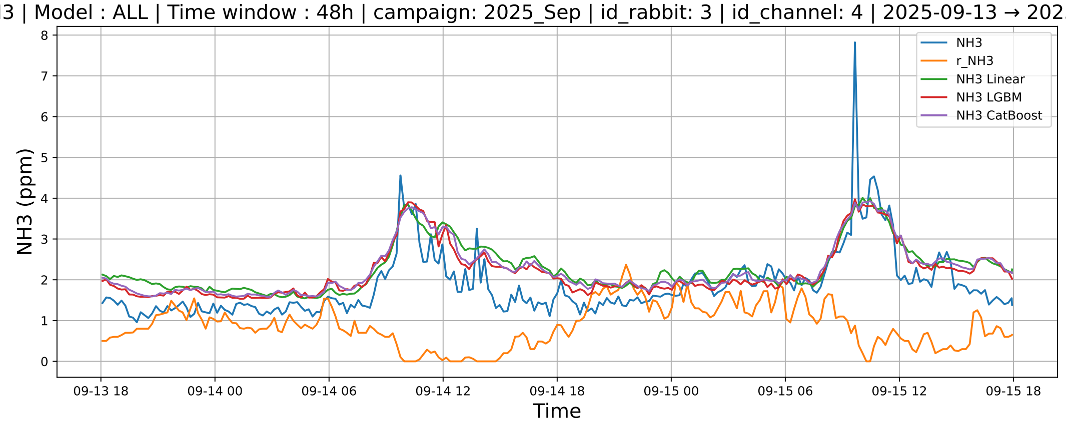
4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.2.6.1 Time window : 24



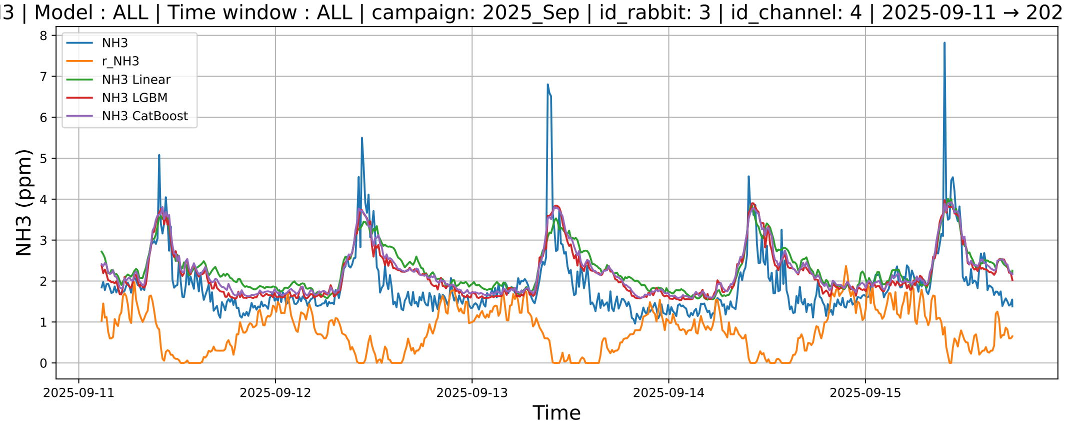
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

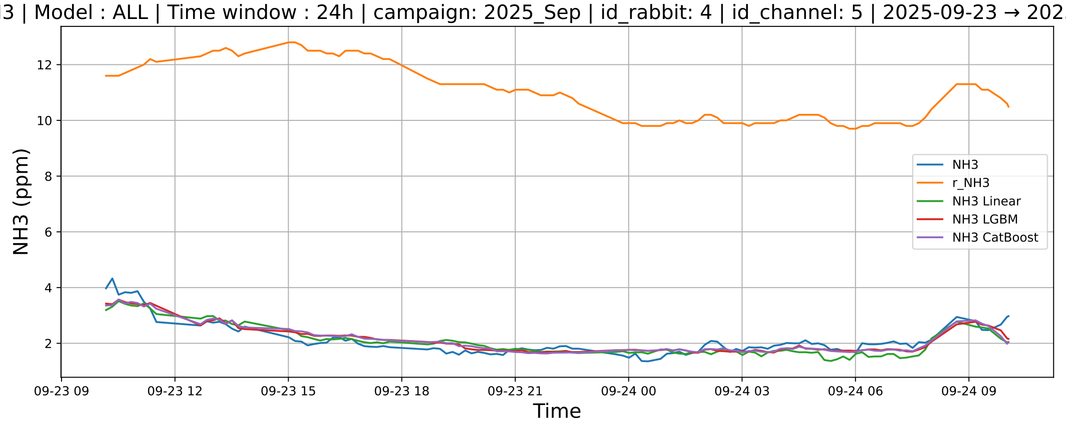
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

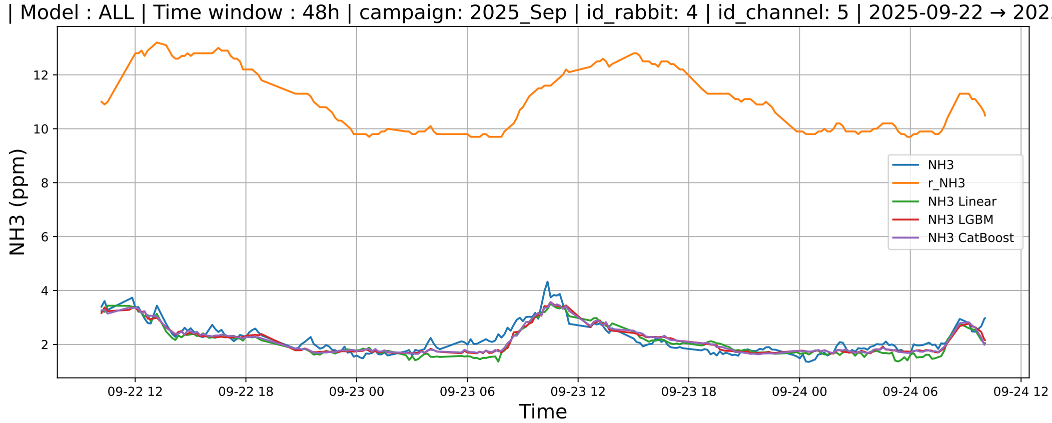
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



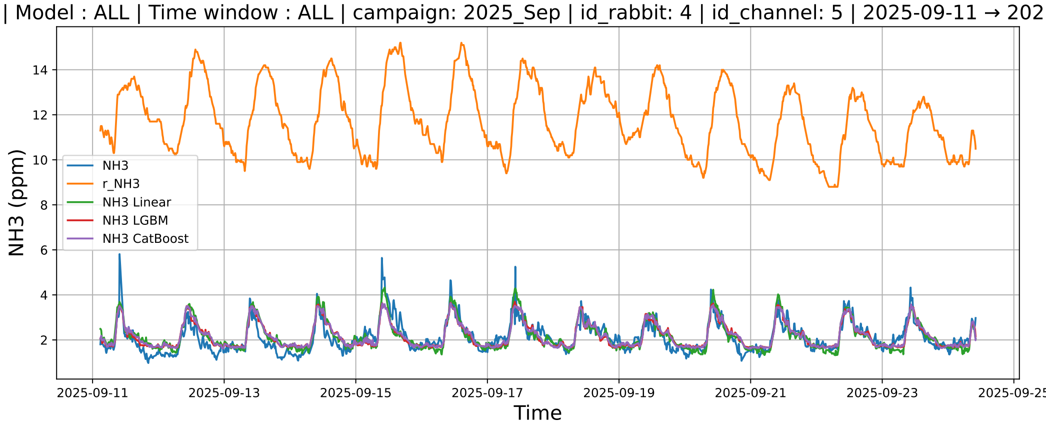
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



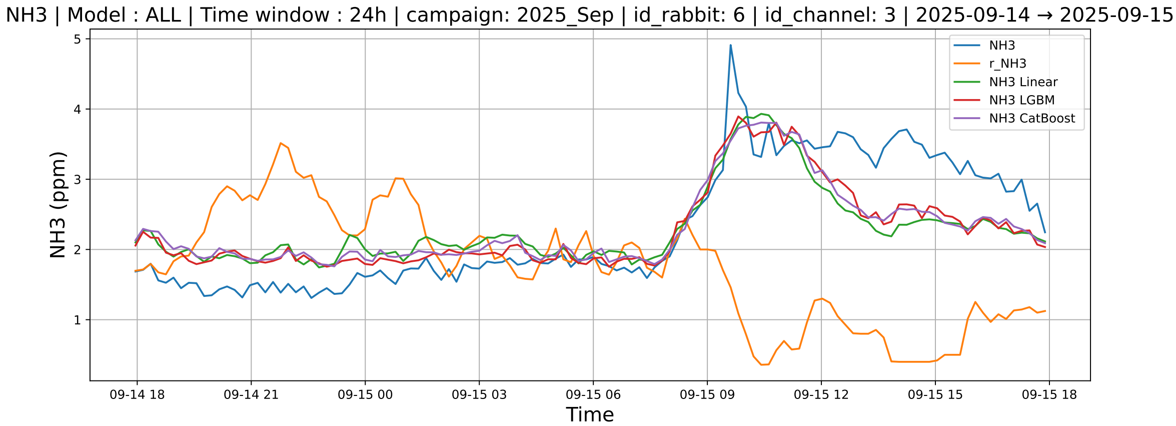
4.2.7.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

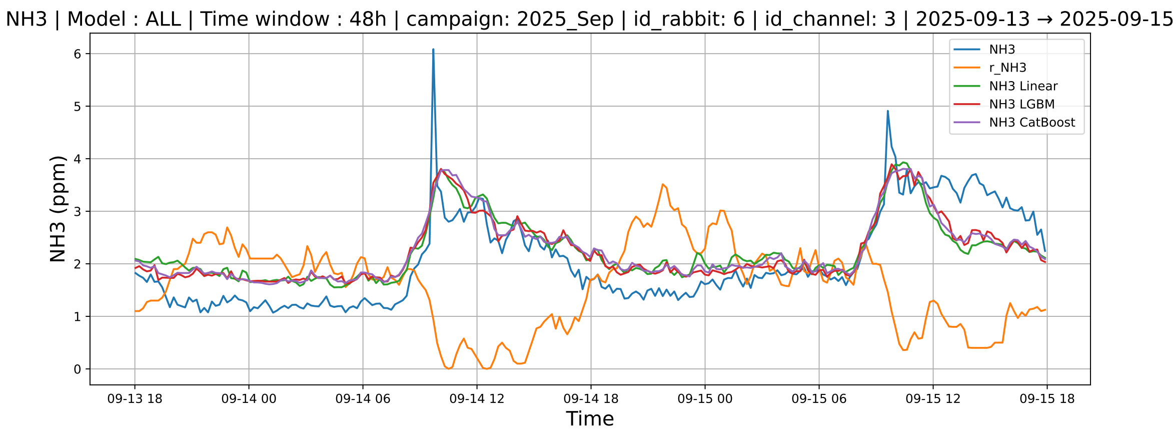


4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

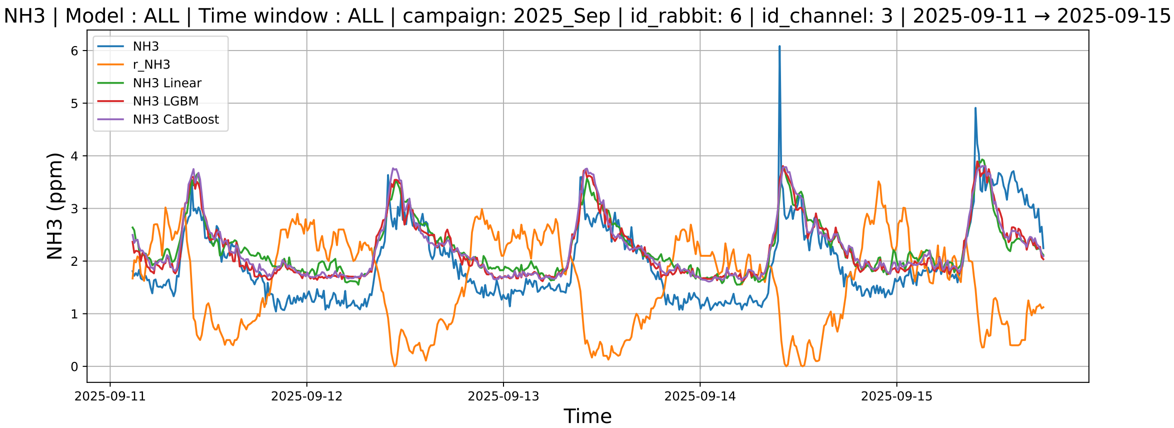
4.2.8.1 Time window : 24



4.2.8.2 Time window : 48



4.2.8.3 Time window : ALL


NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15
NH3
r_NH3
NH3 Linear
NH3 LGBM
NH3 CatBoost
NH3 (ppm)
0
1
2
3
4
5
6
2025-09-11
2025-09-12
2025-09-13
2025-09-14
2025-09-15
Time